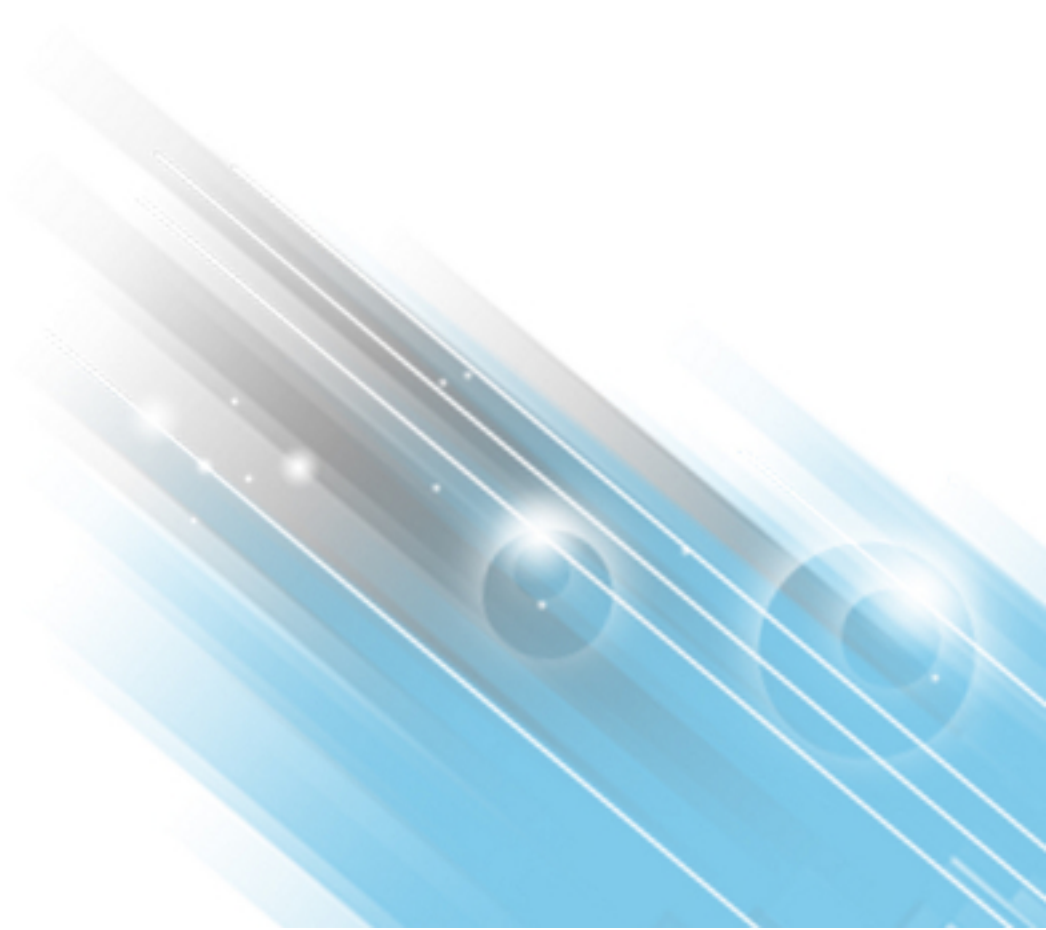




HP LoadRunner

Software Version: 12.02

HP LoadRunner Tutorial



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Welcome to the LoadRunner Tutorial

Performance Testing Overview

We recommend that you watch How HP LoadRunner Works on the HP Video Channel.

LoadRunner Basics

For an introduction to using LoadRunner for performance testing, see a short LoadRunner Product Walk-Through. This will give you an understanding of some of the LoadRunner basics and terminology.

The HP Enterprise Business Channel provides many videos about Load Testing..



What will you learn during this tutorial?

The LoadRunner tutorial is a self-paced printable guide, designed to lead you through the process of load testing and familiarize you with the LoadRunner testing environment.

- This lesson provides an introduction to LoadRunner and testing concepts.
- Lessons 1 through 4 describe how to develop a Vuser script using VuGen - the Virtual User Generator.
- Lessons 5 and 6 explain how to design and run load tests using the LoadRunner Controller.
- Lesson 7 introduces the Analysis tool, showing you how to create graphs and reports which will help you analyze your load test.

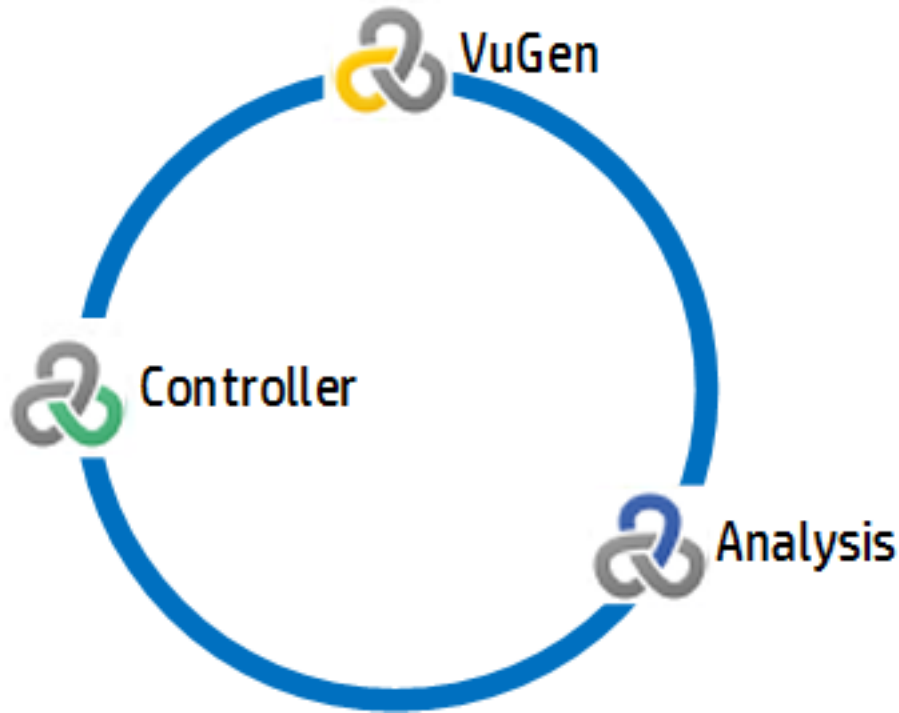
At the conclusion of this tutorial, you will be ready to design, run, and monitor a simple load test on your own system. It is recommended that you work through the tutorial in the order in which the information is presented.

What are the LoadRunner components?

LoadRunner contains the following components:

- The **Virtual User Generator** or **VuGen** records end-user business processes and creates an automated performance testing script, known as a Vuser script.

- The **Controller** organizes, drives, manages, and monitors the load test.
- **Analysis** helps you view, dissect, and compare the results of the load tests.



- **Load Generators**, computers that run Vusers to generate a load on the system.

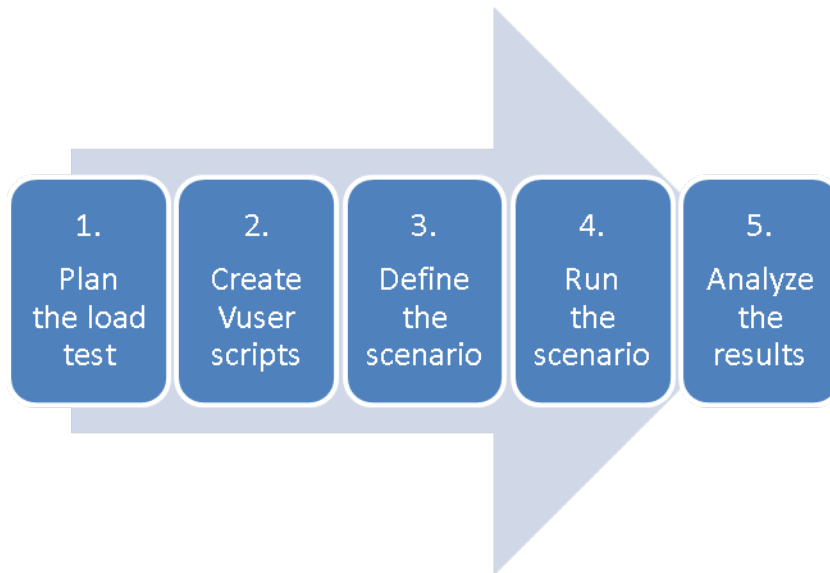
LoadRunner Terminology

Term	Description
Scenario	Defines the events that occur during a testing session, based on performance requirements.
Virtual Users or Vusers	Vusers emulate the actions of human users working on your system. A scenario can contain tens, hundreds, or even thousands of Vusers.
Vuser Script	The recorded actions of a business process performed in your application.
Protocol	A protocol is the method of communication between a client and the server.
Transaction	To measure the performance of your system, you define transactions . A transaction represents one or more end-user business processes. A transaction enables you to measure how long it takes to perform these business processes.

Term	Description
Script footprint	Defined by the quantities of the various resources that are required on a load generator in order to execute the Vuser script. Typical resources include memory, CPU power, and disk space.

What is the LoadRunner load testing process?

Load testing with LoadRunner typically consists of five phases: planning, script creation, scenario definition, scenario execution, and results analysis.



1. **Plan Load Test.** Define your performance testing requirements, for example, number of concurrent users, typical business processes, and required response times.
2. **Create Vuser Scripts.** Use VuGen to capture the end-user activities into automated scripts.
3. **Define a Scenario.** Use the Controller to set up the load test environment.
4. **Run a Scenario.** Use the Controller to drive, manage, and monitor the load test.
5. **Analyze the Results.** Use LoadRunner Analysis to create graphs and reports, and evaluate the system performance.

Starting HP Web Tours

To illustrate LoadRunner as a solution for load testing, this tutorial uses performance requirements for a sample application. The sample application, *HP Web Tours*, is a web-based travel agency system. *HP Web Tours* users connect to a web server, search for flights, book flights, and check flight itineraries.

While LoadRunner supports over 50 types of applications, this tutorial demonstrates how to load test a Web-based application. If you are load testing applications that are not Web-based, please contact HP for assistance.

In this section of the tutorial, you will learn how to start and log on to *HP Web Tours*.

1. Start the sample Web server.

Select **Start > All Programs > HP Software > HP LoadRunner > Samples > Web > Start HP Web Tours Server**. In icon-based desktops, such as Windows 8, search for "**Start HP**" and select **Start HP Web Tours Server** from the results.

The Start Web Server dialog box opens. Keep this dialog box open while you access the Web Tours application.

2. Open HP Web Tours.

Select **Start > All Programs > HP Software > HP LoadRunner > Samples > Web > HP Web Tours Application**. In icon-based desktops, such as Windows 8, search for "**HP Web**" and select **HP Web Tours Application** from the results. A browser opens and displays the *HP Web Tours* home page.

Note: Ensure that LoadRunner is installed in the default folder on your computer. If LoadRunner is installed in a non-default folder, the HP Web Tours application will not open.

Note: The HP Web Tours application requires a browser with Java installed. For more information, refer to the relevant Java documentation.

3. Log on to HP Web Tours.
 - a. Type in the following information:
 - User name: **jojo**
 - Password: **bean**
 - b. On the left pane, click **Login**. HP Web Tours welcomes you to the application.
4. Reserve a flight.
 - a. On the left pane click **Flights**. The Find Flight page opens.
 - b. Change the **Arrival city** to **Los Angeles**.
 - c. Click **Continue**.
 - d. In the Find Flight page, accept the default flight selection and click **Continue**.

- e. In the Payment Details page, click **Continue**. The Invoice page is displayed, summarizing your flight reservation.
5. End your HP Web Tours session.

On the left pane, click **Sign Off** to log off.

Defining the Application Performance Requirements

Now that you are familiar with HP Web Tours, imagine that you are the performance engineer responsible for signing off that HP Web Tours meets the needs of your business. Your project manager has given you 4 criteria for release:

1. HP Web Tours must successfully handle 10 concurrent travel agents.
2. HP Web Tours must be able to process 10 simultaneous flight bookings with response time not exceeding 90 seconds.
3. HP Web Tours must be able to handle 10 travel agents running simultaneous itinerary checks with response time not exceeding 120 seconds.
4. HP Web Tours must be able to handle 10 agents signing in and signing out of the system with response time not exceeding 10 seconds.

This tutorial will walk you through the process of building load tests that validate each business requirement so that you can attach a pass or fail before release.

Where To Go From Here

Now that you have been introduced to the LoadRunner basics, you can proceed to "Lesson 1: Building a Vuser Script" on page 11.

Lesson 1: Building a Vuser Script

To generate a load on your system, you first build a Vuser script that you can run to emulate real user behavior. You use VuGen to create and build Vuser scripts.

In this lesson you will cover the following topics:

- ["Introducing VuGen - the Virtual User Generator" below](#)
- ["Creating a Blank Vuser Script" below](#)
- ["Recording a Vuser script" on the next page](#)
- ["Viewing the Vuser script" on page 16](#)

Introducing VuGen - the Virtual User Generator

In a performance testing environment, LoadRunner replaces human users with virtual users, also known as **Vusers**. Vusers generate a load on a system by emulating actions of typical users - in a repeatable and predictable manner.

You use VuGen (LoadRunner's Virtual User Generator) to create Vuser scripts. VuGen works on a record-and-playback principle. As you walk through a business process on your application, VuGen records your actions and transforms these actions into steps in a Vuser script. These Vuser scripts form the foundation of your load tests.

Creating a Blank Vuser Script

To develop a Vuser script, you first open VuGen and create a blank script. Thereafter, you can enhance the blank script by recording events and adding manual enhancements to the script.

In this section, you will open VuGen and create a blank Vuser script that is based on the **Web - HTTP/HTML** protocol.

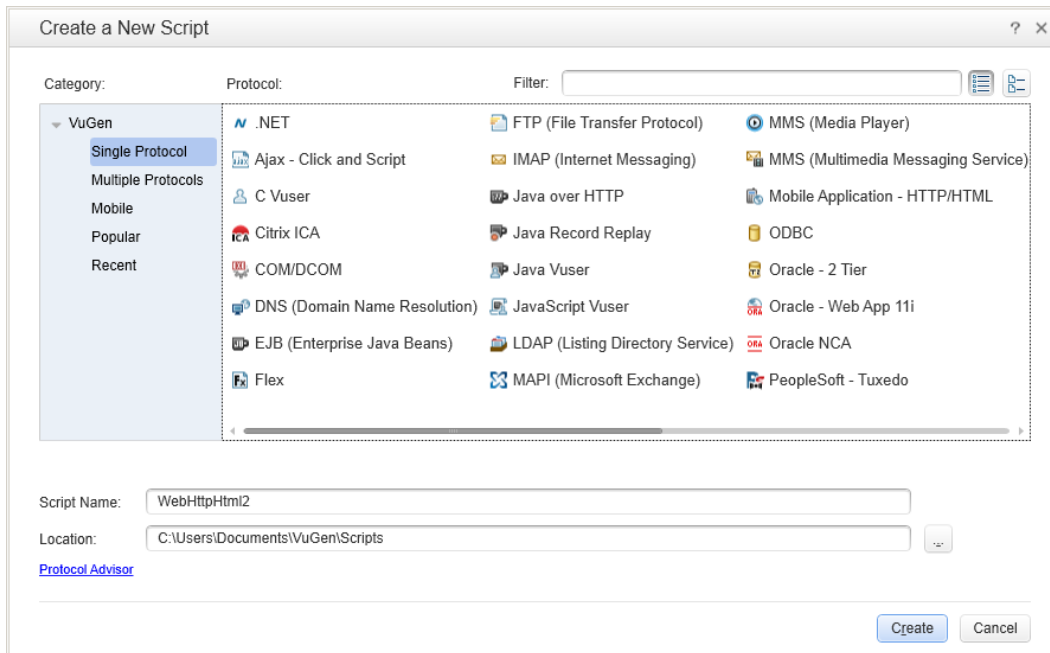
Definition: A protocol is the method of communication between a client and a server.

To create a blank Vuser script:

1. Double-click the **Virtual User Generator** shortcut icon on the desktop to open VuGen.



2. Click **File > New Script and Solution** or click the **Add New Script** button  on the VuGen toolbar. The Create a New Script dialog box opens.



3. Make sure the **Category** is **Single Protocol**. VuGen displays a list of the protocols that are available for a single-protocol script.
4. From the list of available protocols, select **Web - HTTP/HTML** and then click **Create**. VuGen creates a blank Vuser script and displays the script in the VuGen Editor.

Recording a Vuser script

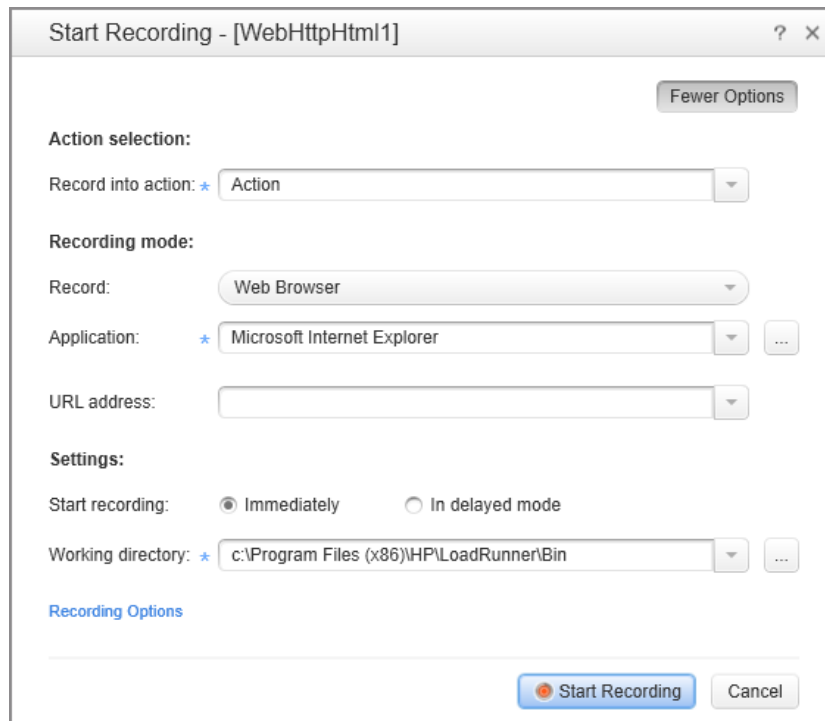
The next step in developing the Vuser script is to record the actions performed by a real user. In the previous section, you created an empty **Web - HTTP/HTML** Vuser script. Now you can begin to record actions directly into the script. In this section, you will track the actions that a travel agent performs to reserve a flight from Denver to Los Angeles, and check the flight itinerary.

Note: To enable VuGen to record the actions that you perform in the Web Tours application, click **Record > Recording Options**. In the Recording Options dialog box, select **General > Script**, and then under **Scripting Options**, make sure that the **Track processes created as COM local servers** check box is cleared.

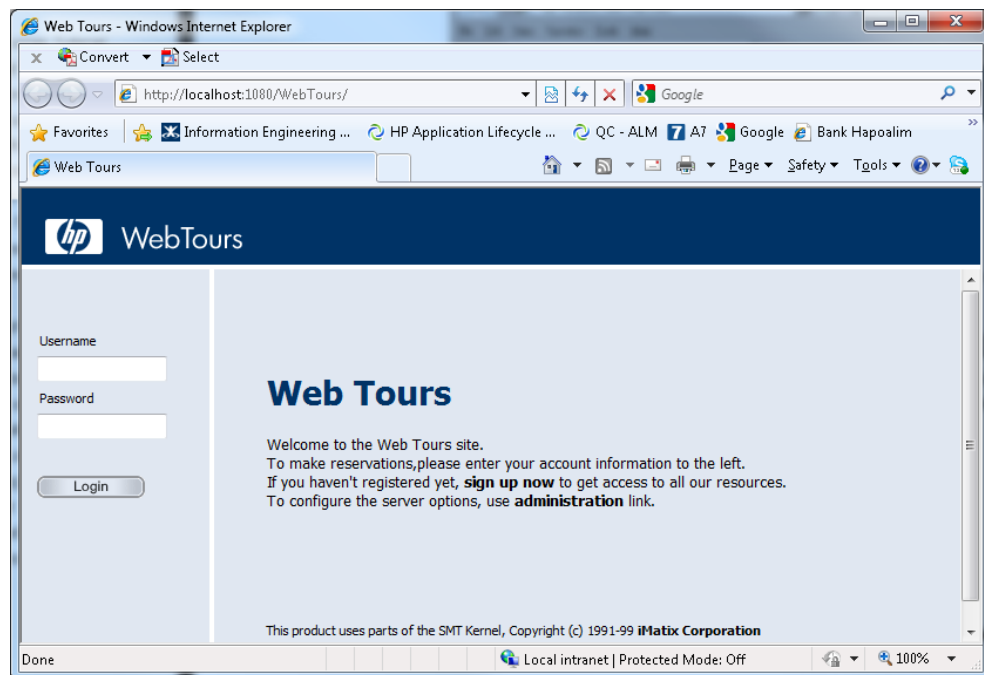
To record the Vuser script:

1. Start recording on the HP Web Tours web site.

- a. Click **Record > Record** or click the **Record** button  on the VuGen toolbar. The Start Recording dialog box opens.



- b. In the **Record into action** box, select **Action**.
 - c. From the **Record** list, select **Web Browser**.
 - d. In the **Application** box, make sure that **Microsoft Internet Explorer** appears.
 - e. In the **URL address** box, type **http://localhost:1080/WebTours**.
 - f. Click **Start Recording**. A new web browser opens and displays the HP Web Tours home page.



Note: If there is an error opening HP Web Tours, make sure that the Web Tours Server is running. The presence of the **Start Web Tours** dialog box indicates that the server is running. To start the server, select **Start > All Programs > HP Software > HP LoadRunner > Samples > Web > Start Web Server**. In icon-based desktops, such as Windows 8, search for "Start HP " and select **Start HP Web Tours Server** from the results.

VuGen's floating Recording toolbar opens.



2. Log on to HP Web Tours.
 - a. Type in the following information:
 - User name: **jojo**
 - Password: **bean**
 - b. On the left pane, click **Login**. HP Web Tours welcomes you to the application.
3. Enter the flight details.

- a. Click **Flights**. The Find Flight page opens.
 - b. From **Departure City**, select **Denver** (default).
 - c. **Departure Date**: Keep the default date.
 - d. From **Arrival City**, select **Los Angeles**.
 - e. **Return Date**: Keep the default date.
 - f. From **Seating Preference**, select **Aisle**.
 - g. Keep the rest of the default settings and click **Continue**. The Find Flight page opens.
4. Select a flight.

Click **Continue** to accept the default flight selections. The Payment Details page opens.

5. Enter payment information and book the flight.
- a. In the **Credit Card** box, enter **12345678**.
 - b. In the **Exp Date** box, enter **06/14**.
 - c. Click **Continue**. The Invoice page opens, displaying your invoice.
6. In the left pane, click **Itinerary**. The Itinerary page opens.
7. In the left pane, click **Sign Off**.

8. Close the browser, and then click the **Stop Recording** button  on the VuGen floating toolbar to stop the recording process.

VuGen generates the required code and inserts the code into the Vuser script.

If the Design Studio opens, click **Close** to close the Design Studio.

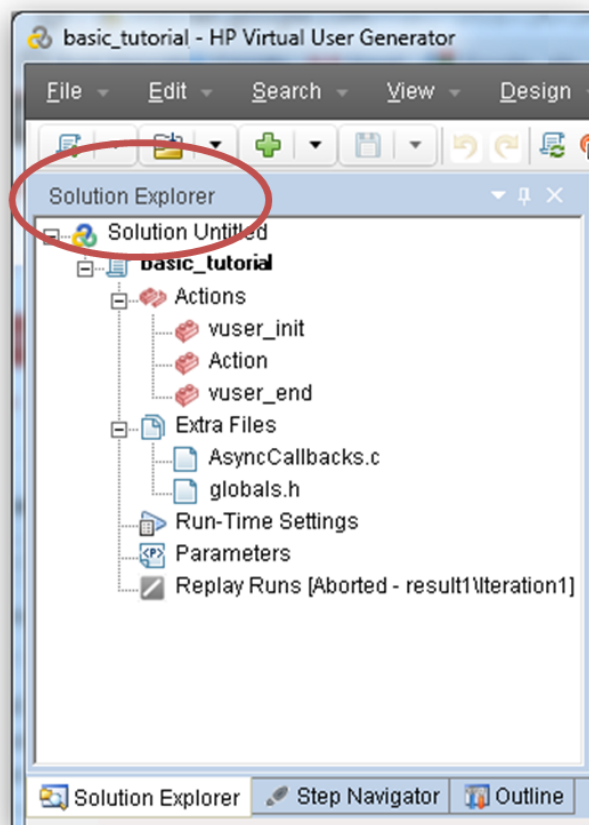
9. Save the Vuser script.
- a. Select **File > Save Script As**.
 - b. Navigate to **<LoadRunner Installation>\tutorial**, create a new folder named **Scripts**, and then navigate to the new **Scripts** folder.
 - c. In the **File name** box, type **basic_tutorial**.
 - d. Click **Save**. VuGen saves the script and displays the script name in the VuGen title bar.

Viewing the Vuser script

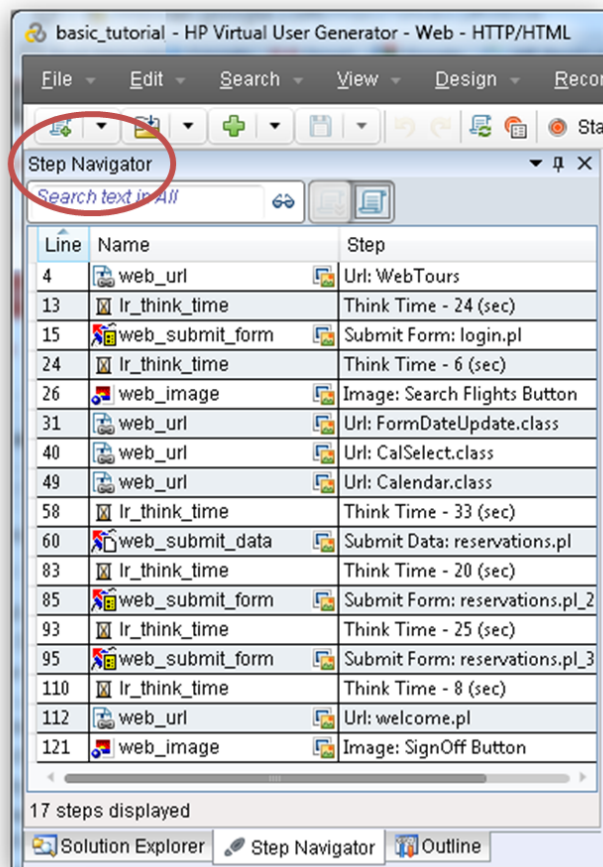
You have now recorded the actions of a travel agent logging in, booking a flight, checking the itinerary, and logging off. VuGen recorded your steps from the moment you clicked the **Start Recording** button to the moment you clicked the **Stop Recording** button.

You can now use VuGen to view the script. VuGen lets you see the Vuser script in a number of formats:

- The **Solution Explorer** gives you structured access to the various parts of a Vuser script, as well as to a number of files that are associated with the Vuser script.



- The **Step Navigator** displays an icon-based view of the script that lists the actions of the Vuser as steps. For each action you performed during recording, VuGen generated a corresponding step in the Step Navigator.



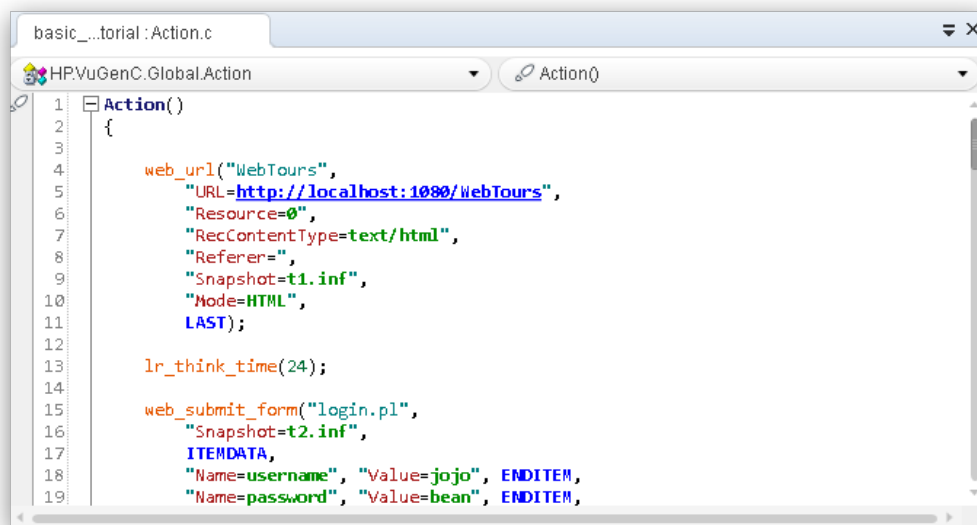
The Step Navigator displays a snapshot icon  to indicate that a specific step contains a snapshot.



To view the script in the Step Navigator, select **View > Step Navigator**, or click the **Step**

Navigator button  on the VuGen toolbar. Double-click any step in the Step Navigator to display the corresponding function in the Editor.

- The VuGen **Editor** displays a text-based view of the script. In the Editor, the actions of the Vuser are listed as API functions. In the Editor, VuGen uses color-coding to show the functions and their argument values in the script. You can type C or LoadRunner API functions, as well as control flow statements, directly into the script.



The screenshot shows the HP VuGen Global Action editor. The title bar reads "basic_tutorial : Action.c". Below the title bar, there's a toolbar with a "Global Action" icon and a dropdown menu showing "Action()". The main area contains a script for an Action() function. The script is as follows:

```
1 Action()
2 {
3
4     web_url("WebTours",
5             "URL=http://localhost:1080/WebTours",
6             "Resource=0",
7             "RecContentType=text/html",
8             "Referer=",
9             "Snapshot=t1.inf",
10            "Mode=HTML",
11            LAST);
12
13     lr_think_time(24);
14
15     web_submit_form("login.pl",
16                    "Snapshot=t2.inf",
17                    ITEMDATA,
18                    "Name=username", "Value=jojo", ENDITEM,
19                    "Name=password", "Value=bean", ENDITEM,
```

Where To Go From Here

Now that you are familiar with recording and viewing a basic Vuser script, you can proceed to "Lesson 2: Replaying Your Vuser Script" on page 19.

Lesson 2: Replaying Your Vuser Script

In the previous lesson, you recorded a set of typical user actions, and VuGen transcribed these actions into a Vuser script. You can now replay your recorded script to verify that the script runs properly, before you incorporate the script into a load testing scenario.

Before replaying the script, you must configure the runtime settings for the script, which define the behavior of the Vuser.

In this lesson you will cover the following topics:

- [How do I set the runtime behavior?](#)
- [How do I run my Vuser script?](#)
- [Where can I view information about the replay?](#)
- [How do I know if my script replayed successfully?](#)
- [How do I search or filter the results?](#)

How do I set the Vuser's runtime behavior?

LoadRunner runtime settings let you emulate different kinds of user activity and behavior. For example, you could emulate a user who responds immediately to output from the server, or a user who stops and thinks before each response. You can also configure runtime settings to specify how many times the Vuser should repeat a set of actions in the Vuser script.

There are general runtime settings, and settings that are specific to certain Vuser protocols only. For example, for a Web emulation, you could instruct your Vusers to replay your script in Firefox, Chrome, or Internet Explorer. Runtime settings for specific protocols will be covered in "[Lesson 4: Preparing a Vuser Script for Load Testing](#)" on page 36. This lesson describes some of the general runtime settings - settings that apply to all Vuser protocols. The general runtime settings include:

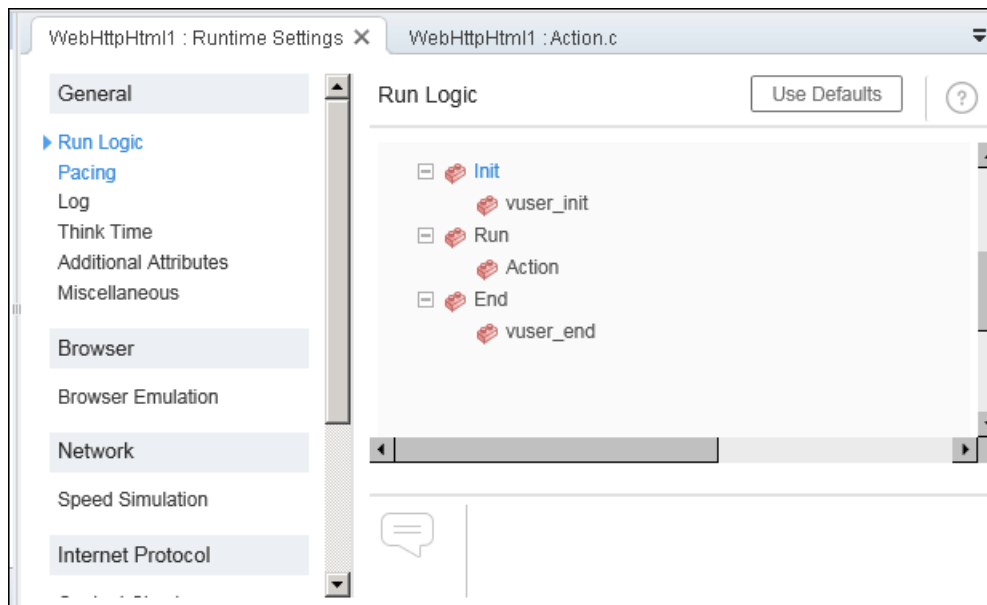
- **Run Logic.** The number of times that a Vuser repeats various sections of the Vuser script.
- **Pacing.** The time to wait between repetitions.
- **Think Time.** The time the Vuser stops to think between steps in the script.
- **Log.** The level of information that you want to gather during playback.

Note: This lesson describes how to use **VuGen** to modify the runtime settings. A later lesson describes how to use the **LoadRunner Controller** to modify the runtime settings.

To set the runtime settings:

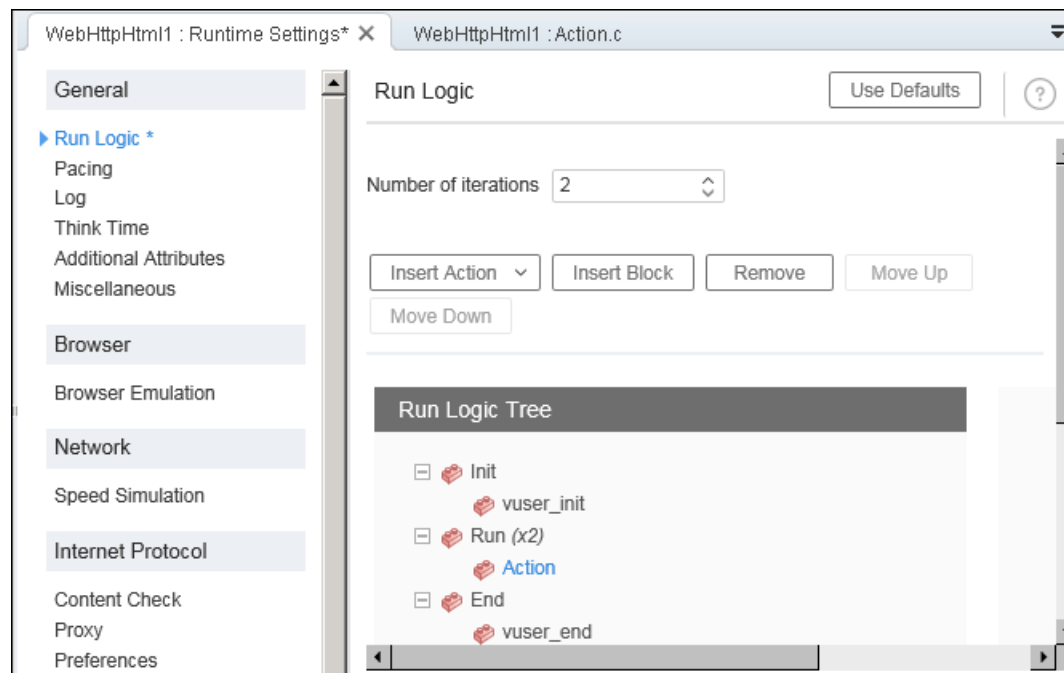
1. Open the Vuser script that you saved in the previous lesson.
2. Open the Runtime Settings dialog box.

Click **Replay > Runtime Settings**. The Runtime Settings view opens in the VuGen editor.



3. Set the Run Logic settings.

In the left pane, under **General**, click **Run Logic**.

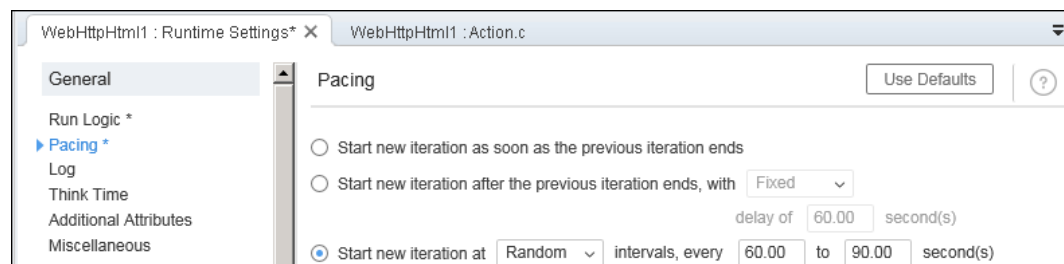


The **Run Logic** settings enable you to set the number of iterations for the Vuser script. This is the number of times to repeat the Action section of the Vuser script when the script is replayed.

Set the **Number of iterations** to 2.

4. Set the Pacing settings.

In the left pane, under **General**, click **Pacing**.



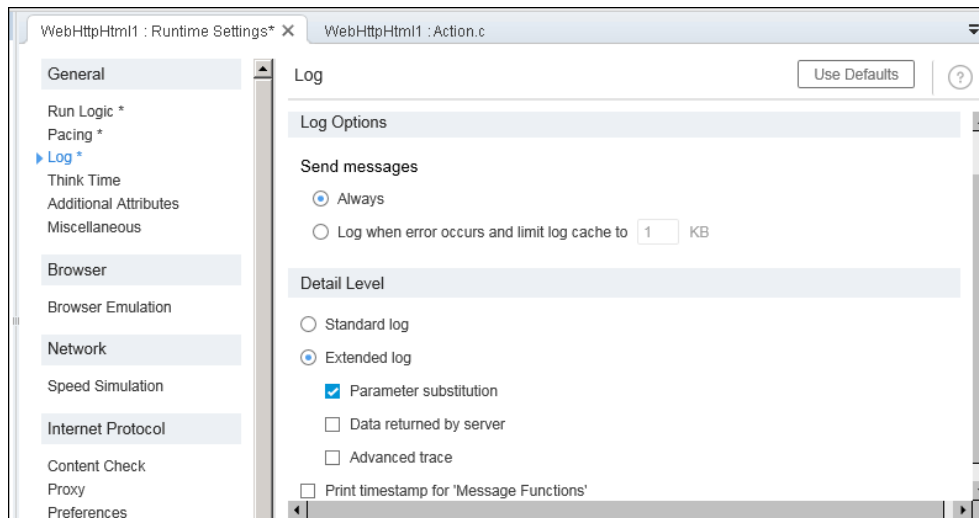
The **Pacing** settings enable you to control the time between iterations. You will specify a random time. This accurately emulates a real-life setting where the user waits between actions, but at random intervals. For example, you do not see real users always waiting exactly 60 seconds between repeated actions.

Select the third radio button, and select the following:

Start new iteration at Random intervals, every 60.000 to 90.000 second(s).

5. Set the Log settings.

In the left pane, under **General**, select **Log**.

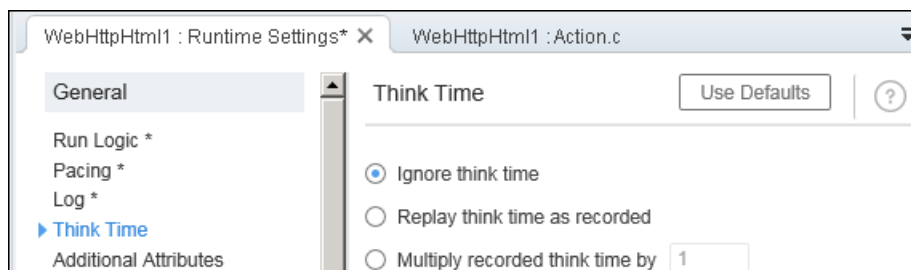


The **Log** settings indicate how much information to log while running the Vuser Script. While developing a Vuser script, you may want to enable some logging for debugging purposes, but once you verify that your script is functional, you can enable logging for errors only, or even disable logging.

Select **Extended log** and enable **Parameter substitution**. This option will be relevant for the following lesson at which point it will be discussed.

6. View the Think Time settings.

In the left pane, under **General**, click **Think Time**.

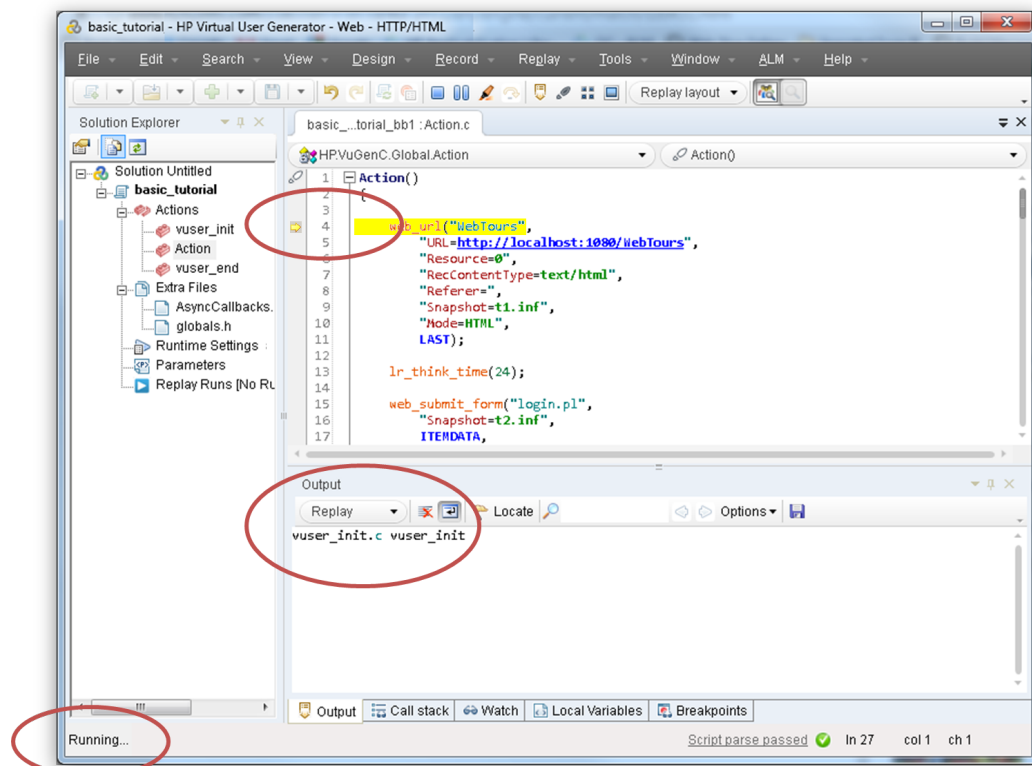


Keep the default think time setting - **Ignore think time**. You will set the think time from the Controller. Keep in mind that when you run the script in VuGen, it will run quickly since it will not include think time.

How do I run my Vuser Script?

After recording the Vuser script and setting the runtime settings, you are ready to run the script. VuGen provides a number of indicators that your script is running.

1. The message **"Running"** appears in the lower left corner of VuGen.
2. A yellow marker appears in the Editor and indicates the specific line that is being replayed.
3. The Output pane displays messages as the replay progresses.



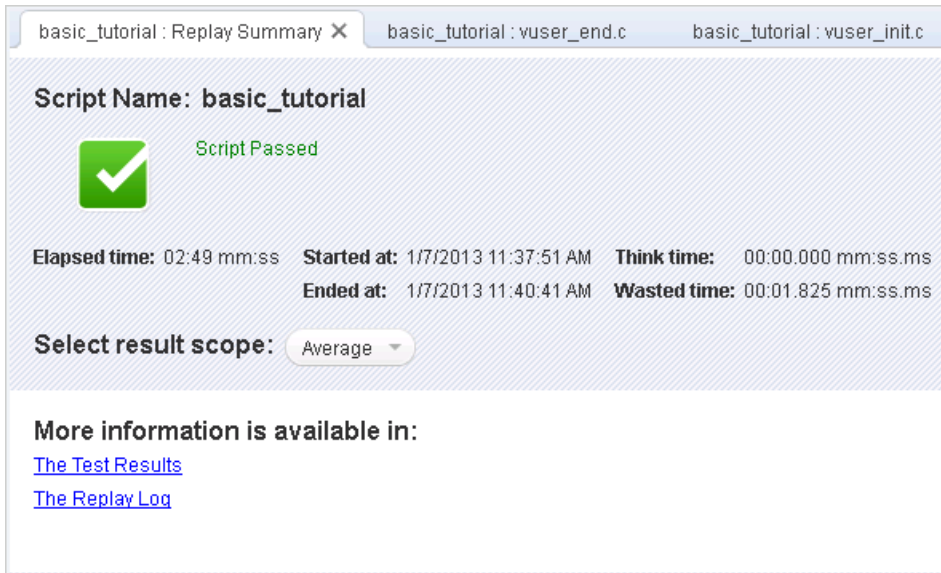
To run a Vuser script:

1. Open the Vuser script in VuGen.
2. Click **Replay > Run** or click the **Replay** button  on the VuGen toolbar.

After the replay ends, a message box may prompt you to scan for correlations. Click **No**.

Where can I view information about the replay?

When the Vuser script stops running, you can view a summary of the replay. The replay summary is displayed in the **Replay Summary** tab.



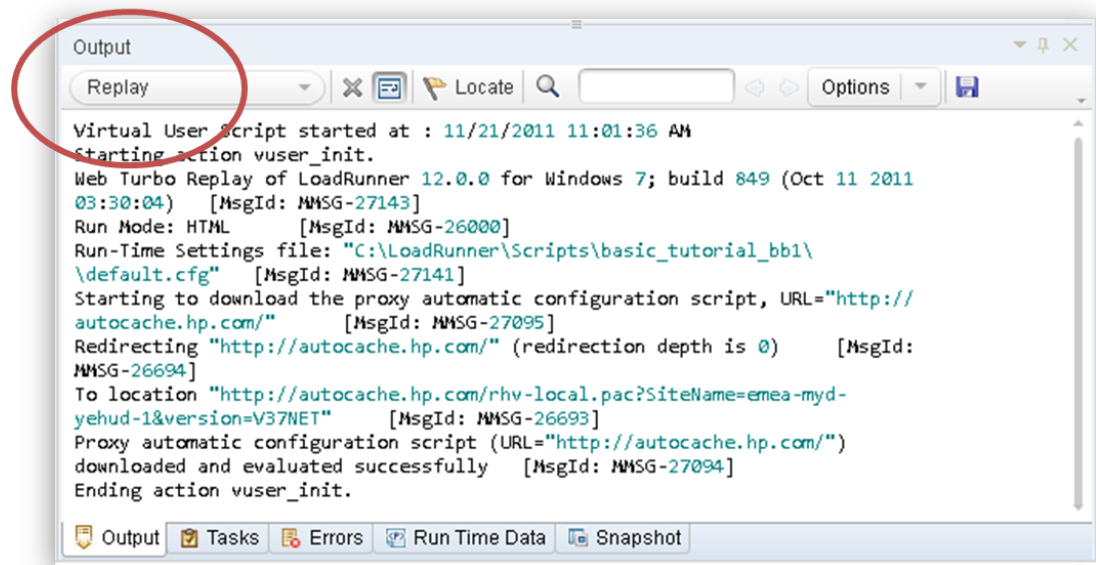
The Replay Summary tab lists basic information about the script run, such as the duration of the replay, and the start and end times of the replay. In addition, the Replay Summary tab provides you with a link that displays a log of the script events, and another link that displays detailed results of the script run.

The Replay log - a log of the events that occurred during the replay of the script - is displayed in VuGen's Output pane. The Output pane uses color-coding to display a textual summary of the events that occurred during the replay.

In this section of the tutorial, you will open the Replay log and locate specific events and notifications within the log.

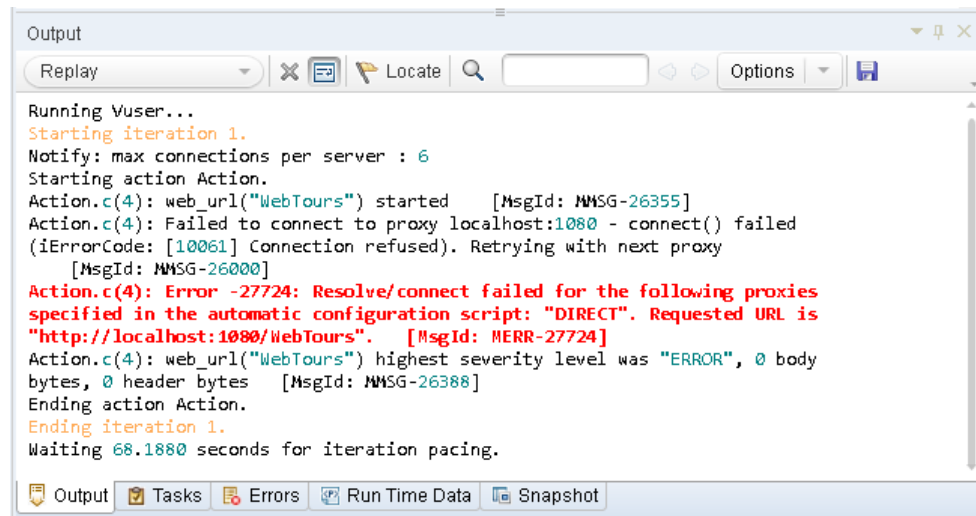
To view the Replay log:

1. After replaying a Vuser script, click **View > Output** or click the **Output** button  on the VuGen toolbar. Alternatively, click the **Replay Log** link in the Replay Summary tab.
2. Make sure that **Replay** is selected in the Output pane.



3. From the VuGen menu, click **Search > Quick Find** to open the Search dialog box.
4. From **Scope**, select **Current Script**.
5. Click **Include in search**, and then select the **Logs** check box.
6. Use the Search dialog box to locate the following items in the Replay log:
 - a. "Virtual User Script Started" - the beginning of the script run.
 - b. "Vuser Terminated" - the end of the script run.
 - c. "iteration" - the beginning and the end of each iteration and the iteration number. (This text appears in orange lettering.)

Note: The Output pane displays successful steps in green and errors in red. For example, if the Vuser was unable to connect to the server, the Output pane would display the error text in red, and indicate the line number in the script where the error occurred.

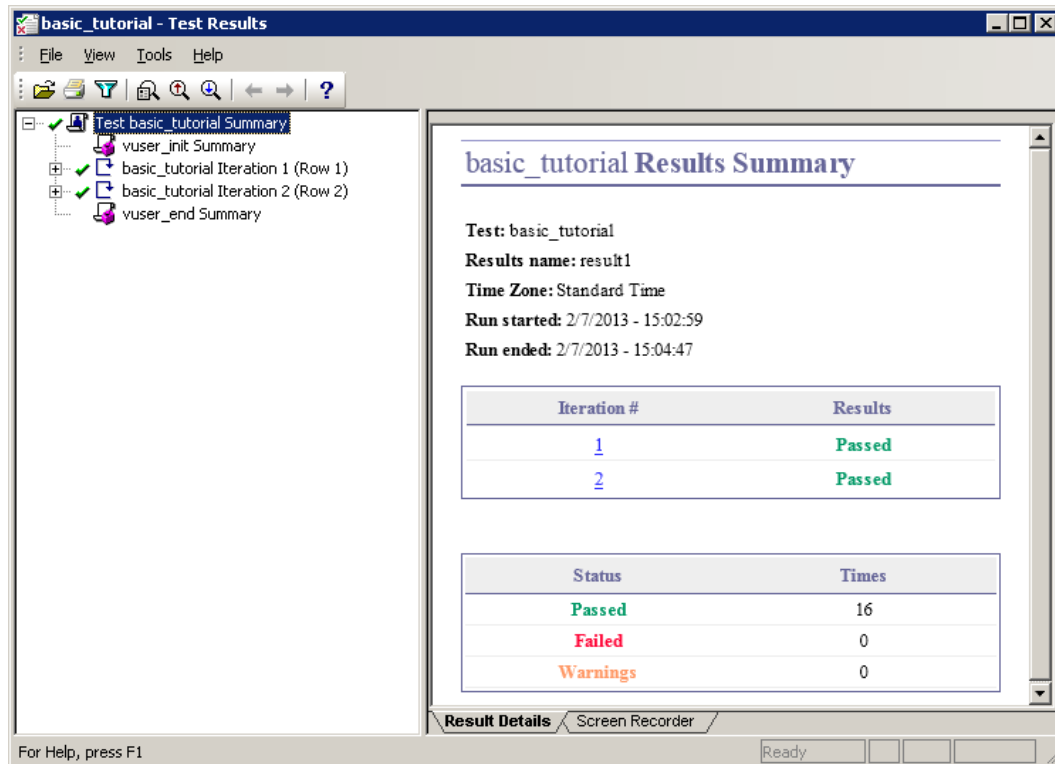


Note: If you double-click a line in the Output pane, VuGen indicates the corresponding step in the script in the VuGen Editor.

How do I know if my script replayed successfully?

After you replay the events that you recorded, you need to look at the replay results to see if the script replayed successfully. If something failed, you want to know when and why it failed.

In this section you will view and analyze the results of your script run. VuGen summarizes the results of the replay in the Test Results window.



When the Test Results window first opens, it contains two panes: The Tree pane (on the left) and the Results Summary pane (on the right).

- The **Tree pane** contains the results tree. Each iteration in the tree is numbered.
- The **Results Summary pane** contains the details of the script replay, as well as screen recorder movies, if any. The top table indicates which iterations passed and which failed. The test is considered to have passed when the Vuser successfully navigates through the HP Web Tours site according to the original recording. The bottom table indicates whether transactions and checkpoints passed or failed. You will add these features to your test later on in the tutorial.

To view replay results:

1. Click the **Replay Summary** tab.
2. Click **Replay > Test Results**. The Test Results window opens and displays a summary of the replay results.

In the next section, you will drill down into the replay results to determine if the script reached the intended Web pages during replay.

How do I search or filter the replay results?

If your replay results indicate that something failed, you can drill down and locate the point of failure.

In the Tree pane of the Test Results window, you can expand the test tree and view the results of each step separately. The Summary pane shows a snapshot of the replay during that iteration.

1. Expand an iteration node in the tree.
 - a. In the results tree, expand the **basic_tutorial Iteration 1** node.
 - b. Expand the **Action Summary** node. The expanded node shows a list of the steps performed in that iteration.

2. View a result snapshot.

Click the **Submit Form:login.pl** node. The Summary pane displays the replay snapshot associated with that step.

3. View the step summary.

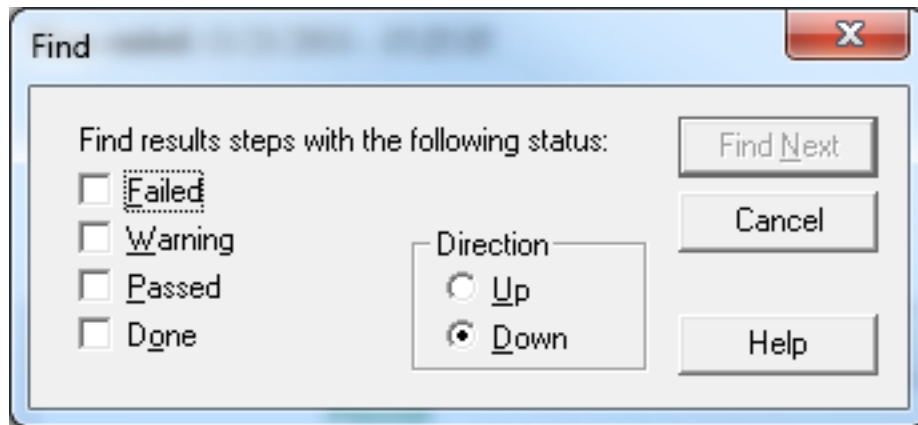
The Summary pane displays the step summary information: the object or step name, the details about whether the page loaded successfully, the result (**Passed**, **Failed**, **Done**, or **Warning**), and the time that the step was performed.

4. Search for a result status.

You can search the replay results for the words **Passed** or **Failed**.

This is helpful, as in a case where the overall Results Summary indicates that the replay failed, you can determine where it failed.

- a. To search the replay results, select **Tools > Find** or click the **Find** button  on the Test Results toolbar. The Find dialog box opens.



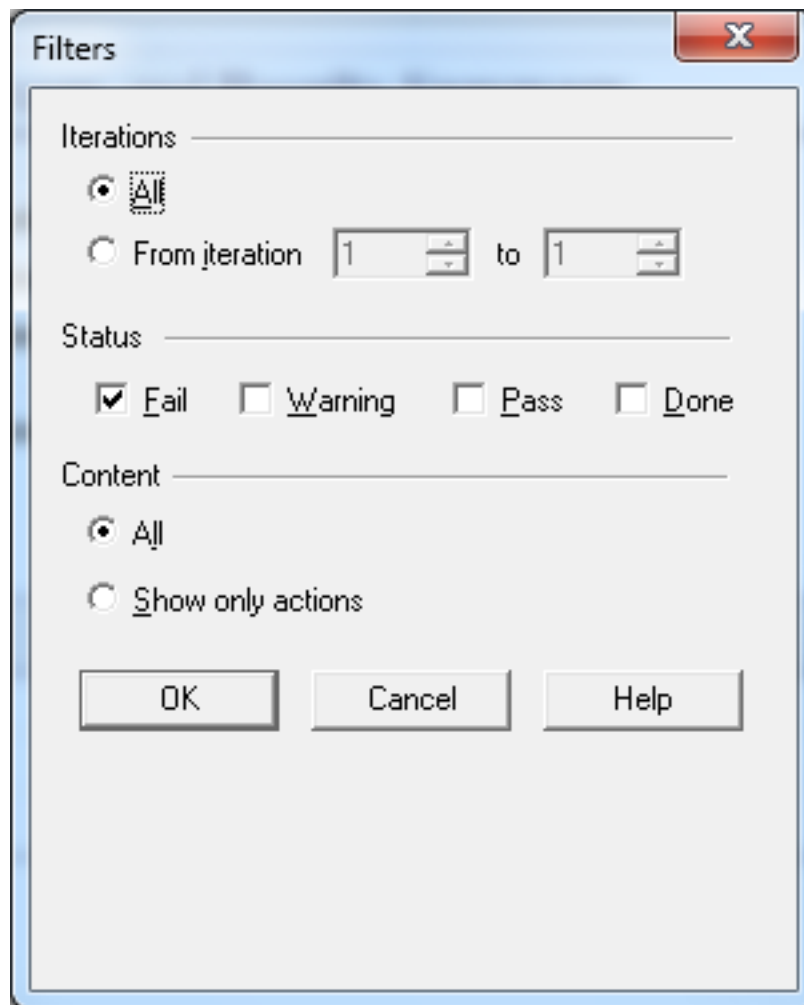
- b. Select the **Passed** check box, make sure no other options are selected, and click **Find Next**. The Test Tree pane highlights the first step in which the status is **Passed**.

Note: If no steps of the selected status can be found, no step is highlighted.

5. Filter the results.

You can filter the Test Tree pane to display a specific iteration or status. For example, you can filter it to show only Failed status.

- a. Select **View > Filters** or click the **Filters** button  on the Test Results toolbar. The Filters dialog box opens.



- b. Under **Status**, select **Fail** and clear all the other options.
 - c. Under **Content**, select **All** and click **OK**. The left pane becomes empty, since there were no failures.
6. Close the Test Results window.

Click **File > Exit**.

Where To Go From Here

You have successfully replayed your Vuser script that emulates a user operating the *HP Web Tours* application. You can now proceed to "Lesson 3: Solving Common Replay Problems" on page 31.

Lesson 3: Solving Common Replay Problems

After you create a Vuser script, you validate it by running it from VuGen. Sometimes a replay will fail, even though the recording of the same actions succeeded.

Many applications use dynamic values that change each time you use the application. For example, some servers assign a unique session ID for every new session. When you try to replay a recorded session, the application creates a new session ID that differs from the recorded session ID. Dynamic values, such as these unique session IDs, may pose difficulties when you replay specific types of Vuser scripts. For example, dynamic session IDs often pose problems when replaying Web-HTTP/HTML scripts, but not when replaying Web-TruClient scripts.

LoadRunner use correlation to address the issue of dynamic values. Correlation saves the changing values, in our case the session ID, to a parameter. When running the Vuser script, the Vuser does not use the recorded value—instead, it uses the new session ID, assigned to it by the server.

In this lesson you will observe how LoadRunner solves the issue of dynamic values when running Web-HTTP/HTML Vuser scripts.

In this lesson you will cover the following topics:

- [Preparing HP Web Tours for replay errors](#)
- [How do I work with unique server values?](#)

Preparing HP Web Tours for replay errors

To illustrate a common replay failure, you need to modify a setting in the *HP Web Tours* application. This setting tells the *HP Web Tours* server to issue and require unique session IDs.

1. Open HP Web Tours.

Select **Start > All Programs > HP Software > HP LoadRunner > Samples > Web > HP Web Tours Application**. In icon-based desktops, such as Windows 8, search for "**HP Web**" and select **HP Web Tours Application** from the results. A browser opens with the *HP Web Tours* home page.

2. Change the server options.
 - a. Click the **administration** link on the HP Web Tours home page. The Administration Page opens.
 - b. Select the **Set LOGIN form's action tag to an error page** checkbox. This setting tells the server not to allow duplicate session IDs.
 - c. Scroll down to the bottom of the page and click **Update**.

- d. Scroll down to the bottom of the page and click the **Return to the Web Tours Homepage** link.

How do I work with unique server values?

In the modified configuration of HP Web Tours, the server assigns a unique session ID to each Vuser. If you try to replay the unmodified Vuser script that you recorded in "[Lesson 1: Building a Vuser Script](#)" on page 11, the replay will fail.

To overcome this issue, you use VuGen to detect the need to correlate the session ID. You will instruct VuGen to insert a step that saves the original session ID to a parameter. In each subsequent replay session, VuGen saves the new unique session ID to a parameter. As the Vuser executes the steps in the Vuser script, the Vuser uses the saved session ID value instead of the originally recorded value.

1. Record a new Vuser script with dynamic values.
 - a. Record a new Vuser script with the same steps that you recorded previously in "[Lesson 1: Building a Vuser Script](#)" on page 11.
 - b. Save the script as **basic_tutorial_Cor**.
2. Replay the new script.

Click **Replay > Run** or click the **Replay** button  on the VuGen toolbar.

VuGen runs the new Vuser script. You may notice several error messages in the Replay Log in the Output pane, indicated by the red-colored text.

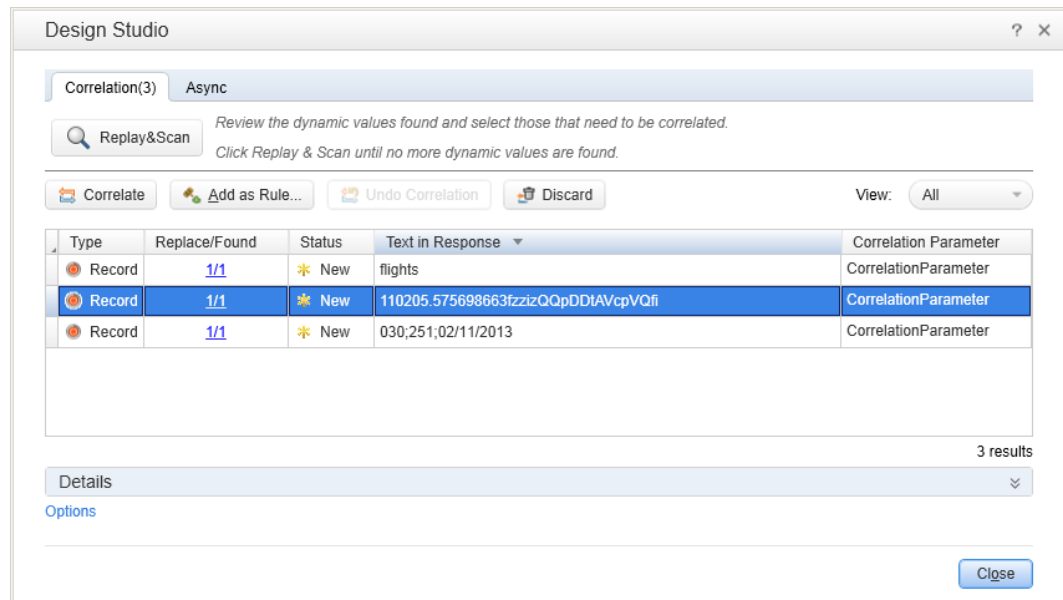
After the replay ends, a message box may prompt you to scan for correlations. Click **No**.

3. View the Replay Summary.

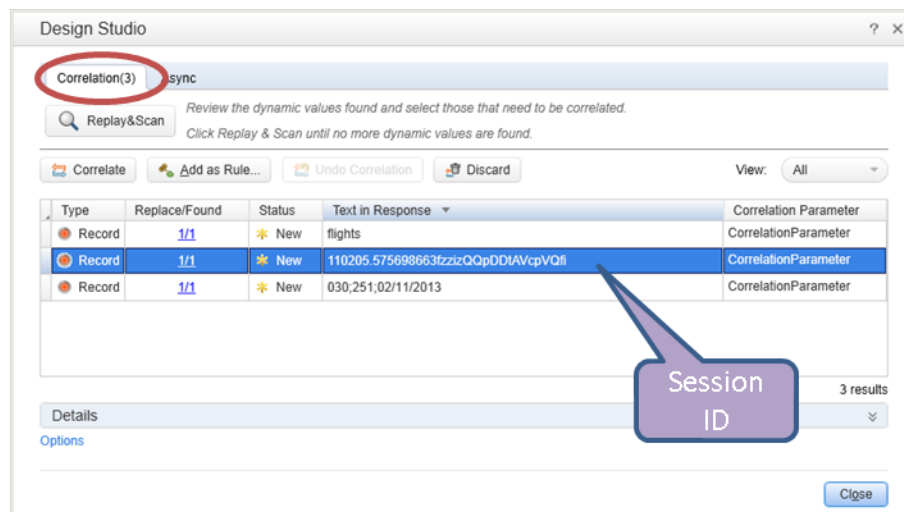
Look at the Replay Summary tab. The summary shows that the replay of your script failed.

4. Scan the script for correlations.

Select **Design > Design Studio**.



VuGen scans the script and its associated data, searching for possible dynamic values. The Correlation tab of the Design Studio lists three dynamic values that may require correlation. The longest of the three values is the Session ID.



5. Correlate the Session ID.
 - a. Select the Session ID entry in the Correlation tab, and click **Correlate**. VuGen changes the status of the Session ID to **Applied**, and inserts a new function at the top of the Vuser script. The new function saves the original session ID to a parameter.

In each subsequent replay session, VuGen saves the new unique session ID to a parameter. When the Vuser runs, the Vuser uses the saved ID value instead of the

originally recorded value.

- b. Click **Close** to close the Design Studio.
6. Examine the syntax of the correlation statement.

In the VuGen Editor, locate the statement that VuGen added to the script. The new statement looks similar to the following:

```
web_reg_save_param_regexp(  
  "ParamName=CorrelationParameter",  
  "RegExp=userSession\\ value=(.*?)>\\n<table\\ border",  
  "SEARCH_FILTERS",  
  "Scope=Body",  
  "IgnoreRedirections=No",  
  "RequestUrl=*/nav.pl*",  
  LAST);
```

The statement instructs VuGen to save the first occurrence of the value contained in the regular expression (the unique session ID) to a parameter called **CorrelationParameter**.

7. Play the script again.
- a. Click **Replay > Run** or click the **Replay** button  on the VuGen toolbar to replay the script again. When the replay ends, look in the Replay Log in the Output pane. Notice that VuGen no longer issues the red-colored error messages.
 - b. In the script, right-click **web_reg_save_param_regexp** and select **Go to step in Replay Log**. VuGen places the cursor at the corresponding line in the Replay log. The log indicates that function **web_reg_save_param_regexp** succeeded, indicating that the correlation worked successfully.
8. Reset the HP Web Tours server to ignore unique session IDs.
- a. Click **Start > All Programs > HP LoadRunner > Samples > Web > HP Web Tours Application** to open HP Web Tours. In icon-based desktops, such as Windows 8, search for "HP Web" and select **HP Web Tours Application** from the results.
 - b. On the HP Web Tours home page, click the **administration** link.
 - c. In the **Administration Page**, clear the **Set LOGIN form's action tag to an error page** check box.
 - d. Scroll down to the bottom of the page and click **Update**.

Where To Go From Here

Now that you are familiar with some of the common replay problems, you can proceed to "Lesson 4: Preparing a Vuser Script for Load Testing" on page 36.

Lesson 4: Preparing a Vuser Script for Load Testing

In the previous lessons, you verified that the replay of your Vuser script produced an accurate emulation of a real user. The next step is to prepare the script for load testing. How will the system work with many users working simultaneously? Will the system slow down to an unacceptable level?

In this lesson you will learn about different methods to enhance the script and to make it more effective for the load testing process.

In this lesson you will cover the following topics:

- [How do I measure the duration of a business process?](#)
- [How do I emulate multiple users?](#)
- [How do I verify Web page content?](#)
- [How can I produce debugging information?](#)
- [Did the replay of my test succeed?](#)

How do I measure the duration of a business process?

When preparing an application for deployment, you need to measure the duration of specific business processes—how long does it take to log on, book a flight, and so on. Each business process is normally made up of one or more steps or actions in your script. In a Vuser script, you designate a series of actions you want to measure by including these actions in a **transaction**.

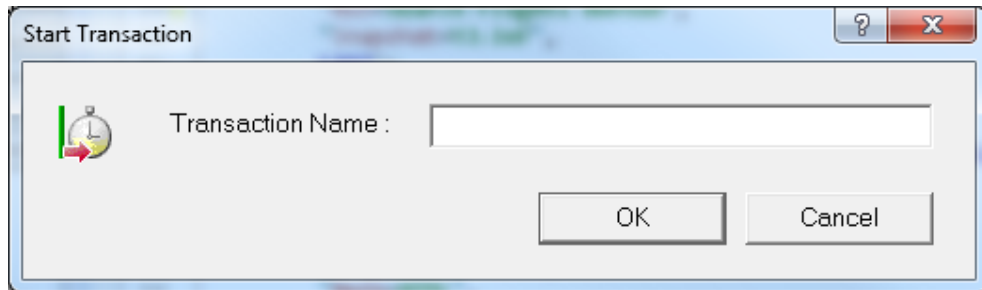
When you run a Vuser script that includes a transaction, LoadRunner gathers information about the time it takes to perform the transaction, and displays the results in color-coded graphs and reports. You use this information to help determine if the application meets the performance requirements.

You can manually insert a transaction anywhere in a Vuser script. To mark a set of steps as a transaction, insert a **start_transaction** marker before the first step and an **end_transaction** marker after the last step.

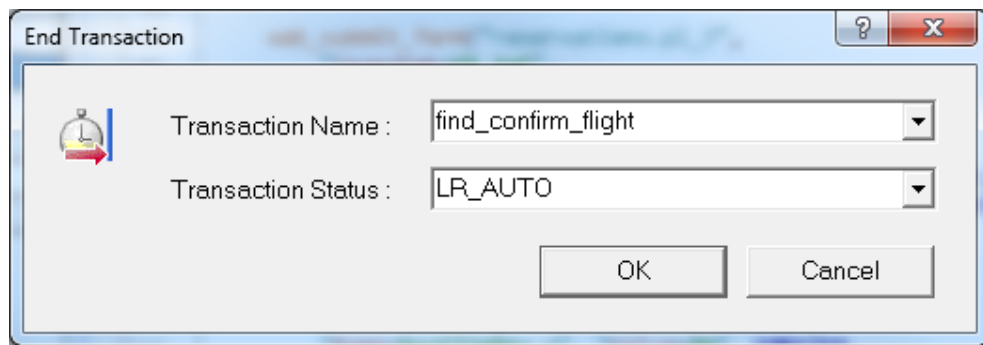
In this section you will insert a transaction into your script to measure the amount of time it takes for the user to find and confirm a flight.

To insert a transaction into a Vuser script:

1. In VuGen, open the **Basic_Tutorial** script which you created in "[Lesson 1: Building a Vuser Script](#)" on page 11. If it is still open, you can select the tab displaying its name. Otherwise you can open it from the **File** menu.
2. Click the **Step Navigator** tab to display the Step Navigator.
3. Insert a start transaction marker.
 - a. In the Step Navigator, locate the step **Image: Search Flights Button**.
 - b. Double-click the **Image: Search Flights Button** step to display the corresponding **web_image** step in the Editor.
 - c. Click **View >Steps Toolbox**. The steps Toolbox appears in the right pane.
 - d. In the Steps Toolbox, under **Common**, select **lr_start_transaction**, drag it into the Editor, and release it before the **web_image** step. The Start Transaction dialog box opens.



- e. In the **Transaction Name** box, enter **find_confirm_flight** and click **OK**. VuGen inserts an **lr_start_transaction** step in the Step Navigator, and a corresponding **lr_start_transaction** function in the Editor.
4. Insert an end transaction marker.
 - a. In the Step Navigator, locate the step **Submit Data: reservations.pl_2**.
 - b. Double-click the **Submit Data: reservations.pl_2** step to display the corresponding **web_submit_data** step in the Editor.
 - c. In the Steps Toolbox, under **Common**, select **lr_end_transaction**, drag it into the Editor, and release it after the **web_submit_data** step. The End Transaction dialog box opens.



- d. Make sure that **find_confirm_flight** appears in the **Transaction Name** box, and then click **OK**. VuGen inserts an **lr_end_transaction** step in the Step Navigator, and a corresponding **lr_end_transaction** function in the Editor.

 A screenshot of the 'Step Navigator' window. It has a search bar at the top with the text 'Search text in All'. Below is a table with three columns: 'Line', 'Name', and 'Step'. The table contains 18 rows of steps. Two rows are circled in red: line 27, 'lr_start_transaction' (Start Transaction - find_confirm_flight) and line 97, 'lr_end_transaction' (End Transaction - find_confirm_flight). The row for line 99, 'lr_think_time' (Think Time - 25 (sec)), is highlighted in blue.

Line	Name	Step
4	web_url	Url: WebTours
13	lr_think_time	Think Time - 24 (sec)
15	web_submit_form	Submit Form: login.pl
24	lr_think_time	Think Time - 6 (sec)
27	lr_start_transaction	Start Transaction - find_confirm_flight
29	web_image	Image: Search Flights Button
34	web_url	Url: FormDateUpdate.class
43	web_url	Url: CalSelect.class
52	web_url	Url: Calendar.class
61	lr_think_time	Think Time - 33 (sec)
63	web_submit_data	Submit Data: reservations.pl
86	lr_think_time	Think Time - 20 (sec)
88	web_submit_form	Submit Form: reservations.pl_2
97	lr_end_transaction	End Transaction - find_confirm_flight
99	lr_think_time	Think Time - 25 (sec)
101	web_submit_form	Submit Form: reservations.pl_3
116	lr_think_time	Think Time - 8 (sec)
118	web_url	Url: welcome.pl
127	web_image	Image: SignOff Button

You have now defined the **find_confirm_flight** transaction.

How do I emulate multiple users?

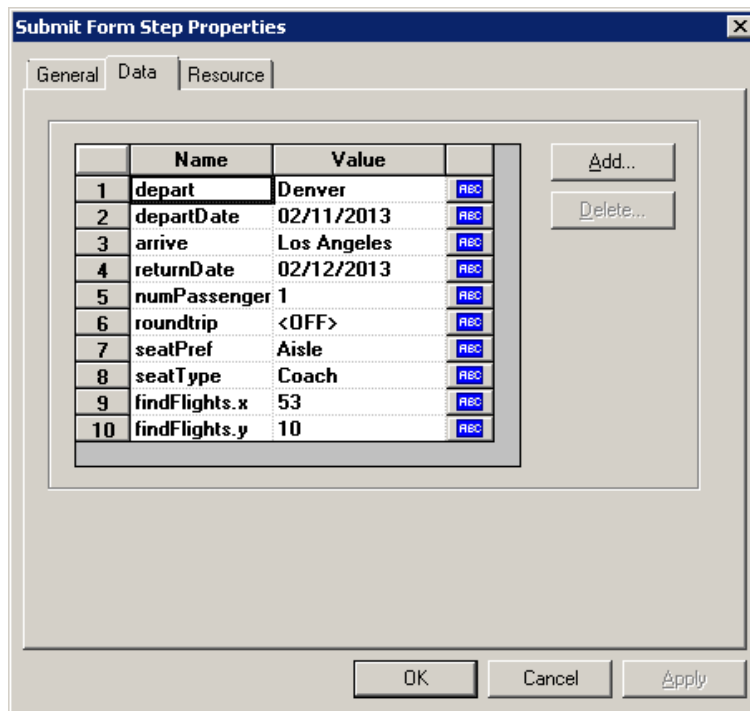
In your emulation, you tracked a user booking a flight and choosing an aisle seat. In a real-life setting, however, different users will have varying preferences. To improve your test, you need to

check if the booking will work when users select different seating preferences (**Aisle**, **Window**, or **None**).

To accomplish this, you will **parameterize** the script. This means that you take the recorded value, **Aisle**, and replace it with a parameter. You will place values for the parameter in a parameter file. When you run the script, the Vuser will use values from the parameter file (**Aisle**, **Window**, or **None**) thereby emulating a true travel agency environment.

To parameterize your script:

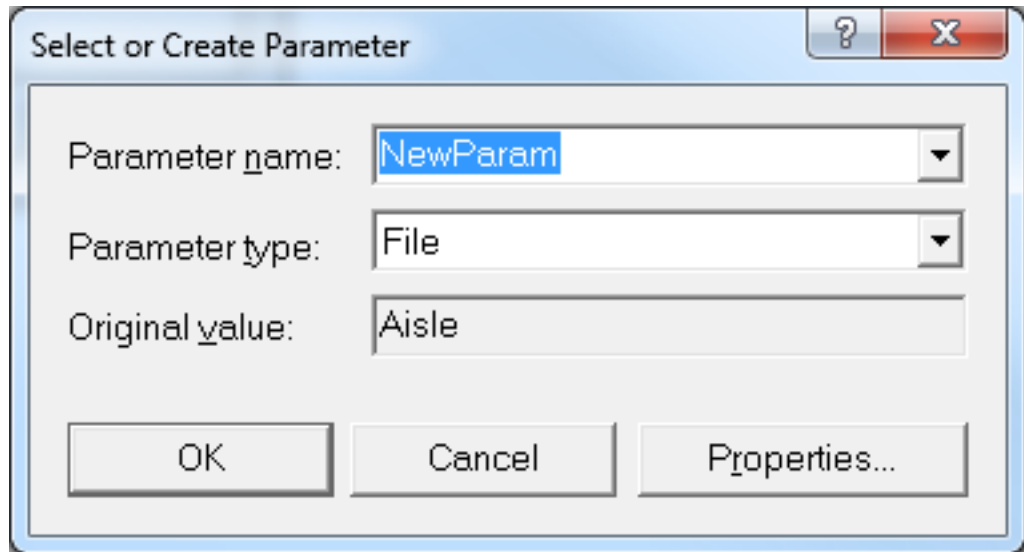
1. Find the section where you want to vary the data.
 - a. Select **View > Step Navigator** to show the Step Navigator in the left pane.
 - b. In the Step Navigator, locate the **Submit Data: reservations.pl** step.
 - c. Right-click the **Submit Form: reservations.pl** step and select **Show Arguments**. The Submit Form Step Properties dialog box opens.



The ABC icon  to the right of each argument in the grid indicates that the argument has a fixed value.

2. Change the fixed value to a varying value.
 - a. In the Submit Form Step Properties dialog box, select **seatPref** in the seventh row of the grid.
 - b. Click the **ABC** icon  adjacent to the **seatPref** argument. The Select or Create

Parameter dialog box opens.



3. Create a parameter.
 - a. In the **Parameter name** box, type **seat**.
 - b. Click **OK**. In the Submit Form Step Properties dialog box, VuGen replaces the **ABC** icon  with a **Parameter** icon .

Submit Form Step Properties

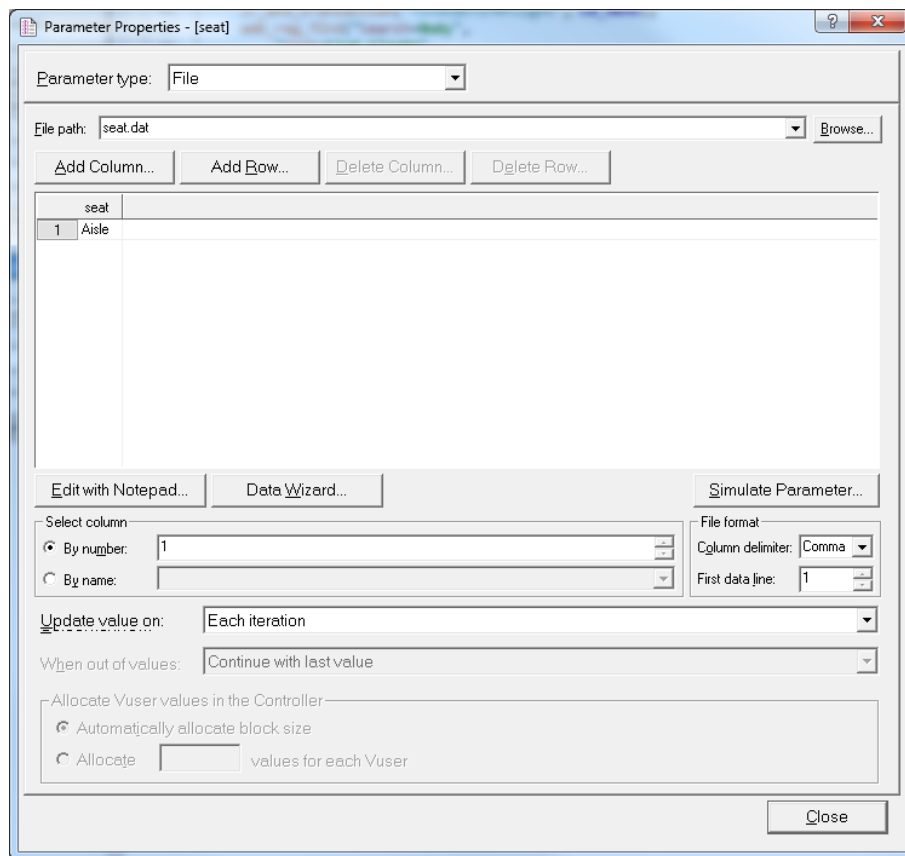
General Data Resource

	Name	Value	
1	depart	Denver	REC
2	departDate	02/11/2013	REC
3	arrive	Los Angeles	REC
4	returnDate	02/12/2013	REC
5	numPassenger	1	REC
6	roundtrip	<OFF>	REC
7	seatPref	{seat}	
8	seatType	Coach	REC
9	findFlights.x	53	REC
10	findFlights.y	10	REC

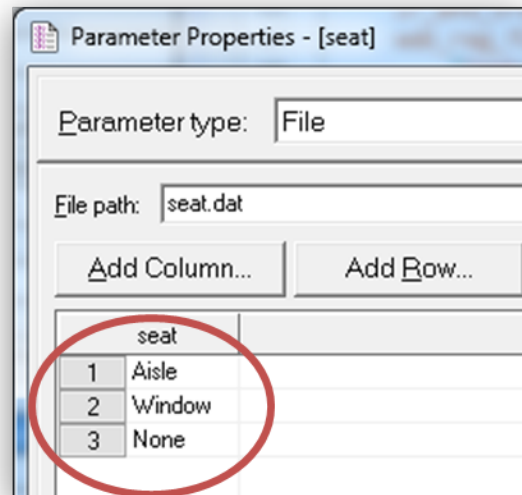
Add...
Delete...

OK Cancel Apply

- c. Click the **Parameter** icon  adjacent to **{seat}** and select **Parameter Properties**. The Parameter Properties dialog box opens.



4. Specify values for the parameter.
 - a. Click **Add Row**. VuGen adds a row to the table.
 - b. Replace the word **Value** with **Window**.
 - c. Click **Add Row**. VuGen adds another row to the table.
 - d. Replace the word **Value** with **None**.



Note: The values are not case sensitive.

- e. Keep the default settings in the **Select column** and **File format** sections of the dialog box.
5. Define how the test will vary the data.
 - a. Keep the default setting that instructs VuGen to take a new value for each iteration:
Update value on: Each iteration.
 - b. Click **Close** to close the Parameter Properties dialog box.
 - c. Click **OK** to close the Submit Form Step Properties dialog box.

You have now created a parameter for the seating preference. When you run the load test, the Vusers will use the parameter values instead of the recorded value, **Aisle**.

When you run the script, the Replay log will show the parameter substitution that occurs for each iteration. The Vuser will use **Aisle** for the first iteration, **Window** for the second iteration, and **None** for the third iteration.

How do I verify Web page content?

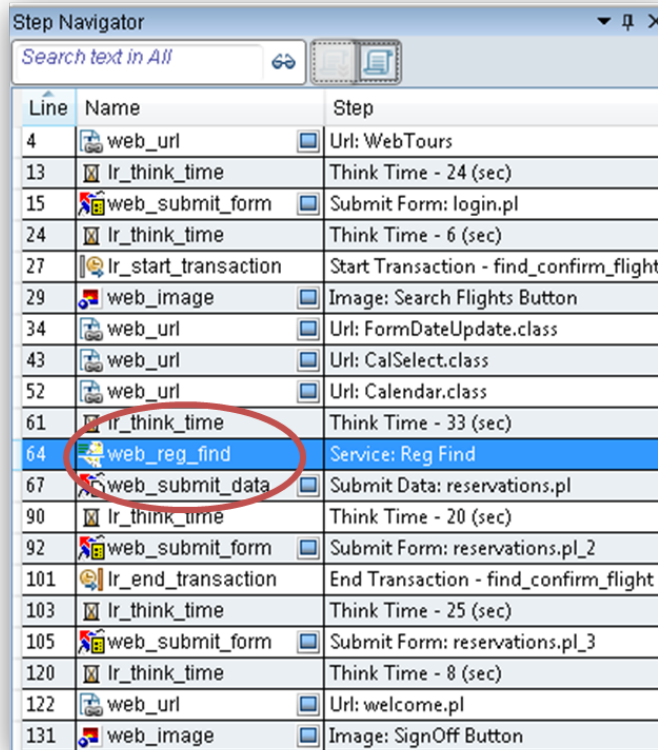
When running a test, you often need to verify if certain content is found on a returned page. A content check verifies that expected information appears on a web page while the script is running. You can insert two types of content checks:

- **Text check.** Checks that a text string appears on a web page.
- **Image check.** Checks for an image on a web page.

In this section, you will insert a text check that checks if the phrase **Find Flight** appears on the Reservations page of HP Web Tours.

Inserting a text check

1. Click the **Step Navigator** tab to display the Step Navigator.
2. In the Step Navigator, locate the **Submit Form: reservations.pl** step.
3. On the VuGen toolbar, click the **Show Snapshot pane** button  to display the Snapshot pane.
4. In the Step Navigator, double-click the snapshot icon in the **Submit Form: reservations.pl** step. The corresponding snapshot appears in the Snapshot pane.
5. From the VuGen menu, click **View > Steps Toolbox**. The Steps Toolbox pane appears.
6. In the Steps Toolbox - Search box, type in **web_reg**, and then locate the **web_reg_find** step in the **Filter Results**.
7. Drag the **web_reg_find** step from the toolbox into the Editor, before the **web_submit_form** function. The Find Text dialog box opens.
8. In the Find Text dialog box, type **Find Flight** into the **Search for specific Text** box, and then click **OK**.
9. VuGen inserts a **web_reg_find** step into the Step Navigator and a corresponding **web_reg_find** function into the Editor.



Line	Name	Step
4	web_url	Url: WebTours
13	lr_think_time	Think Time - 24 (sec)
15	web_submit_form	Submit Form: login.pl
24	lr_think_time	Think Time - 6 (sec)
27	lr_start_transaction	Start Transaction - find_confirm_flight
29	web_image	Image: Search Flights Button
34	web_url	Url: FormDateUpdate.class
43	web_url	Url: CalSelect.class
52	web_url	Url: Calendar.class
61	lr_think_time	Think Time - 33 (sec)
64	web_reg_find	Service: Reg Find
67	web_submit_data	Submit Data: reservations.pl
90	lr_think_time	Think Time - 20 (sec)
92	web_submit_form	Submit Form: reservations.pl_2
101	lr_end_transaction	End Transaction - find_confirm_flight
103	lr_think_time	Think Time - 25 (sec)
105	web_submit_form	Submit Form: reservations.pl_3
120	lr_think_time	Think Time - 8 (sec)
122	web_url	Url: welcome.pl
131	web_image	Image: SignOff Button

When you replay the script, VuGen will look for the text **Find Flight** and indicate in the Replay log whether or not the text was found.

How can I produce debugging information?

At certain points during a test run, you may want to generate and send messages that contain information relating to the script execution. These output messages will appear in both the Replay log in the Output pane, and in the Controller's Output window. You can send a standard output message or a message that indicates that an error occurred.

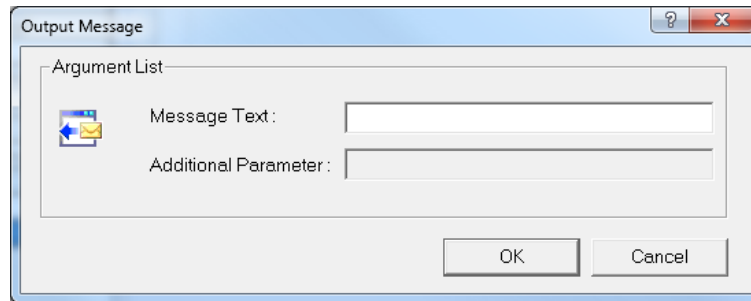
The recommended way to work with error messages is to check for a **Failed** status. If the status is **Failed**, you instruct VuGen to issue an error message. For details, refer to the examples in the HP LoadRunner Function Reference.

In this section of the tutorial, you will instruct VuGen to insert an output message after the application completes a full booking.

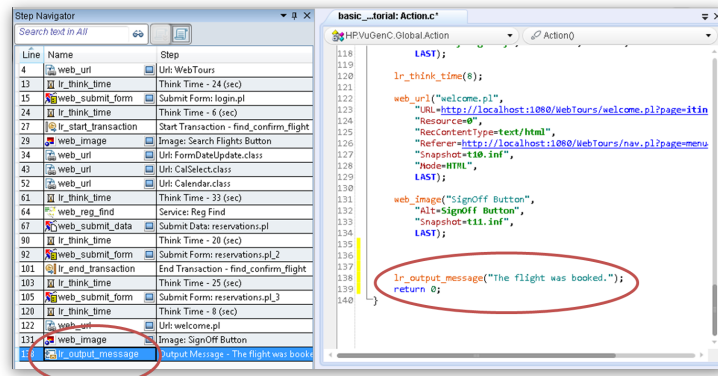
To insert an output message:

1. Click the **Step Navigator** tab to display the Step Navigator.
2. In the Step Navigator, locate the last step, **Image: SignOff Button**.

3. Double-click the **Image: SignOff Button** step. The Editor displays the corresponding **web_image** function.
4. In the Steps Toolbox, under **Common**, locate the **lr_output_message** function.
5. Select the **lr_output_message** function, drag it to the Editor, and release it after the **web_image** function. The Output Message dialog box opens.



6. In the Message Text box, type **The flight was booked.**
7. Click **OK**. VuGen adds an **lr_output_message** to the script, and a corresponding step in the Step Navigator.



8. Click the **Save** button  on the VuGen toolbar to save the script.

Note that to insert an error message you would repeat the same process, except that in the Steps Toolbox, you would select an **lr_error_message** function instead of the **lr_output_message** function.

Did the replay of my test succeed?

In this section, you will run the enhanced script and search the Replay log for text checks. You will view the results of the text checks, as well as details of the transactions and parameterization.

By default, image and text checks are disabled during playback since they require more memory. If you want to perform an image or text check, you need to enable checks in the runtime settings.

1. Enable image and text checks.
 - a. Click **Replay > Runtime Settings** to open the Runtime Settings dialog box.
 - b. Click **Internet Protocol > Preferences**.
 - c. Select **Enable image and text check**.
 - d. Click **OK** to close the Runtime Settings dialog box.
2. Run the script.



Click the **Replay** button on the VuGen toolbar. VuGen begins running the script, generating entries in the Replay log in the Output pane.

Wait for the script to finish running.

3. Locate the text check.
 - a. Click the Output pane, and select **Replay**.
 - b. Click in the Replay log, and then press Ctrl+F to open the Search dialog box.
 - c. Search for **web_reg_find**.

The first instance says as follows:

```
web_reg_find started
```

Click **Find Next** to display the next instance of **web_reg_find**. The second instance says as follows:

```
Registering web_reg_find was successful.
```

This is not the actual text check—it only prepares VuGen to check for the text after the form submission.

Click **Find Next** to display the next instance of **web_reg_find**. This instance indicates:

Registered web_reg_find successful for "Text=Find Flight" (count=1).

This verifies that the text was found. If someone changes the Web page and removes the phrase **Find Flight**, then in subsequent runs, the output will indicate that the text was not found.

4. Locate the beginning of a transaction.
 - a. In the Replay log, press Ctrl+F to open the Search dialog box.
 - b. Search for the word **Transaction**. This notification is shown in blue.
5. View the parameter substitution.
 - a. In the Replay log, press Ctrl+F to open the Search dialog box.
 - b. Search for the word **Parameter**. The log contains a notification "**seat**" = "**Aisle**".
 - c. Search again (F3) for the next substitution. Note how VuGen takes a different value for each iteration.
6. Select **File > Save** or click the **Save** button  on the VuGen toolbar.

Where To Go From Here

Now that you have created a script and adapted it for load testing, you can proceed to "Lesson 5: Creating a Load Testing Scenario" on page 49.

Lesson 5: Creating a Load Testing Scenario

In the previous lesson, you used VuGen to validate your Vuser script. In this lesson, you will evaluate your system under the load of multiple Vusers. You will emulate the actions of ten travel agents concurrently using the flight reservation system, and observe the behavior of the system under the load of these users. To design and run this test, you use the LoadRunner Controller.

Scenario Objectives

In this lesson, the objective is to create a scenario that emulates the behavior of ten travel agents simultaneously logging on, searching flights, purchasing flights, checking itineraries, and logging off.

In this lesson you will cover the following topics:

- [Introducing the LoadRunner Controller](#)
- [How do I start the Controller?](#)
- [The Controller at a glance](#)
- [How do I modify the script details?](#)
- [How do I generate a load on the system?](#)
- [How do I emulate real load behavior?](#)
- [How do I emulate different types of users?](#)
- [How do I monitor the system under load?](#)

Introducing the LoadRunner Controller

Load testing means testing your system under typical working conditions. For example, you may test the system while many travel agents simultaneously reserve flights on the same flight reservation system.

You design the test to emulate real-life situations. To do this, you need to be able to generate a load on an application and schedule when the load is applied (because users do not log on and off the system at precisely the same time). You also need to emulate different types of user activity and behavior. For example, some users may use Firefox to access the system, whereas other users use Internet Explorer. Users may also employ different network connections to access the system, such as modem, DSL, or cable. You create and save these settings in a scenario.

The Controller provides you with all the tools you need to help you build and run tests to accurately emulate your working environment.

How do I start the Controller?

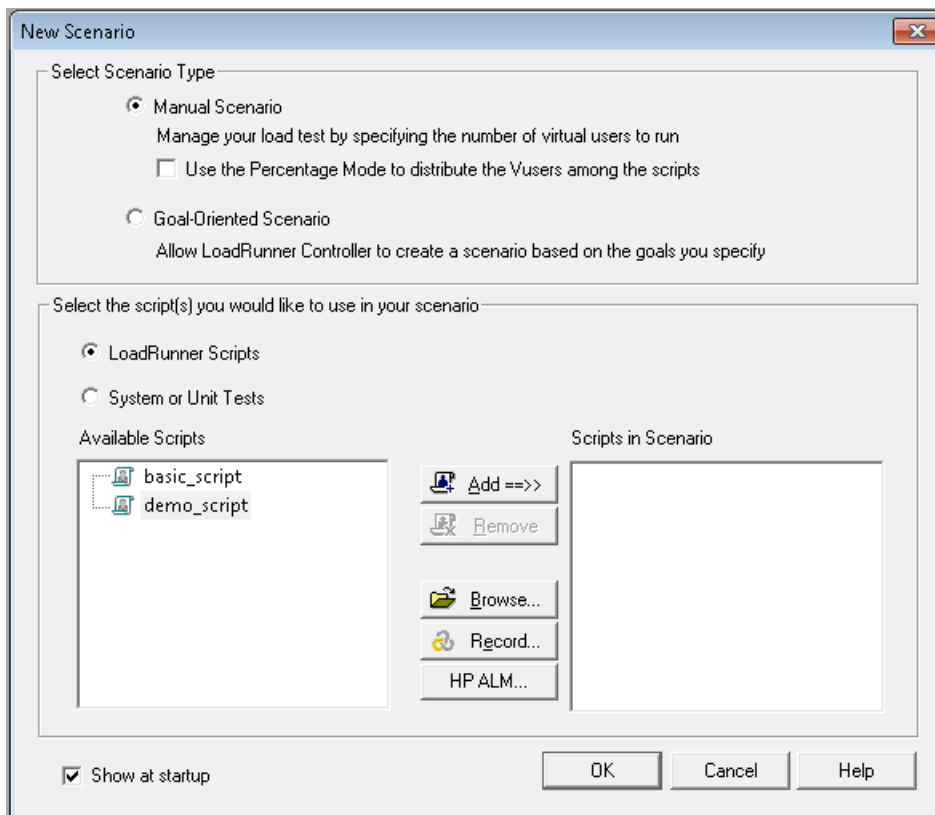
To begin developing a scenario, you open the LoadRunner Controller.

1. Open the HP LoadRunner Controller.

Click the **Controller** shortcut icon on your desktop.



The HP LoadRunner Controller opens and displays the New Scenario dialog box.



2. Select a scenario type.

There are two scenario types:

A **Manual Scenario** gives you control over the number of running Vusers and the times at which they run, and lets you test how many Vusers your application can run simultaneously.

You can use Percentage Mode to distribute the total number of Vusers between the scripts based on a percentage specified by your business analyst. The Percentage Mode check box is selected by default the first time you start LoadRunner following installation. If it is selected, clear it.

A **Goal-Oriented Scenario** is used to determine if your system can achieve a particular goal. You determine the goal based on, for example, a specified transaction response time or number of hits/transactions per second, and LoadRunner automatically builds a scenario for you based on these goals.

- Click **Manual Scenario**.

3. Add a Vuser script to the load test.

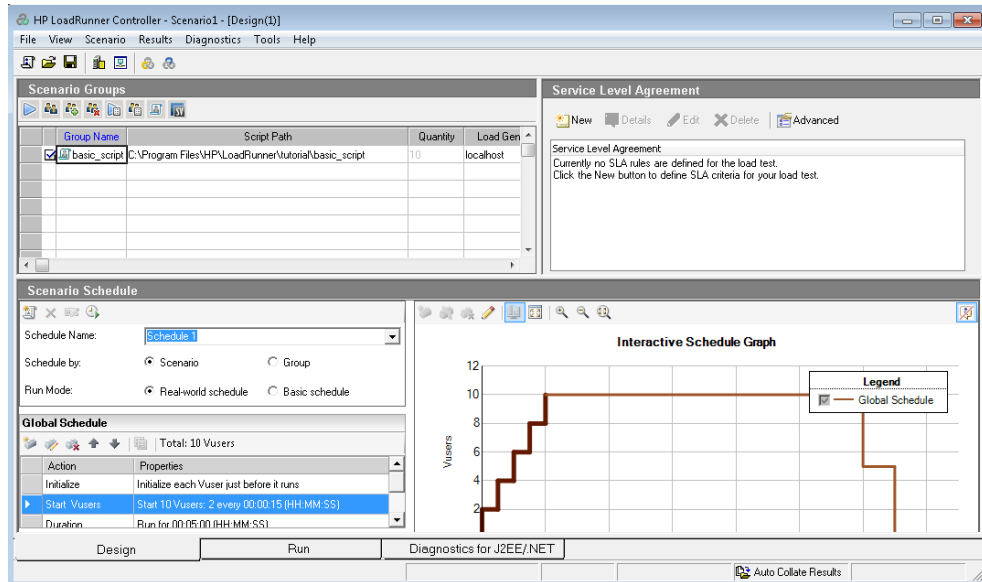
In this tutorial, you will use only one Vuser script to model a single group of users performing identical actions. To more accurately emulate a real-world scenario with more versatile user profiles, you would create a number of different Vuser groups, with each group running several scripts with different user settings.

The script that you previously recorded in VuGen contains the business processes that you want to test. They include logging on, searching for a flight, buying a ticket, checking the flight itinerary, and then logging off the site. You will add a similar script to the scenario, and configure the scenario to emulate eight travel agents simultaneously performing these actions on the flight reservation system. You will add two more Vusers during the test.

For this purpose, a sample script is provided that is similar to the one you created. We recommend that you use the sample script.

- a. If **basic_script** is already in the **Available Scripts** pane, select it and click the **Add** button to move the script to the **Scripts in Scenario** pane .
- b. If **basic_script** is not in the **Available Scripts** pane, click the **Browse** button. Locate **basic_script** in the <LoadRunner Installation>\Tutorial folder. Click **Open**. Click the **Add** button to move the script to the **Scripts in Scenario** pane.
- c. Click **OK**. The LoadRunner Controller opens and displays the **Design** tab of your new

scenario.



Note: The control for the Design tab is in the lower left corner of the Controller.

The Controller at a Glance

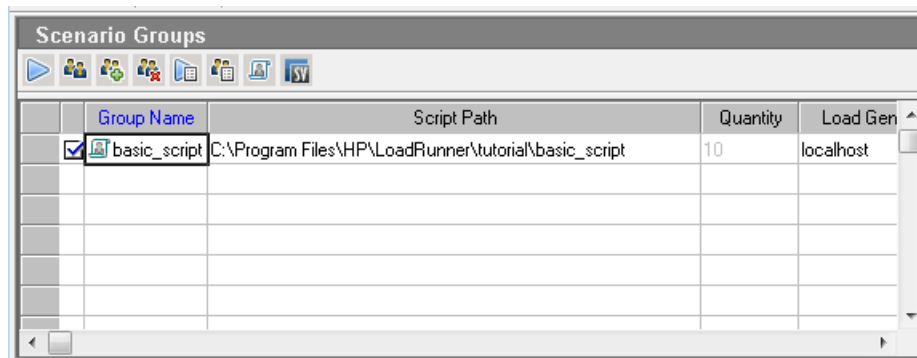
The **Design** tab of the Controller is the main interface for designing your load test. The **Design** tab is divided into three panes:

1. **Scenario Groups pane.** You configure the Vuser groups in the Scenario Scripts pane. You create different groups to represent typical users of your system and specify the number of Vusers that will run, and the machine that they will run on.
2. **Service Level Agreement pane.** When you design a load test scenario, you can define goals or SLAs (Service Level Agreements) for the performance metrics. When you run the scenario, LoadRunner gathers and stores performance-related data. When you analyze the run, Analysis compares this data against the SLAs and determines SLA statuses for the defined measurements.
3. **Scenario Schedule pane.** In the Scenario Schedule pane, you set the load behavior to accurately portray real-world user behavior. You define actions according to which the Vusers will run, the rates at which load is applied to the application, the load test duration, and how the load is terminated

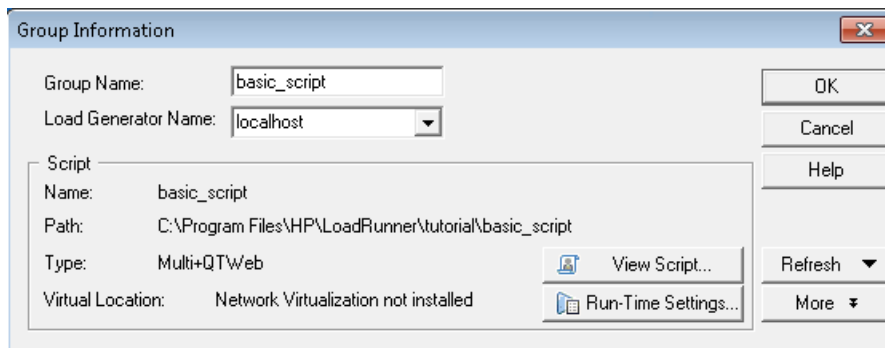
How do I modify the script details?

You modify the script details as follows:

1. Check to see that **basic_script** appears in the **Group Name** column of the **Scenario Groups** pane.



2. Change the **Group Name**.
 - a. In the **Scenario Groups** pane, select **basic_script** and click the **Details** button . The Group Information dialog box opens.



- b. In the **Group Name** box, enter a more meaningful name, for example **travel_agent**.
- c. Click **OK**. The new name is displayed in the **Scenario Groups** pane of the **Design** tab.

How do I generate a load on the system?

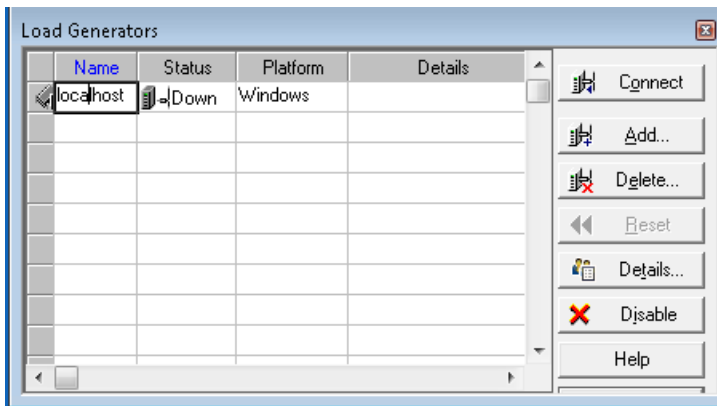
After you have added your Vuser scripts to the scenario, you configure the load generators, the computers that generate the load on the system.

Definition: A load generator is a computer that runs multiple Vusers in order to generate a load on the system. You can use a number of load generators, each generator hosting multiple Vusers.

In this section, you will learn about adding load generators to the scenario, and testing the load generator connections.

Add a load generator.

Click the **Load Generators** button  on the Controller toolbar. The Load Generators dialog box opens.



The Load Generators dialog box enables you to view and configure the load generators that are defined in the scenario. The Load Generators dialog box shows details for the load generator called **localhost**. The status of the **localhost** load generator is **Down**. This indicates that the Controller is not connected to the **localhost** load generator.

In this tutorial, you will use your local computer as the load generator.

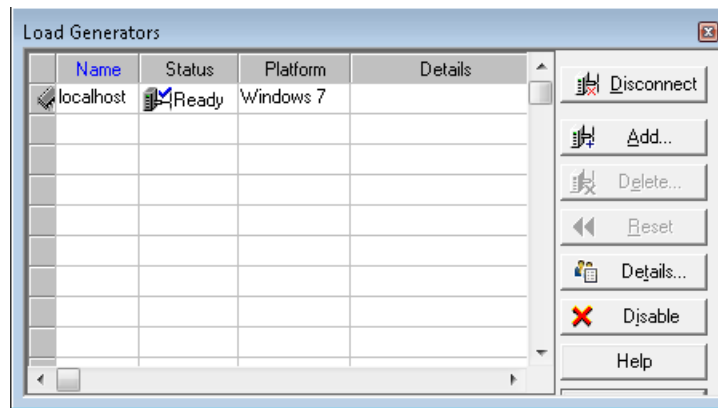
Note: In a typical operational system, you would have several load generators, each hosting multiple Vusers.

Test the load generator connection.

When you run a scenario, the Controller connects to the load generators automatically. However, you can test the connections before trying to run the scenario.

1. In the Load Generators dialog box, select **localhost** and click **Connect**.

The Controller attempts to connect to the load generator machine. When a connection has been made, the **Status** of the load generator changes from **Down** to **Ready**.



2. Click **Close**.

How do I emulate real load behavior?

After you have added your load generators, you are ready to configure load behavior.

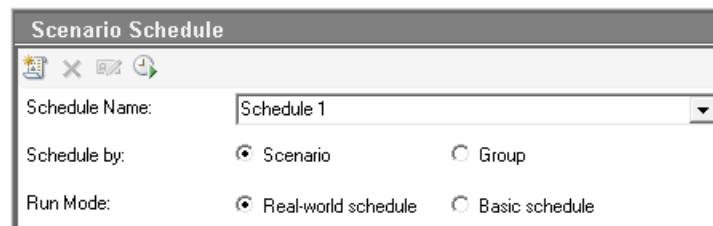
Typical users do not log on and off the system at precisely the same time. LoadRunner allows users to gradually log on to and off the system. It also lets you determine the duration of the scenario, and the way in which the scenario terminates. The scenario that you will configure below will be relatively simple. However, when designing a scenario that more accurately reflects a real life scenario, you can define more true-to-life Vuser activity.

You configure the load behavior for a manual scenario in the **Scenario Schedule** pane of the Controller. The Scenario Schedule pane is divided into three sections: the **Schedule Definition** area, the **Actions** grid, and the **Interactive Schedule** graph.

You will now change the default load settings and configure a scenario schedule.

1. Select schedule type and run mode.

In the **Scenario Schedule** pane, make sure that **Schedule by: Scenario** and **Run Mode: Real-world schedule** are selected.



2. Set the scheduled action definitions.

You can set the **Start Vusers**, **Duration**, and **Stop Vusers** actions for the scenario schedules from either the **Global Schedule** grid, or by manipulating the **Interactive**

Schedule graph. When you set the definitions from the graph, the **Global Schedule** grid properties adjust accordingly.

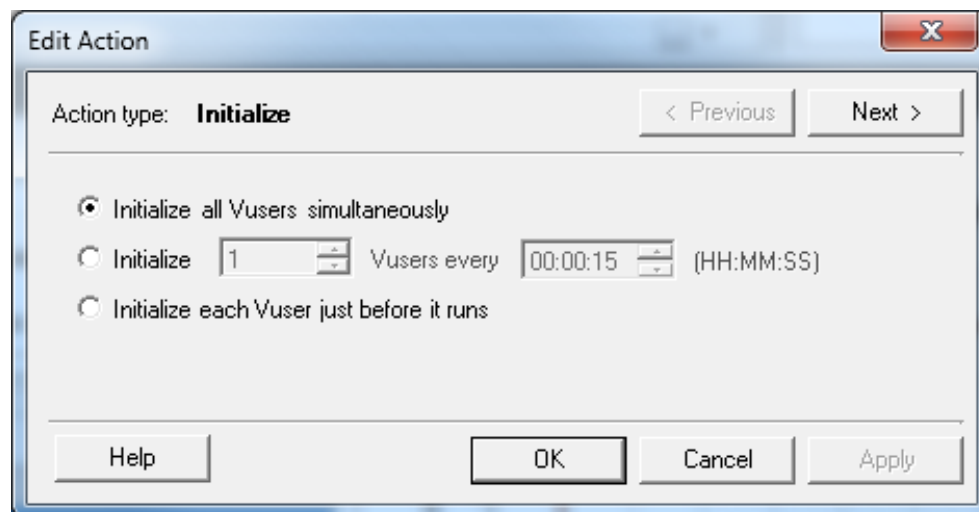
You will now set the definitions so that the **Global Schedule** grid will look as below.

Global Schedule		
		
Action	Properties	
▶ Initialize	Initialize all Vusers simultaneously	
Start Vusers	Start 8 Vusers: 2 every 00:00:30 (HH:MM:SS)	
Duration	Run for 00:10:00 (HH:MM:SS)	
Stop Vusers	Stop all Vusers: 2 every 00:00:30 (HH:MM:SS)	
*		

- a. Set up the Vuser Initialization.

Initialization means preparing the Vusers and the load generators for a load test run by running the **vuser_init** action in the script. Depending on your system configuration, initializing Vusers before they start running may provide more realistic results.

- i. Double-click **Initialize** in the **Global Schedule** grid. The Edit Action dialog box opens displaying the **Initialize** action.



Edit Action

Action type: **Initialize**

 < Previous
 Next >

☒ Initialize all Vusers simultaneously

☐ Initialize Vusers every (HH:MM:SS)

☐ Initialize each Vuser just before it runs

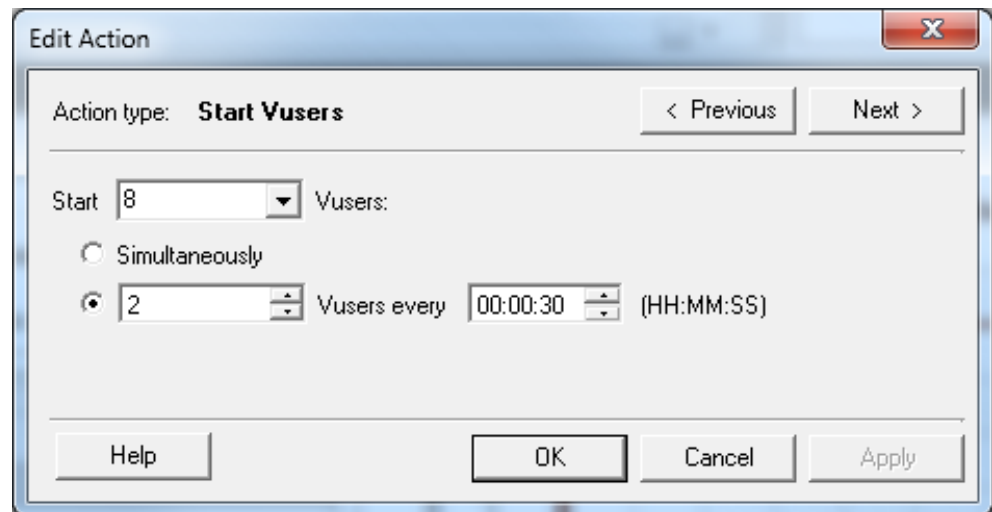
Help
 OK
 Cancel
 Apply

- ii. Select **Initialize all Vusers simultaneously**.
 - iii. Click **OK**.

b. Specify a gradual start for the Vusers

Starting Vusers at intervals allows you to examine the gradual increase of Vuser load on the application under test over time, and helps you pinpoint exactly when the system response time slows down.

- i. Double-click **Start Vusers** in the **Global Schedule** grid. The Edit Action dialog box opens displaying the **Start Vusers** action.



- ii. In the **Start X Vusers** box, enter **8**, and select the second option— **2 Vusers every 00:00:30** (30 seconds).

- iii. Click **OK**.

c. Schedule the duration.

You specify a duration to make sure that the Vusers continue performing the schedule action for a specific period so you can measure continuous load on the server. If you set a duration, the script will run for as many iterations as necessary during that period, disregarding the number of iterations set in the script's runtime settings.

- i. Make sure that the Interactive Schedule Graph is in Edit mode by clicking the **Edit**

Mode button  on the Interactive Schedule Graph toolbar.

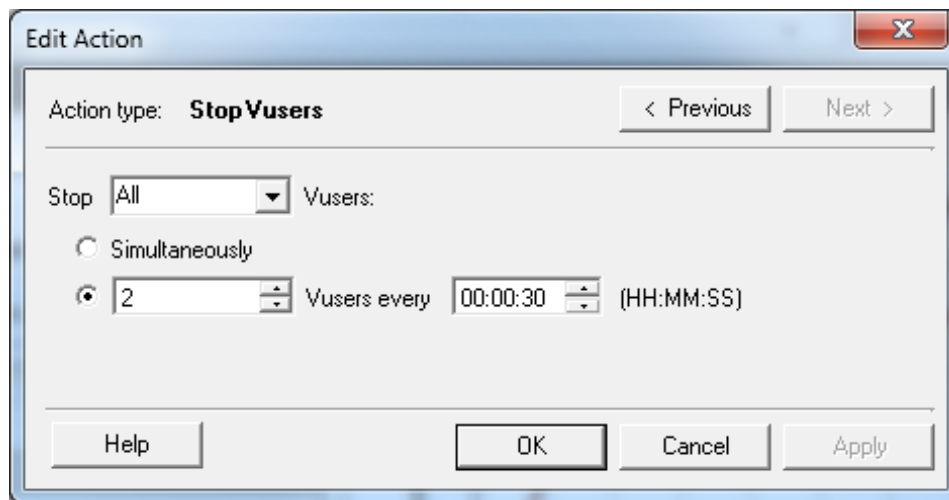
- ii. In the Interactive Schedule Graph, click the horizontal line which represents Duration. The line is highlighted and a dot and a diamond are displayed at its endpoints.

Note: The Legend is displayed on top of the diamond, click the **Hide Legend** button  on the Interactive Schedule Graph toolbar to show the diamond.

- iii. Drag the diamond shaped endpoint to the right until the time in brackets reads **00:11:30**. You have just set the Vusers to run for a period of 10 minutes.
- d. Schedule a gradual closure.

Gradually stopping Vusers is recommended to help detect memory leaks and check system recovery, after the application has reached a threshold.

- i. Double-click **Stop Vusers** in the **Global Schedule** grid. The Edit Action dialog box opens displaying the **Stop Vusers** action.



- ii. Select the second option and enter the following values—**2 Vusers every 00:00:30** (30 seconds).
- iii. Click **OK**.

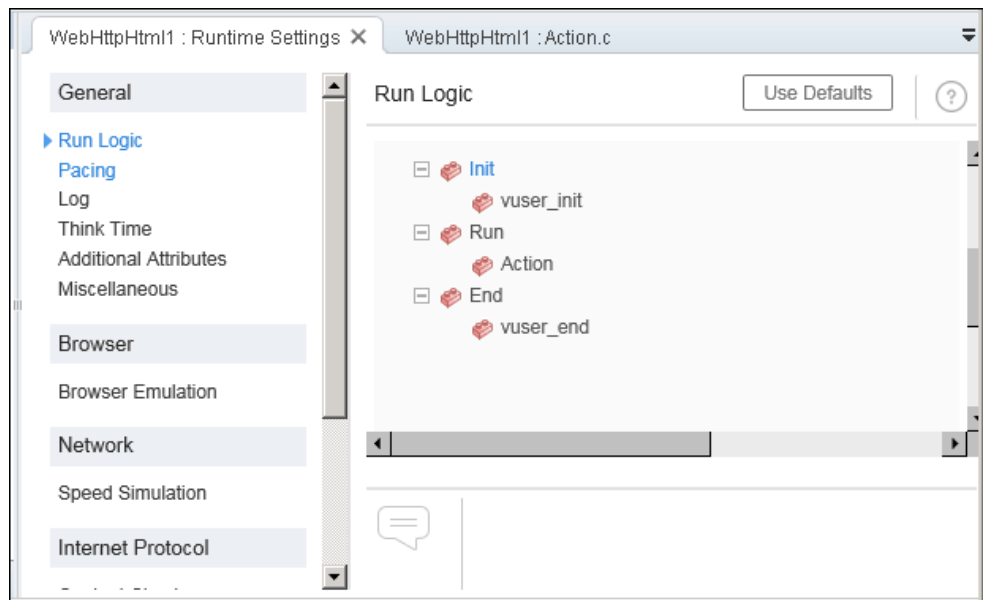
How do I emulate different types of users?

Now that you have configured a load schedule, you will specify how the Vusers will behave during the test.

When emulating a real user, you need to consider the user's actual behavior. Behavior refers to the time that a user takes to pause between actions, the number of times a user repeats an action, and so on.

In this section, you will learn more about LoadRunner's runtime settings, and you will enable Think Time and Logging.

1. Open the runtime settings.
 - a. In the Controller, click the **Design** tab.
 - b. In the **Scenario Groups** pane, select the **travel_agents** group.
 - c. Click the **Runtime Settings** button  on the Scenario Groups toolbar. The Runtime Settings dialog box opens.



The runtime settings let you emulate different kinds of user activity and behavior. They include:

Run Logic. The number of times a Vuser repeats a set of actions.

Pacing. The time to wait before repeating the action.

Log. The level of information that you want to gather during the test. The first time you run a scenario, it is recommended to generate log messages to make sure that you have debugging information in case the first run fails.

Think Time. The time the user stops to think between steps. Since users interact with the application according to their experience level and objectives, more technically proficient users may work more quickly than new users. Vusers can be made to emulate their real-world counterparts more accurately during a load test by enabling think time.

Speed Simulation. Users using different network connections such as modem, DSL, and cable.

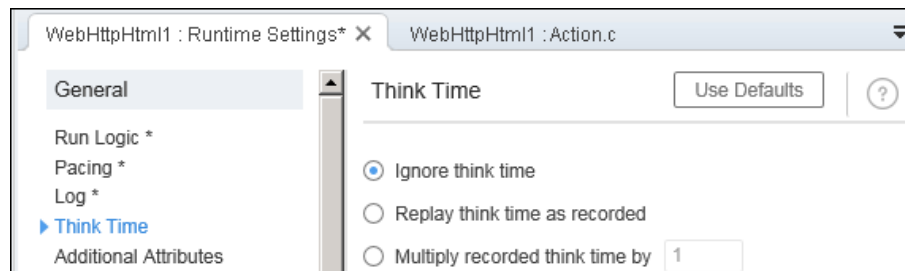
Browser Emulation. Users using different browsers to see their application's performance.

Content Check. For automatically detecting user-defined errors.

Suppose that your application sends a custom page when an error occurs. This custom page always contains the words ASP Error. You need to search all of the pages returned by the server and see if the text ASP Error appears.

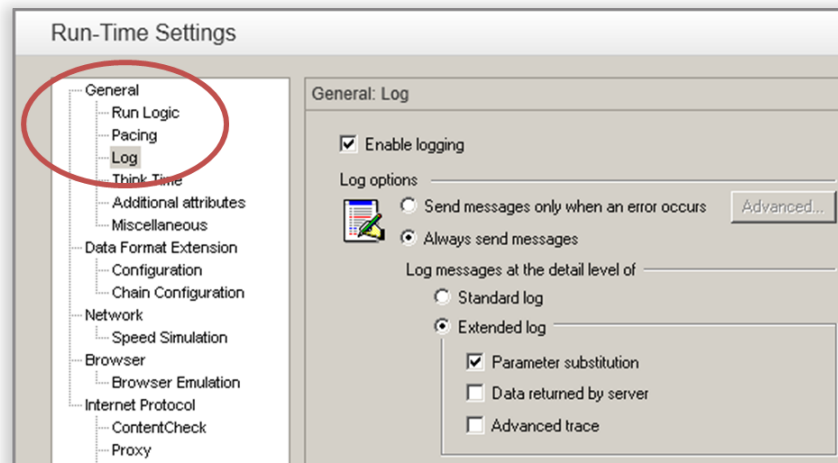
You can set up LoadRunner to automatically look for this text during the test run, using the Content Check runtime settings. LoadRunner searches for the text and generates an error if it is detected. During the scenario run, you can identify the content check errors.

2. Enable think time.
 - a. In the Runtime Settings dialog box, click **General > Think Time**.
 - b. Select **Replay think time**, and select **Use random percentage of recorded think time**.
 - c. Specify a minimum of **50%** and a maximum of **150%**.



The above specifications use a random percentage of the recorded think time to emulate users with a varying range of proficiency. For example, if the recorded think time for selecting a flight was 4 seconds, the random think time could be anything between 2-6 seconds (50% to 150% of 4).

3. Enable logging.
 - a. In the Runtime Settings dialog box, click **General > Log**.
 - b. Select **Enable logging**.
 - c. Under **Log options**, select **Always send messages**.
 - d. Click **Extended log**, and select **Data returned by server**.



Note: After the initial debugging run, extended logging is not recommended for a load test. It is enabled only for the purposes of this tutorial to provide information for the Vuser Output log.

- e. Click **OK** to close the Runtime Settings dialog box.
- f. Save the scenario.

How do I monitor the system under load?

Now that you have defined how your Vusers will behave during the test, you are ready to set up your monitors.

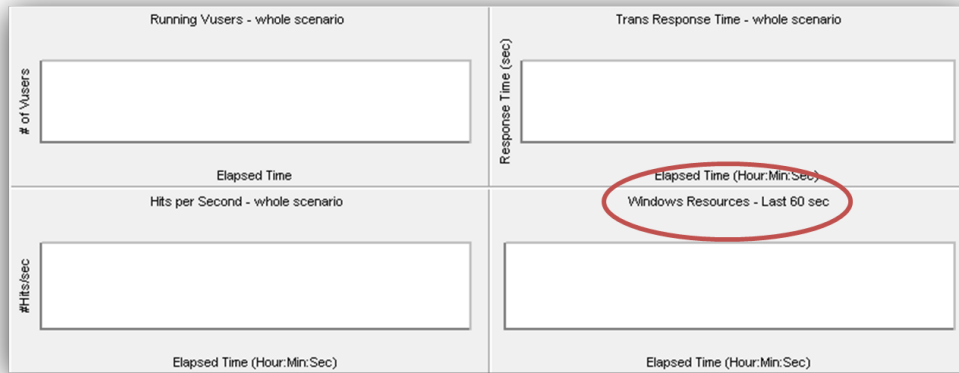
While generating a load on an application, you want to see how the application performs in real time and where potential bottlenecks exist. You use LoadRunner's suite of integrated monitors to measure the performance of every single tier, server, and component of the system during the load test. LoadRunner includes monitors for a variety of major backend system components including Web, application, database, and ERP/CRM servers.

For instance, you can select a Web Server Resources monitor according to the type of Web server that is running. You can purchase a license for the relevant monitor, for example IIS, and use that monitor to pinpoint problems reflected in the IIS resources.

In this section, you will learn how to add and configure the Windows Resources monitor. You can use this monitor to determine the impact of load on your CPU, disk, and memory resources.

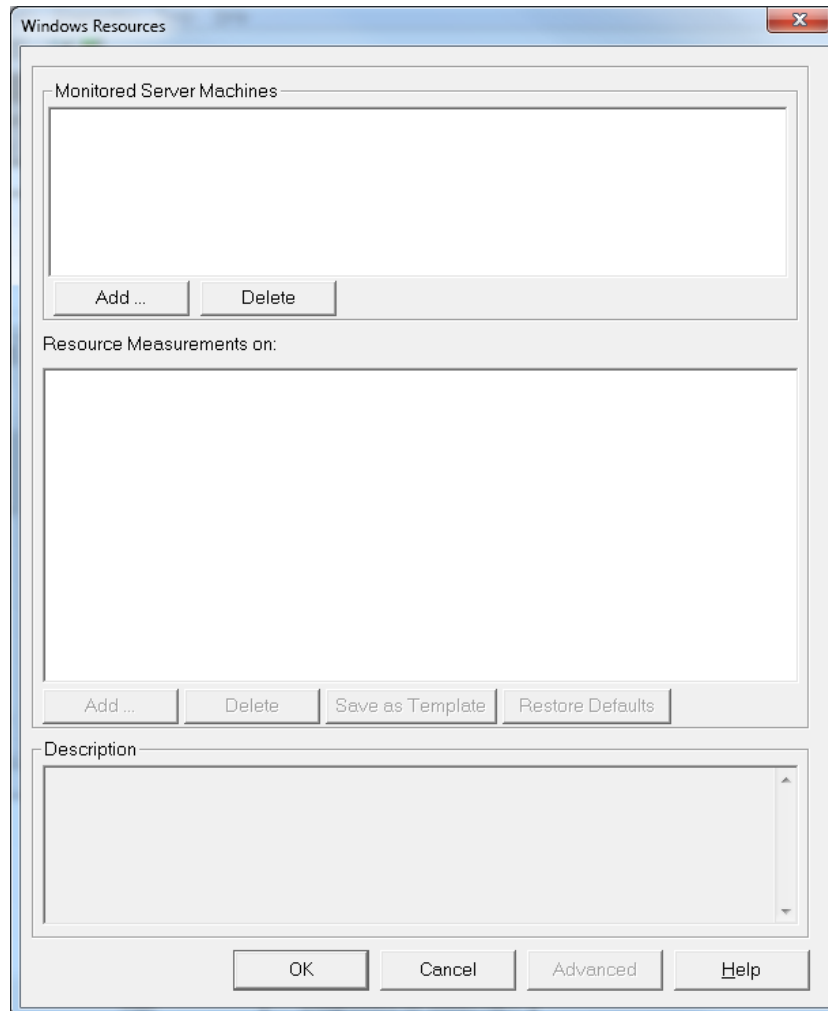
1. Select the Windows Resources Monitor.
 - a. Click the **Run** tab at the bottom of the Controller.

The Windows Resources graph is one of four default graphs that are displayed in the graph viewing area. You will learn how to open other graphs in the next lesson.

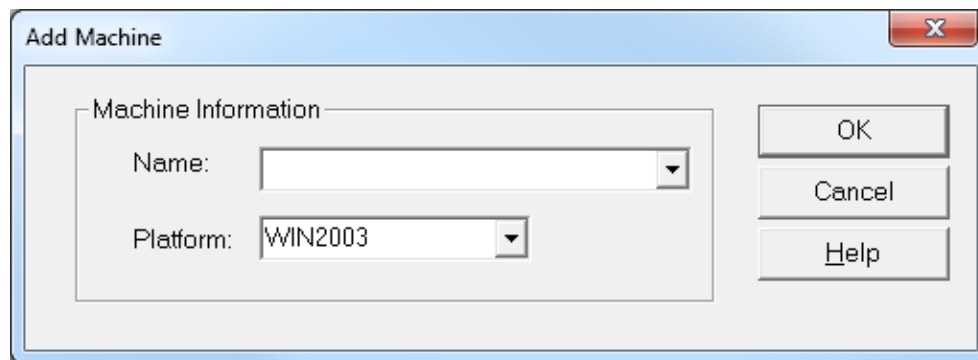


- b. Right-click inside the Windows Resources graph and select **Add Measurements**. The

Windows Resources dialog box opens.

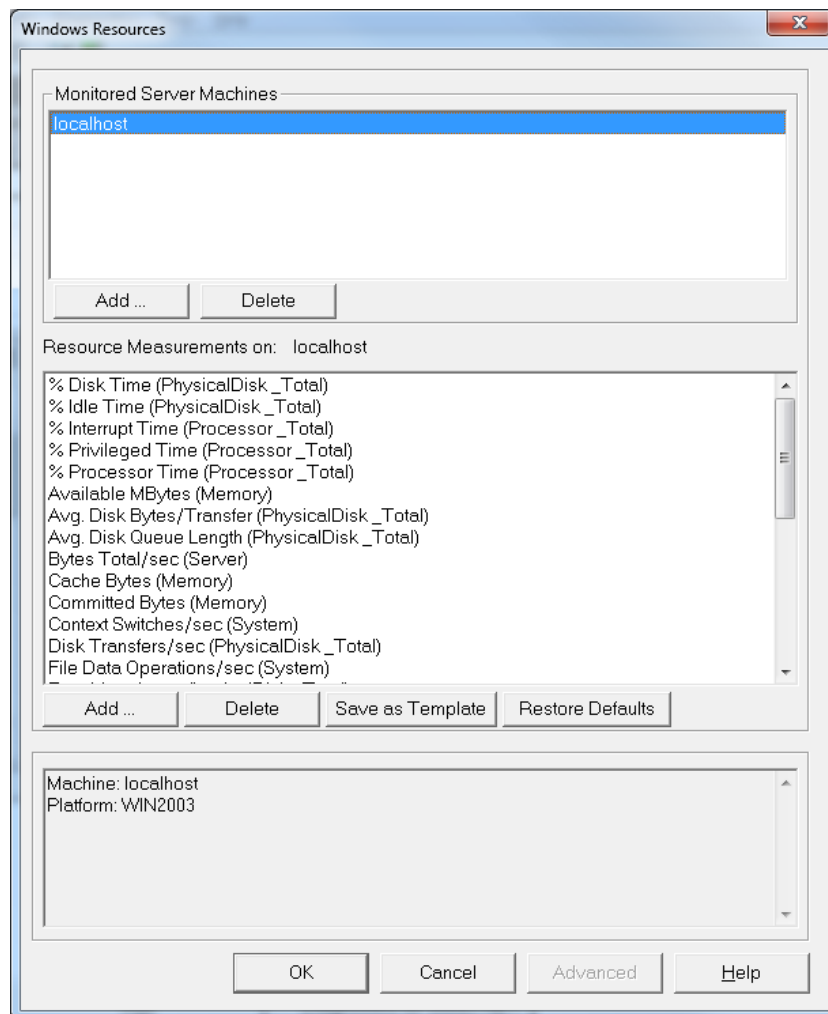


2. Select the monitored server.
 - a. In the **Monitored Server Machines** section of the Windows Resources dialog box, click **Add**. The Add Machine dialog box opens.



- b. In the **Name** box, type **localhost**. (If your load generator was running off a different machine you would type the server name or IP address of that machine.)
- c. From the **Platform** list, select the platform on which the machine runs.
- d. Click **OK**.

The default Windows Resources measurements are listed under the **Resource Measurements on <server machine>**.



3. Activate the monitor.

Click **OK** in the Windows Resources dialog box to close the dialog box and activate the monitor.

Where To Go From Here

Now that you have designed a load test scenario, you can proceed to "Lesson 6: Running the Load Test" on page 66.

Lesson 6: Running the Load Test

When you run a load test, LoadRunner generates a load on the system. You can then use LoadRunner's monitors and graphs to observe the performance of the system under load.

In this lesson you will cover the following topics:

- [The Controller Run view at a glance](#)
- [How do I run a load test scenario?](#)
- [How do I watch a Vuser running in real time?](#)
- [Where can I view a summary of user actions?](#)
- [How can I increase the load during the test?](#)
- [How is the application performing under load?](#)
- [Did the application encounter errors?](#)
- [How do I know that the test finished running?](#)
- [Did the application perform well under load?](#)

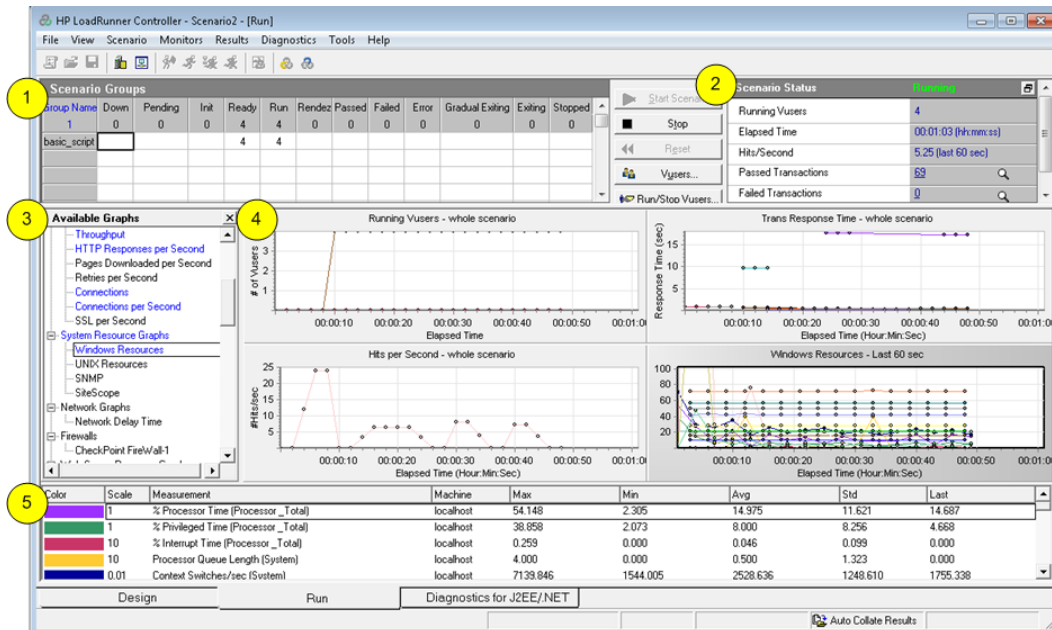
The Controller Run view at a glance

The Run tab in the Controller is the control center from which the scenario is managed and monitored. The **Run** tab (located at the lower section of the Controller window) contains five panes:

1. **Scenario Groups pane.** In the upper-left pane, you can view the status of Vusers in the scenario groups. You use the buttons to the right of this pane to start, stop, and reset the scenario, to view individual Vuser status, and to increase the load on the application during a scenario by manually adding more Vusers.
2. **Scenario Status pane.** In the upper-right pane, you can view a summary of the load test, including the number of running Vusers and the status of each Vuser action.
3. **Available Graphs pane.** In the middle-left pane, you can see a list of the LoadRunner graphs. To open a graph, select a graph in the tree, and drag it into the graph viewing area.
4. **Graph Display pane.** In the middle-right pane, you can customize the display to view between one and eight graphs (**View > View Graphs**).

5. **Graph Legend pane.** In the bottom pane, you can view data from the selected graph.

Note: The control for the Run tab is at the bottom of the Controller.



How do I run a load test scenario?

In this section, you will start the scenario.

1. Open the Controller Run view.

Click the **Run** tab at the bottom of the Controller.

Notice that there are 8 Vusers in the **Down** column of the **Scenario Groups** pane. These are the Vusers you created when you created the scenario.

Scenario Groups												
Group Name	Down	Pending	Init	Ready	Run	Rendez	Passed	Failed	Error	Gradual Exiting	Exiting	Stopped
1	8	0	0	0	0	0	0	0	0	0	0	0
travel_agents	8											

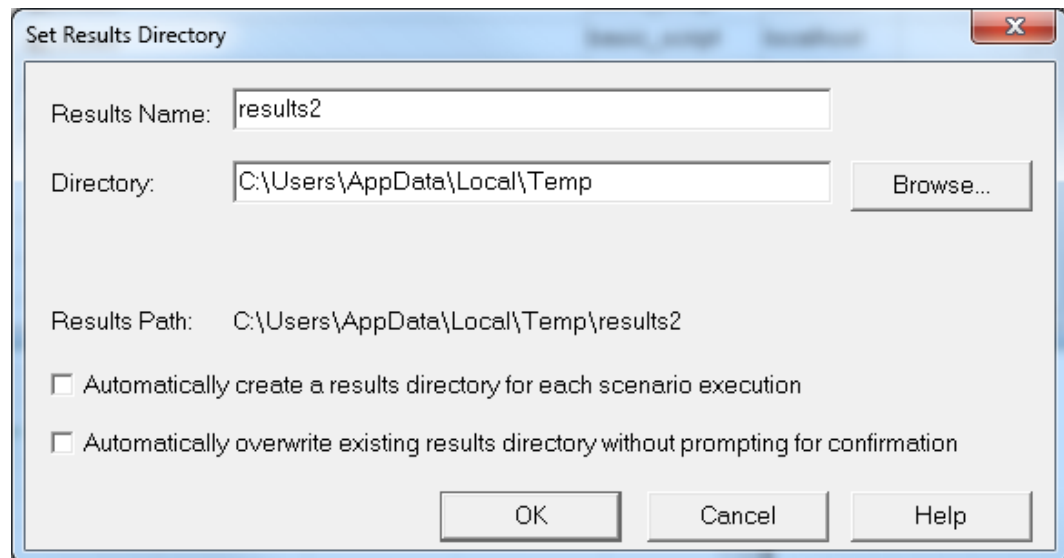
Since the scenario has not yet run, all other counters remain at zero and all the graphs in the graph viewing area (except Windows Resources) are blank. When you start running the scenario in the next step, the graphs and counters will begin to display information.

2. Start the scenario.

Click the **Start Scenario** button or select **Scenario > Start** to start running the scenario.

If you are running the tutorial for the first time, the Controller begins the scenario. The result files are saved automatically to the load generator's **temp** folder.

If you are repeating the test, you will be prompted to overwrite the existing results file. Click **No**, because the results of the first load test should be used as baseline results to be compared with subsequent load test results. The Set Results Directory dialog box opens.



Specify a new results folder. Enter a unique and meaningful name for each results set, because you may want to superimpose the results of several scenario runs when you come to analyze the graphs.

How do I monitor the application under load?

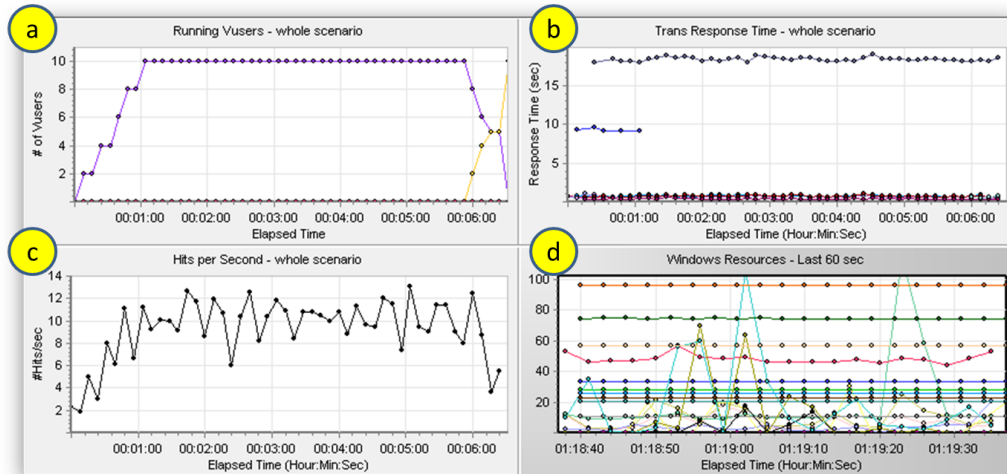
You use the Controller's online graphs to view performance data collected by the monitors. You use this information to isolate potential problem areas in your system environment.

1. Examine the performance graphs.

The Graph Display pane of the Run tab displays the following default graphs:

- a. **Running Vusers - Whole Scenario graph.** Displays the number of Vusers running at a given time.
- b. **Transaction Response Time - Whole Scenario graph.** Shows the amount of time it takes for each transaction to be completed.

- c. **Hits per Second - Whole Scenario graph.** Displays the number of hits (HTTP requests) made to the Web server by Vusers during each second of the scenario run.
- d. **Windows Resources graph.** Displays the Windows resources measured during a scenario.

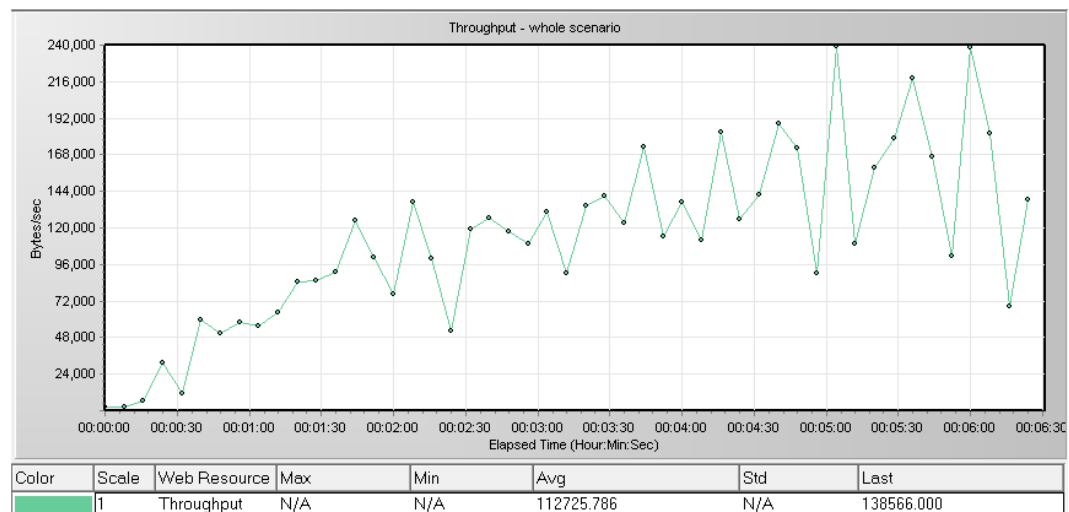


2. Highlight individual measurements.
 - a. Double-click the Windows Resources graph to enlarge it to fit the entire Graph Display pane. Notice that each measurement appears on a color-coded row in the Graph Legend pane. Each row corresponds to a line in the graph with the same color.
 - b. Select a row to highlight the corresponding line in the graph, and vice versa.
 - c. Double-click the graph again to reduce its size.
3. View the throughput information.

In the Available Graphs pane, under **Web Resources Graphs**, select the **Throughput** graph, and drag it into the Graph Display pane. The Throughput graph measurements are displayed in the Graph Display pane and the Graph Legend pane.

The Throughput graph shows the amount of data (measured in bytes) that the Vusers receive from the server in any given second. You can compare this graph with the Transaction Response Time graph to see how throughput affects transaction performance.

If the throughput scales upward as time progresses and the number of Vusers increases, this indicates that the bandwidth is sufficient. If the graph were to remain relatively flat as the number of Vusers increased, it would be reasonable to conclude that the bandwidth is constraining the volume of data delivered.

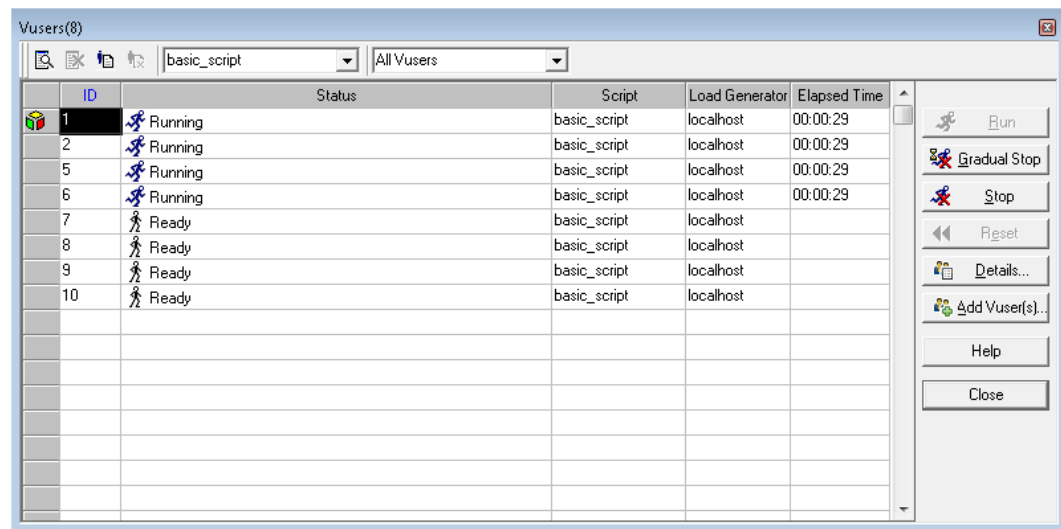


How do I watch a Vuser running in real time?

When emulating users, you should be able to view the Vusers' actions in real time to make sure they are performing the correct actions. The Controller lets you view the actions in real time using the Runtime Viewer.

To visually observe a Vuser's actions:

1. In the Controller's **Run** tab, click the **Vusers** button. The Vusers dialog box opens.



The **Status** column displays the status of each Vuser. In the example above, you can see that four Vusers are running and four are down. The Start Vusers action in the scheduler

instructed the Controller to release two Vusers at a time. As the scenario progresses, Vusers will continue to be added in groups of two at 30-second intervals.

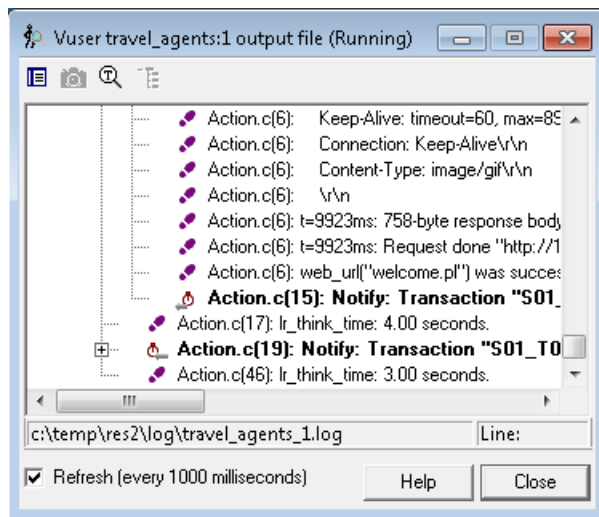
2. Select a running Vuser in the Vuser list.
3. Click the **Show the selected Vusers** button  on the Vusers toolbar. The Runtime Viewer opens and displays the actions performed by the selected Vuser. The Runtime Viewer updates as the Vuser proceeds through the steps of the Vuser script.
4. Click the **Hide the selected Vusers** button  on the Vusers toolbar to close the Runtime Viewer.

Where can I view a summary of Vuser actions?

To see the progress of an individual Vuser during a running test, you can display a log file that contains a text summary of the Vuser's actions.

To review a text summary of the Vuser's actions

1. In the Vusers window, select a running Vuser.
2. Click the **Show Vuser Log**  button on the Vusers toolbar. The Vuser log dialog box opens.



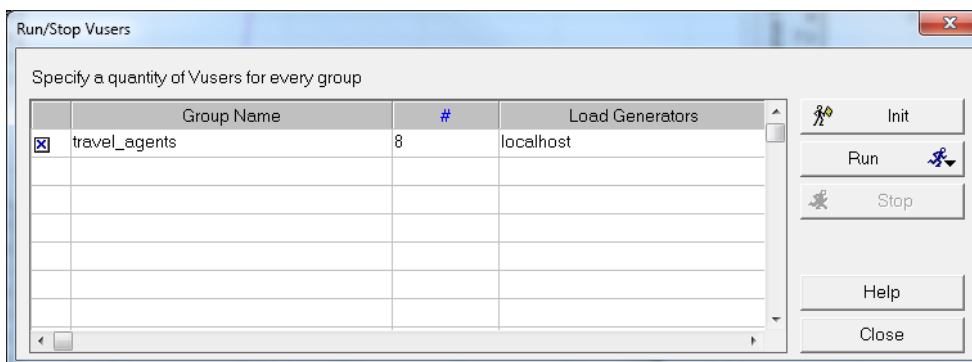
The log contains messages that correspond to the actions of the Vuser. For example, in the window above, the message **Virtual User Script started** indicates the start of the Vuser's run. Scroll to the bottom of the log and watch as new messages are added for each action performed by the Vuser.

3. Close the Vuser Log dialog box and the Vusers dialog box.

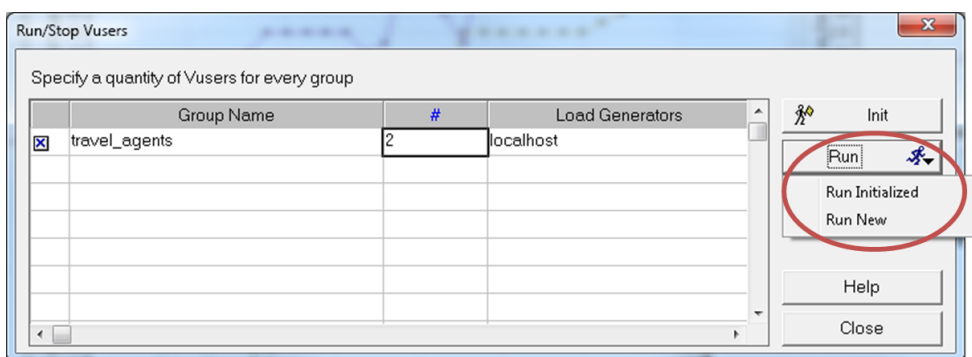
How can I increase the load during the test?

You can manually add more Vusers during a load test to increase the load on the system.

1. To increase the load during a load test:
 - a. Make sure that in the Controller, the **Run** tab is displayed
 - b. Click the **Run/Stop Vusers** button. The Run/Stop Vusers dialog box opens displaying the number of Vusers currently assigned to run in the scenario.



- c. In the **#** column, enter the number of Vusers that you want to add to the group. To run two additional Vusers, replace the number **8** with the number **2**, in the **#** column.
- d. Click **Run** to add the Vusers.
- e. If some of the original Vusers have not yet been initialized, the **Run Initialized** and **Run New** options appear. Select the **Run New** option.



The two additional Vusers are distributed to the **travel_agent** group and are run on the **localhost** load generator. The Scenario Status pane shows that there are now 10 running Vusers.

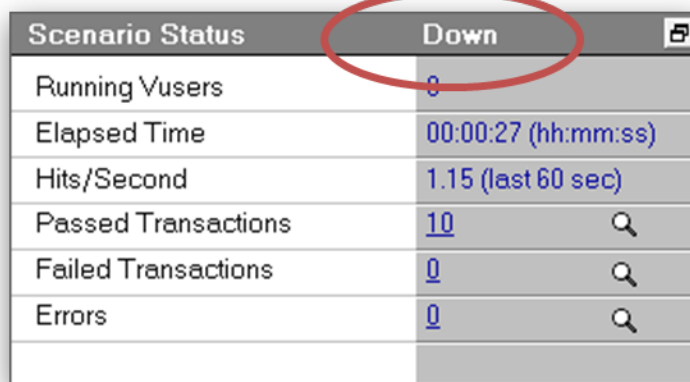
You may get a warning message that the LoadRunner Controller cannot activate additional Vusers. This is because you are using your local machine as a load generator and it has limited memory resources. Generally, use a dedicated machine as a load generator to avoid this issue.

How is the application performing under load?

Check the Scenario Status pane [on the Run tab] for a summary of the running scenario. In the Scenario Status pane, you can drill down to see which Vuser actions are causing the application problems. A high number of failed transactions and errors may indicate that the application is not performing as required under load.

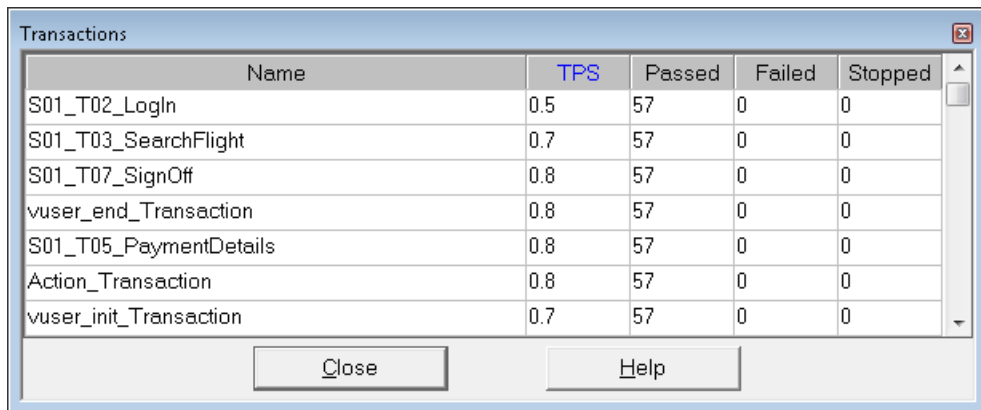
1. View the scenario status.

The header of the Scenario Status pane displays the overall status of the scenario.



Scenario Status	Down	
Running Vusers	0	
Elapsed Time	00:00:27 (hh:mm:ss)	
Hits/Second	1.15 (last 60 sec)	
Passed Transactions	10	🔍
Failed Transactions	0	🔍
Errors	0	🔍

2. View a breakdown of Vuser actions.
 - a. Click **Passed Transactions** in the Scenario Status pane. The Transactions dialog box opens and displays a list of transaction details.



The Transactions dialog box displays a table with the following data:

Name	TPS	Passed	Failed	Stopped
S01_T02_LogIn	0.5	57	0	0
S01_T03_SearchFlight	0.7	57	0	0
S01_T07_SignOff	0.8	57	0	0
vuser_end_Transaction	0.8	57	0	0
S01_T05_PaymentDetails	0.8	57	0	0
Action_Transaction	0.8	57	0	0
vuser_init_Transaction	0.7	57	0	0

Buttons: Close, Help

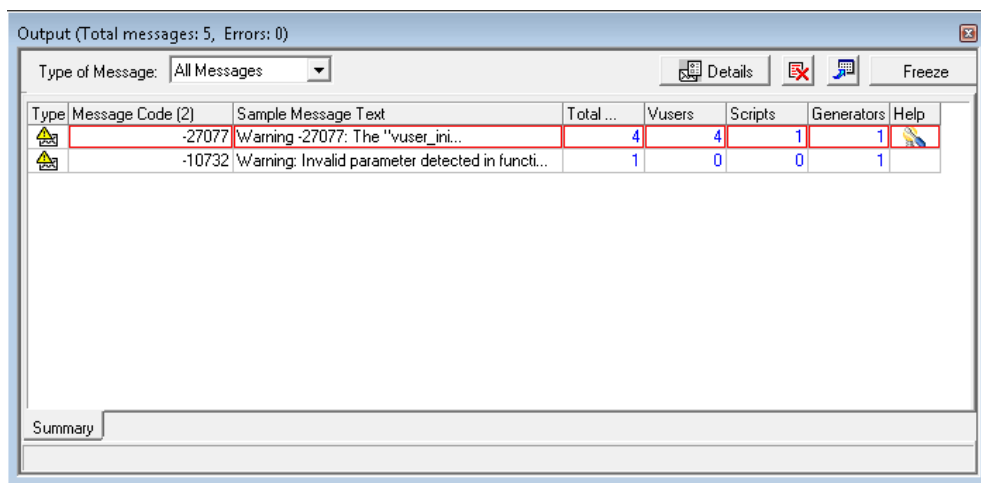
- b. Click **Close** to close the Transactions dialog box.

Did the application encounter errors?

If an application starts to fail under heavy load, you are likely to encounter errors and failed transactions. The Controller displays error messages in the Output window.

1. Check for any error or warning messages.
 - a. Select **View > Show Output**. You can sort the message by type, such as **Warnings** or **Errors**.

The Output dialog box opens and lists a message text, the total number of messages generated, the Vusers and load generators that generated the error, and the scripts in which the errors occurred.



The Output dialog box displays a table with the following data:

Type	Message Code (2)	Sample Message Text	Total ...	Vusers	Scripts	Generators	Help
Warning	-27077	Warning -27077: The "vuser_ini...	4	4	1	1	
Warning	-10732	Warning: Invalid parameter detected in functi...	1	0	0	1	

Buttons: Details, Freeze

Summary

- b. To view a message in detail, select the message and click **Details**. The Detailed Message Text box opens, displaying the complete message text.

2. View log information details.

You can view information about each message, Vuser, script, and load generator associated with an error code by clicking the blue link in the appropriate column.

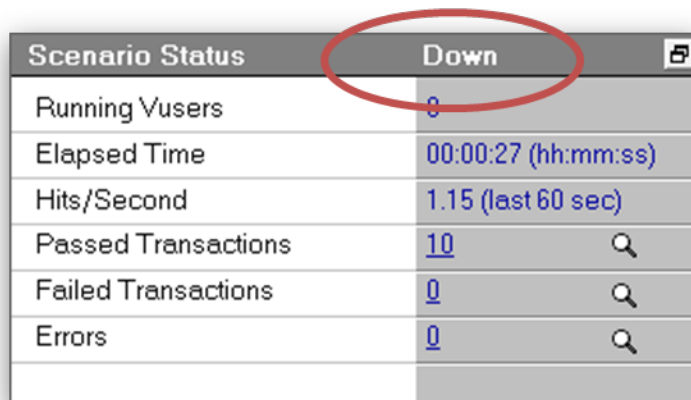
For example, to locate where in the script an error occurred, drill down the Total Messages column. The Output window displays a list of all messages of the selected error code, including the time, iteration number, and line in the script where the error occurred.

3. Drill down the Line Number column.

VuGen opens, displaying the line in the script at which the error occurred. You can use this information to identify transactions with slow response times that are causing the application to fail under load.

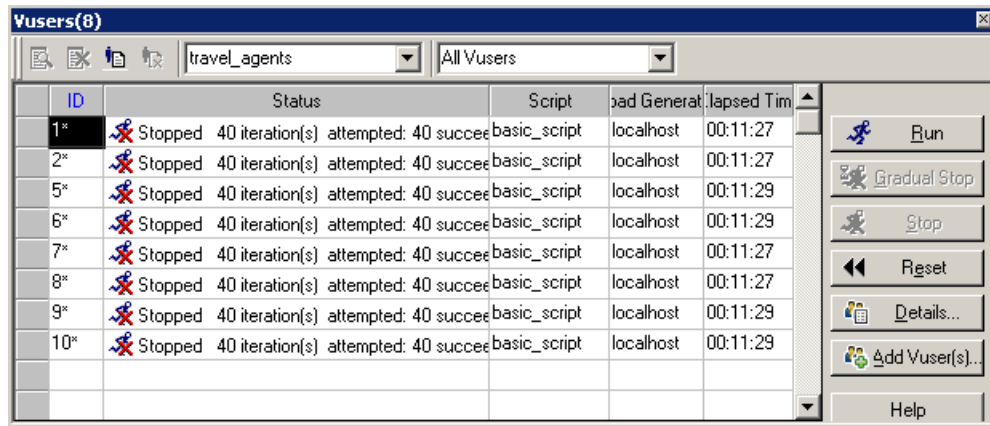
How do I know that the scenario has finished running?

At the conclusion of a scenario run, the header of the Scenario Status pane shows the **Down** status. This indicates that all the Vusers in the scenario have finished running.



Scenario Status	Down	
Running Vusers	0	
Elapsed Time	00:00:27 (hh:mm:ss)	
Hits/Second	1.15 (last 60 sec)	
Passed Transactions	10	Q
Failed Transactions	0	Q
Errors	0	Q

You can open the Vusers dialog box to see the status of each individual Vuser. The Vuser dialog box displays the number of iterations that each Vuser performed, the number of successful iterations, and the elapsed time.



Did the system perform well under load?

To see how well the system performed under load, look at the transaction response times and determine whether the response times were within acceptable limits. If the transaction response times increase during the scenario, you need to look for bottlenecks. You will learn more about this in ["Lesson 7: Analyzing Your Scenario" on page 77](#).

Once a problem has been isolated, a corroborative effort involving developers, DBAs, network, and other systems experts is required to fix the problem. After adjustments are made, the load test is repeated to confirm that the adjustments had the desired effect. You repeat this cycle to optimize system performance.

To save the scenario so that you can run it again with the same settings, select **File > Save** or click the **Save** button  on the Controller toolbar.

Where To Go From Here

Now that you have learned how to run and view a simple load test scenario, you can proceed to ["Lesson 7: Analyzing Your Scenario" on page 77](#).

Lesson 7: Analyzing Your Scenario

In the previous lessons you learned how to design, control, and execute a scenario run. Once you have loaded your server, you want to analyze the run, and pinpoint the problems that need to be eliminated to improve system performance.

The graphs and reports produced during your analysis session present important information about the performance of your scenario. Using these graphs and reports, you can pinpoint and identify the bottlenecks in your application, and determine what changes need to be made to your system to improve its performance.

In this lesson, you will cover the following topics:

- [How does an analysis session work?](#)
- [How do I start my analysis session?](#)
- [The Analysis window at a glance](#)
- [Did I reach my goals? \(Service Level Agreement\)](#)
- [How do I view a summary of performance?](#)
- [How do I graphically view performance?](#)
- [Did my server perform well?](#)
- [How can I pinpoint the source of a problem?](#)
- [What other information can I gather about my scenario run?](#)
- [How can I publish my findings?](#)
- [Conclusion](#)

How does an analysis session work?

The aim of the analysis session is to find the failures in your system's performance and then pinpoint the source of these failures, for example:

- Were the test expectations met? What was the transaction time on the user's end under load? Did the SLA meet or deviate from its goals? What was the average transaction time of the transactions?
- What parts of the system could have contributed to the decline in performance? What was the time of the network and servers?

- Can you find a possible cause by correlating the transaction times and backend monitor matrix?

In the following sections, you will learn how to open LoadRunner Analysis, and build and view graphs and reports that will help you find performance problems and pinpoint the sources of these problems.

How do I start my analysis session?

1. Open HP LoadRunner Analysis.

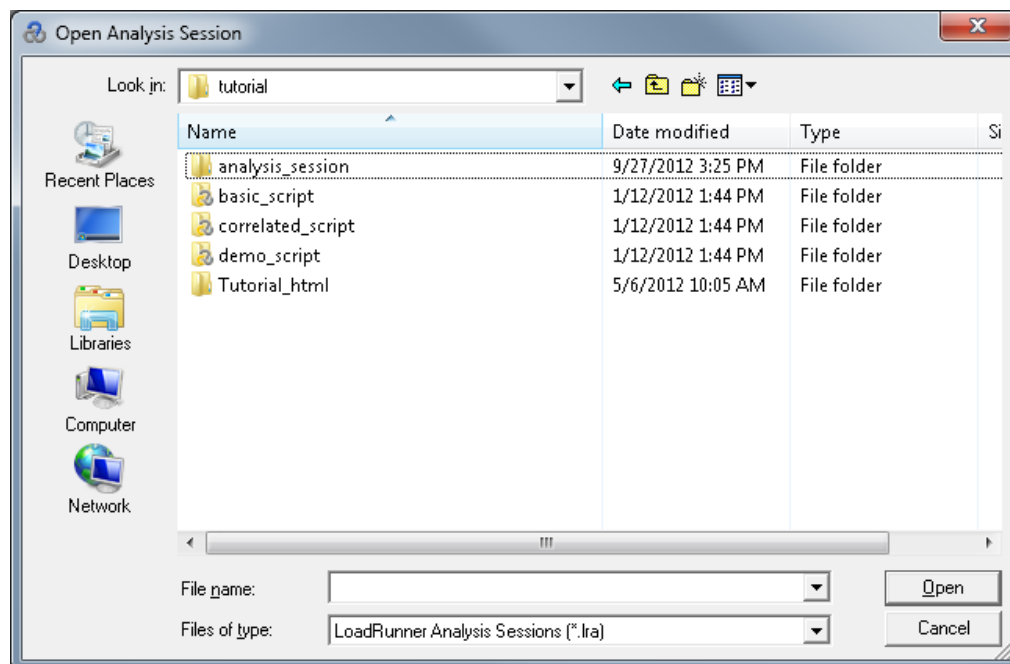
Double-click the **Analysis** icon on the Desktop. LoadRunner Analysis opens.



2. Open the analysis session file.

For the purpose of this section in the tutorial, in order to illustrate more diverse results, we ran a scenario similar to those you ran in the previous lessons. This time, however, the scenario incorporated 70 Vusers rather than 10 Vusers. You will now open the analysis session created from the results of this scenario.

- a. In the Analysis window, select **File > Open**. The Open Analysis Session dialog box opens.

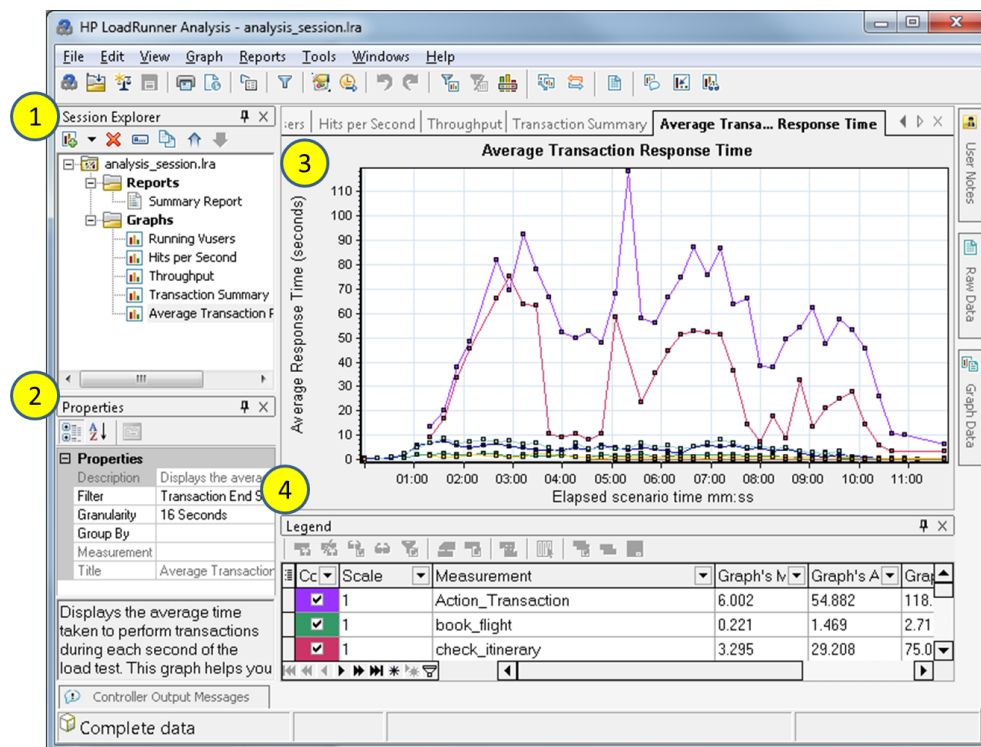


- b. From the <LoadRunner Installation>\tutorial folder, select **analysis_session** and click **Open**. Analysis opens the session file.

The Analysis window at a glance

Analysis contains the following primary panes:

1. Session Explorer
2. Properties pane
3. Graph Viewing pane
4. Legend pane



1. **Session Explorer.** In the upper left pane, Analysis shows the reports and graphs that are open for viewing. From here you can display new reports or graphs that do not appear when Analysis opens, or delete ones that you no longer want to view.
2. **Properties pane.** In the lower left pane, the Properties pane displays the details of the graph or report you selected in the Session Explorer. Fields that appear in black are editable.
3. **Graph Viewing pane.** In the upper right pane, Analysis displays the graphs. By default, the

Summary Report is displayed in this area when you open a session.

4. **Legend pane.** In the lower right pane, you can view data from the selected graph.

Note: There are additional panes that can be accessed from the toolbar. These panes can be dragged and dropped anywhere on the screen.

Did I reach my goals? (Service Level Agreement)

In this section, you will be introduced to the Service Level Agreement, or SLA.

SLAs are specific goals that you define for your load test scenario. Analysis compares these goals against performance-related data that LoadRunner gathers and stores during the run, and then determines the SLA status (**Pass** or **Fail**) for the goal.

For example, you can define a specific goal, or threshold, for the average transaction time of a transaction in your script. After the test run ends, LoadRunner compares the goals you defined against the actual recorded average transaction times. Analysis displays the status of each defined SLA, either **Pass** or **Fail**. For example, if the actual average transaction time did not exceed the threshold you defined, the SLA status will be **Pass**.

As part of your goal definition, you can instruct the SLA to take load criteria into account. This means that the acceptable threshold will vary depending on the level of load, for example, **Running Vusers**, **Throughput**, and so on. As the load increases, you can allow a higher threshold.

Depending on your defined goal, LoadRunner determines SLA statuses in one of the following ways:

- **SLA status determined at time intervals over a timeline.** Analysis displays SLA statuses at set time intervals (for example, every 5 seconds) over a timeline within the run.
- **SLA status determined over the whole run.** Analysis displays a single SLA status for the whole scenario run.

SLAs can be defined either before running a scenario in the Controller, or after in Analysis itself.

In the following section, you will define an SLA using the HP Web Tours example. Assume that the administrator of HP Web Tours would like to know whenever the average transaction times of the **book_flight** and **search_flight** transactions exceed certain values. To do this, you select the transactions and then set threshold values. These threshold values are the maximum amounts of time that would be acceptable as average transaction times.

You will also set these threshold values to take certain load criteria into account; in this case Running Vusers. In other words, as the number of running Vusers increases, the threshold value rises.

This is because although the HP Web Tours administrator would like the average transaction times to be as low as possible, it is understood that at certain times of the year it is reasonable to

assume that the HP Web Tours site will have to handle a higher load than at other times of the year. For example, during peak travel season, a higher number of travel agents log on to the site to book flights, check itineraries, and so on. Given this understandably higher load, at these times a slightly longer average transaction time will be acceptable.

You will set the SLA to take three load scenarios into account: light load, average load, and heavy load. Each scenario will have its own threshold value.

How do I define an SLA?

You will define an SLA in Analysis after the scenario run.

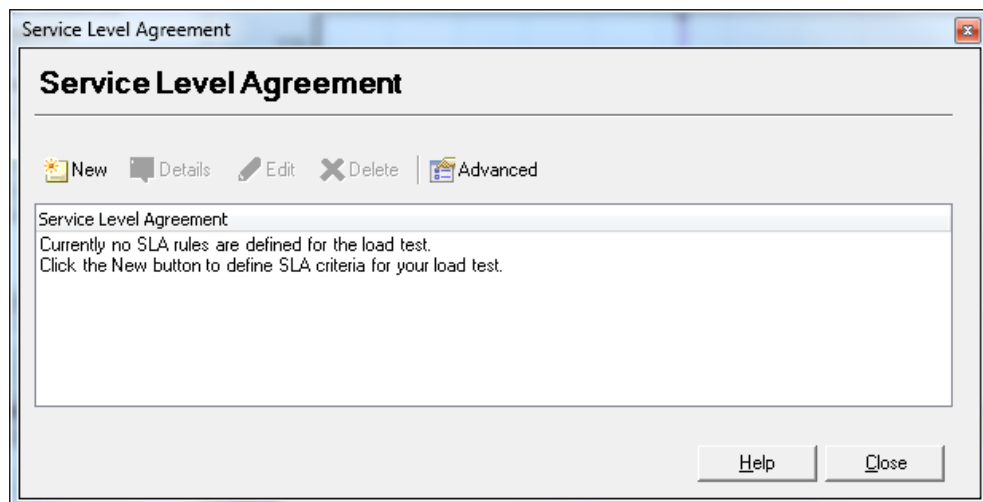
Note: It is preferable to define an SLA in the Controller before a scenario run. However, for the purposes of this tutorial, because you are not analyzing the same test scenario that you ran in previous lessons, you will define the SLA in Analysis. To define an SLA in the Controller, click **New** in the **Service Level Agreement** section of the **Design** tab.

You will now define an SLA that will set specific goals for the average transaction times for the **book_flight** and **search_flight** transactions in the sample session file.

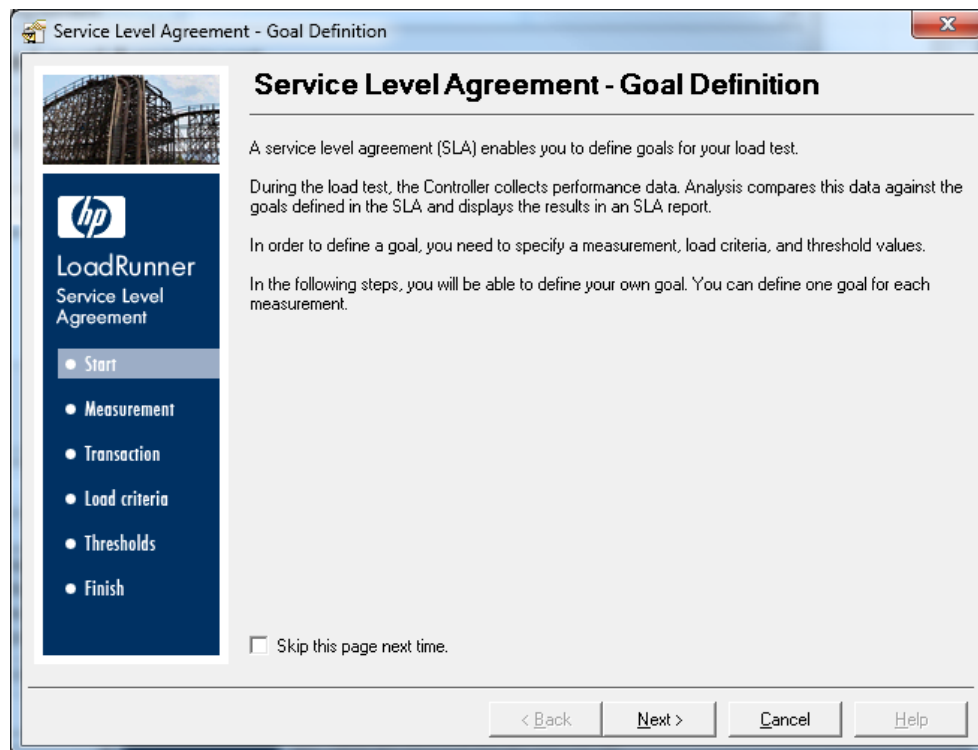
The average transaction times will be measured at set time intervals within the run.

To define an SLA:

1. Open the SLA wizard.
 - a. In LoadRunner Analysis, select **Tools > Configure SLA Rules**. The Service Level Agreement dialog box opens.

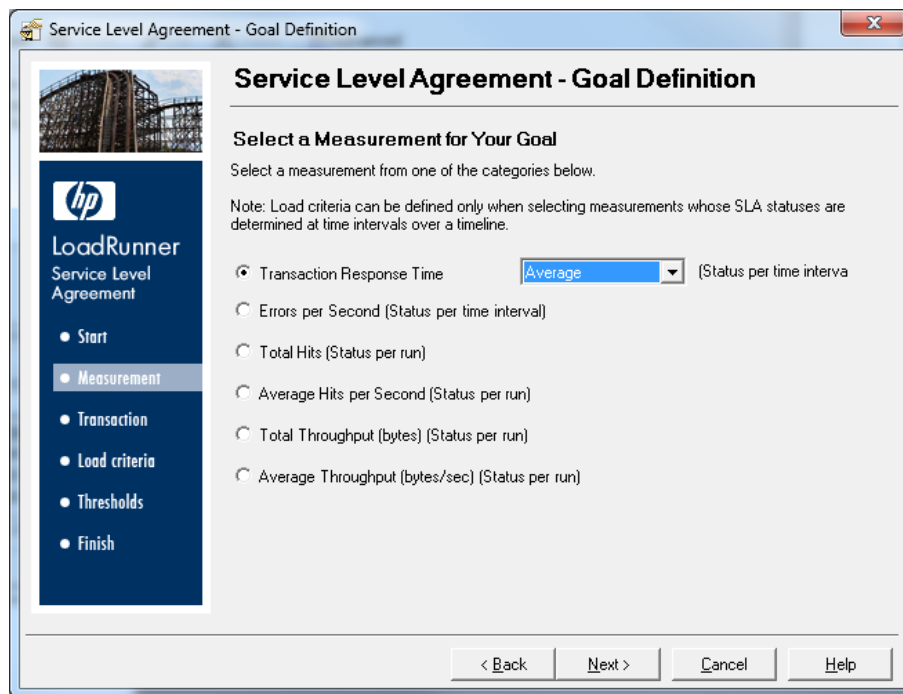


- b. Click **New** to open the Service Level Agreement wizard.



Note that the first time you open the Service Level Agreement wizard, the **Start** page is displayed. If you do not want this page to be displayed the next time you run the wizard, select the **Skip this page next time** check box.

- c. Click **Next**.
2. Select a measurement for your goal.
 - a. In the Select a Measurement for Your Goal page, select **Transaction Response Time: Average**.

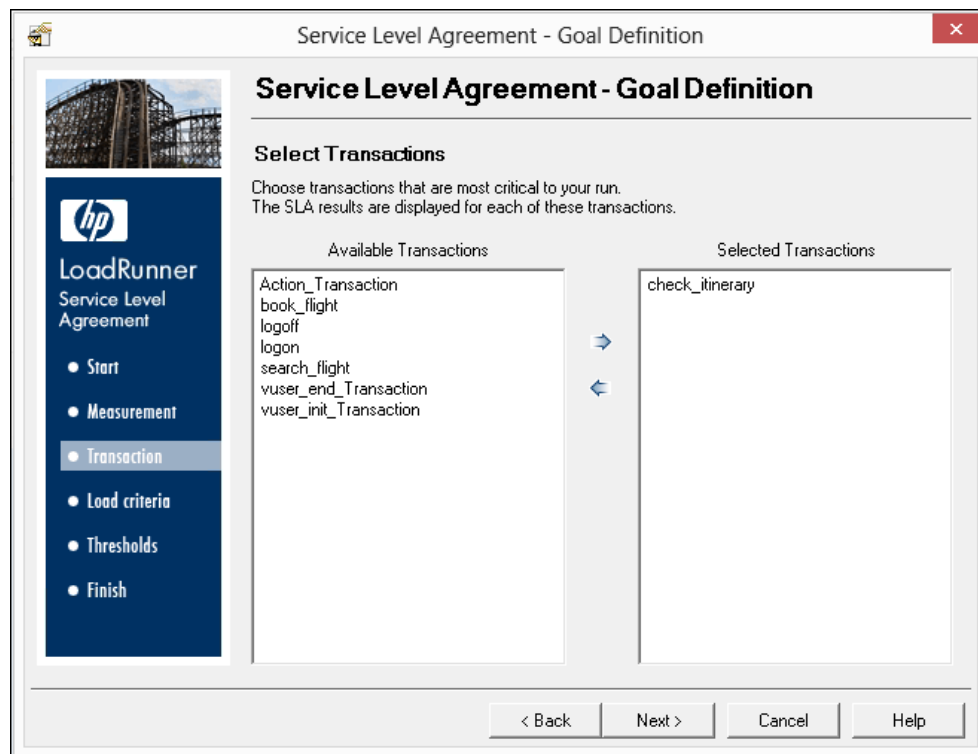


b. Click **Next**.

3. Select the transactions to monitor.

In the Select Transactions page, select a transaction to monitor from the **Available Transactions** list.

a. Double-click the **check_itinerary** transaction to move it to the **Selected Transactions** list.



b. Click **Next**.

4. Set the load criteria.

In the Set Load Criteria page, you instruct the SLA to take different load scenarios into account.

a. Select **Running Vusers** from the **Load Criteria** drop down list.

b. Set the **Load Values** to look like the following example:

Service Level Agreement - Goal Definition

Set Load Criteria

Selected measurement: Transaction Response Time (Average)

Set the load criteria and load values that you want to test.
The threshold values for your goal are examined according to these values.

Note: The load values must cover a continuous and complete range.

Load Criteria: Running Vusers

Load Values:

☒	Less than	20	
☒	Between	20	50
☒	Greater than or equal to	50	

< Back Next > Cancel Help

In the above screen, you set the SLA to define an acceptable average transaction time over three potential load scenarios:

- **Light load.** Between 0 and 19 Vusers
- **Average load.** Between 20 and 49 Vusers
- **Heavy load.** More than 50 Vusers

c. Click **Next**.

5. Set threshold values.

In the Set Threshold Values page, you define the acceptable average transaction times for the **check_itinerary** transaction.

Set the threshold values to look like the following example:

Service Level Agreement - Goal Definition

Set Threshold Values

Selected measurement:

Enter maximum thresholds per load value for each transaction. Results that exceed the threshold values will produce a failed SLA status.

Tip: You can apply one set of thresholds to all transactions in the Apply to all table below.

Transaction Name	<20	≥20 and <50	≥50
check_itinerary	5	10	15

To apply one set of threshold values to all transactions, enter the threshold values and click Apply to all transactions.

Apply to all:	<20	≥20 and <50	≥50
	0	0	0

< Back Next > Cancel Help

You just indicated that the following average transaction times are acceptable:

- **Light load.** 5 seconds or less
- **Average load.** 10 seconds or less
- **Heavy load.** 15 seconds or less

6. Save the SLA.

To save the SLA and close the wizard, click **Next** then **Finish** then **Close** on the pages that follow.

Analysis applies your SLA settings to the Summary Report. The report is then updated to include all the relevant SLA information.

How do I view a summary of performance?

The Summary Report tab displays general information and statistics about the scenario run, as well as all relevant SLA information. For example, what were the worst performing transactions in terms of defined SLAs, how specific transactions performed over set time intervals, and overall SLA statuses. You open the Summary Report from the Session Explorer.

1. What are the overall scenario statistics?

In the Statistics Summary section, you can see that a maximum of 70 Vusers ran in this test. Other statistics such as the total/average throughput, and the total/average hits are also displayed.

Summary Report		Running Vusers	Hits per Second	Throughput	Transaction Summary	Average
Analysis Summary						Period: 28/06/2004 09:5
Scenario Name: C:\Tutorial_Scenario.lrs Results in Session: S:\tutorial_result\tutorial_result.lrr Duration: 11 minutes and 57 seconds.						
Statistics Summary						
.....						
Maximum Running Vusers:				70		
Total Throughput (bytes):				273,489,363		
Average Throughput (bytes/second):				380,904		
Total Hits:				8,789		
Average Hits per Second:				12.241	View HTTP Responses Summary	
Total Errors:				9,226		

2. What were the worst performing transactions?

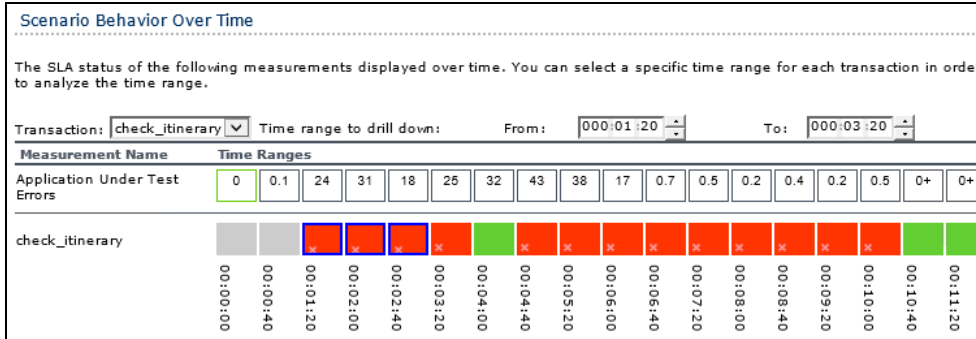
The 5 Worst Transactions table shows you up to five worst-performing transactions for which SLAs were defined.

5 Worst Transactions		
.....		
Transaction Name	Failure Ratio[%] (exceeded time/transaction duration)	Failure Value[%] (response time/SLA)
- check_itinerary	66.4	200.684
Time Interval	Avg exceeding ratio	Max exceeding ratio
00:01:35-00:03:50	258.37	638.59
00:05:10-00:07:25	200.48	398.07
00:07:30-00:08:00	142.07	287.3
00:08:15-00:08:30	61.8	155.9
00:08:45-00:10:25	161.49	356.6

You can see that over the duration of the **check_itinerary** transaction, the SLA threshold was exceeded 66.4% of the time. The average percentage by which it exceeded the SLA threshold over the whole run was 200.684%.

3. Over which time intervals was the SLA threshold exceeded?

The Scenario Behavior Over Time section shows how each transaction performed during different time intervals. The green squares show time intervals where the transaction performed within the SLA threshold, red squares where the transaction failed, and gray squares where no relevant SLA was defined.



You can see that for the transaction for which you defined an SLA, **check_itinerary** exceeded the threshold in most intervals.

4. What was the overall transaction performance?

The Transaction Summary lists a summary of the behavior of each transaction.

Transaction Summary									
Transactions: Total Passed: 1,100 Total Failed: 3,112 Total Stopped: 0					Average Response Time				
Transaction Name	SLA Status	Minimum	Average	Maximum	Std. Deviation	90 Percent	Pass	Fail	Stop
Action Transaction		6.002	58.577	155.399	25.387	89.081	144	3,081	0
book flight		0.219	1.555	3.949	0.817	2.666	175	0	0
check_itinerary		3.295	32.826	119.258	26.407	65.754	147	28	0
logout		0.406	1.005	12.909	1.146	1.67	144	3	0
login		0.444	3.934	9.864	2.161	6.777	175	0	0
search flight		0.464	4.915	11.17	2.365	7.966	175	0	0
vuser_end Transaction		0	0	0	0	0	70	0	0
vuser_init Transaction		0	0.013	0.077	0.024	0.059	70	0	0

Service Level Agreement Legend: Pass Fail No Data

We also see that the **check_itinerary** transaction failed 28 times.

Review the times of each transaction. The **90 Percent** column displays the time of 90% of the executions of a particular transaction. You can see that 90% of the **check_itinerary** transactions that were performed during the test run had a time of 65.754 seconds. This is double its average time, 32.826, which means that the majority of occurrences of this transaction had a very high response time.

Note how the **SLA Status** column shows the relevant overall status for transactions in the SLA: **Fail** for **check_itinerary**.

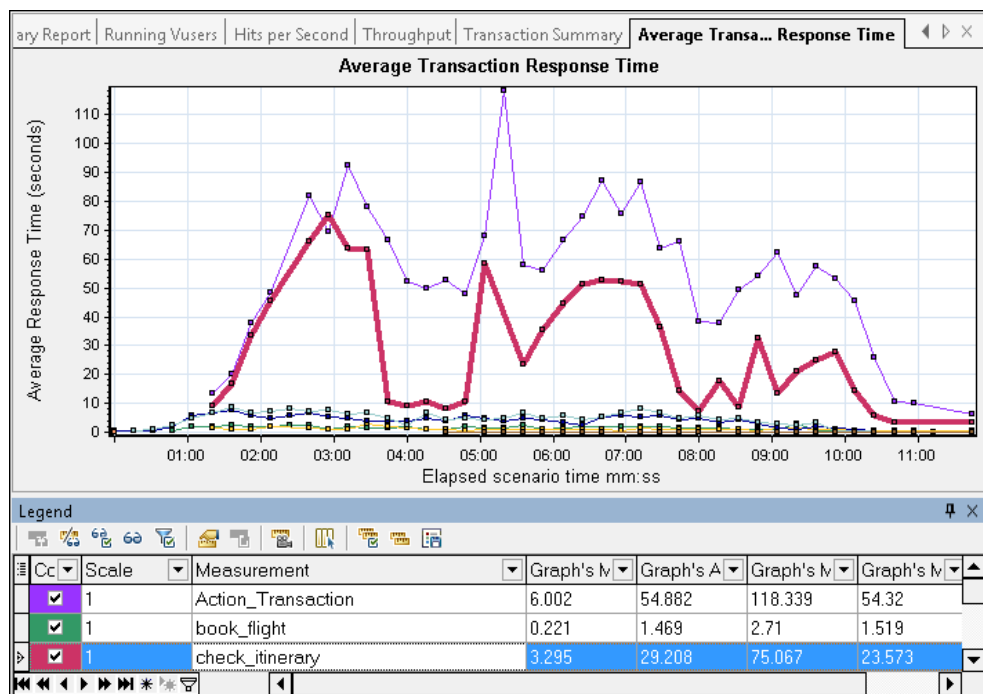
How do I graphically view performance?

You can access available graphs from the Session Explorer pane. You will now view and analyze the Average Transaction Response Time graph.

1. Open the Average Transaction Response Time graph.
 - a. In the Session Explorer under **Graphs**, select **Average Transaction Response Time**. The Average Transaction Response Time graph opens in the graph viewing area.

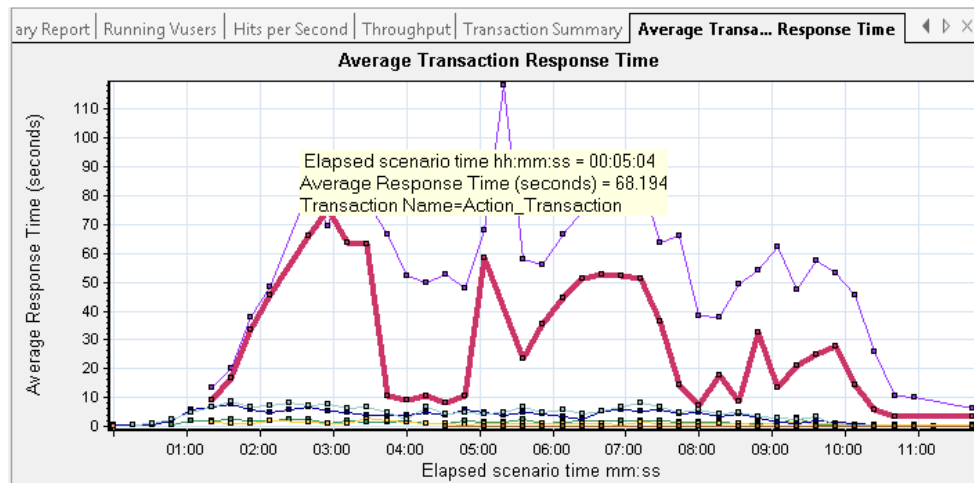
Note: If no graphs are displayed in the Session Explorer pane, right-click the Graphs node and select the Transactions: Average Transaction Response Time node in the Open a New Graph dialog box. Click Open Graph to add the graph to the Session Explorer pane.

- b. In the Legend pane, click the **check_itinerary** transaction. The **check_itinerary** transaction is highlighted in the graph.



The points on the graph represent the average time of a transaction at a specific time during the scenario. Hold your cursor over a point in the graph. A yellow box appears, and

displays the coordinates of that point.



2. Analyze the results.

Note how the average transaction time of the **check_itinerary** transaction fluctuates greatly, and reaches a peak of 75.067 seconds, 2:56 minutes into the scenario run.

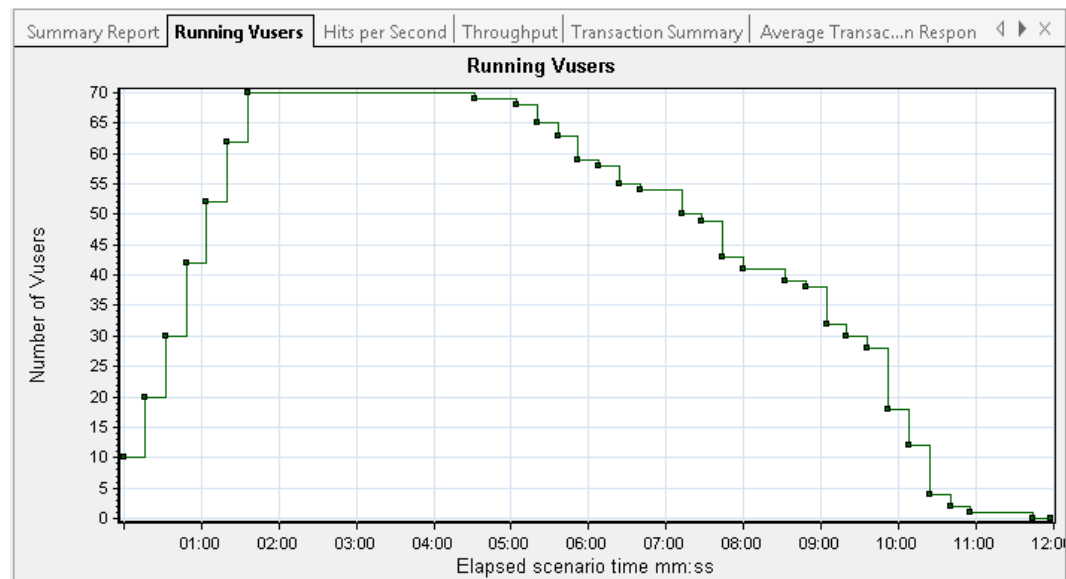
On a well-performing server, the transactions would follow a relatively stable average time. At the bottom of the graph, note how the **logon**, **logoff**, **book_flight**, and **search_flight** transactions have more stable average times.

Did my server perform well?

In the previous section you saw instability in your server's performance. Now you will analyze the effect of 70 running Vusers on the system's performance.

1. Study the behavior of the Vusers.

In the Session Explorer, under **Graphs**, click **Running Vusers**. The **Running Vusers** graph opens in the graph viewing area.



You can see that there was a gradual start of running Vusers at the beginning of the scenario run. Then, for a period of 3 minutes, 70 Vusers ran simultaneously, after which the Vusers gradually stopped running.

2. Filter the graph so that you see only the time slice when all the Vusers ran simultaneously.

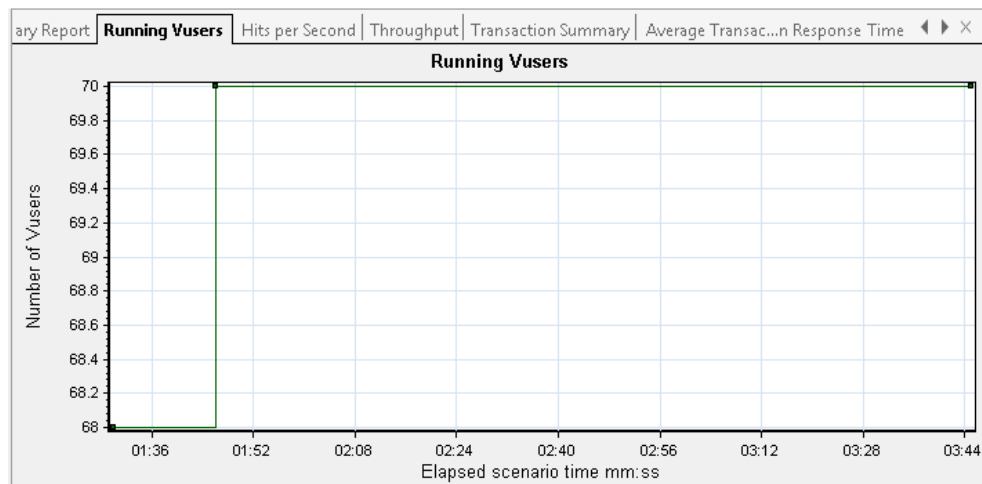
When you filter a graph, the graph data is narrowed down so that only the data for the condition that you specified is displayed. All other data is hidden.

- a. Right-click the graph and select **Set Filter/Group By**, or alternatively, click the **Set**

Filter/Group By button  on the Analysis toolbar.

- b. In the **Filter Condition** area, select the **Values** column of the **Scenario Elapsed Time** row.
- c. Click the down-arrow and specify a time range from **000:01:30** minutes to **000:03:45** minutes.
- d. Click **OK**.
- e. In the Graph Settings dialog box, click **OK**.

The **Running Vusers** graph now displays only those Vusers running between 1:30 minutes and 3:45 minutes of the scenario run. All other Vusers have been filtered out.



Note: To clear the filter, you right-click the graph and select **Clear Filter/Group By**, or alternatively, click the **Clear Filter and Group By** button  on the Analysis toolbar.

3. Correlate the Running Vusers and Average Transaction Response Time graphs to compare their data.

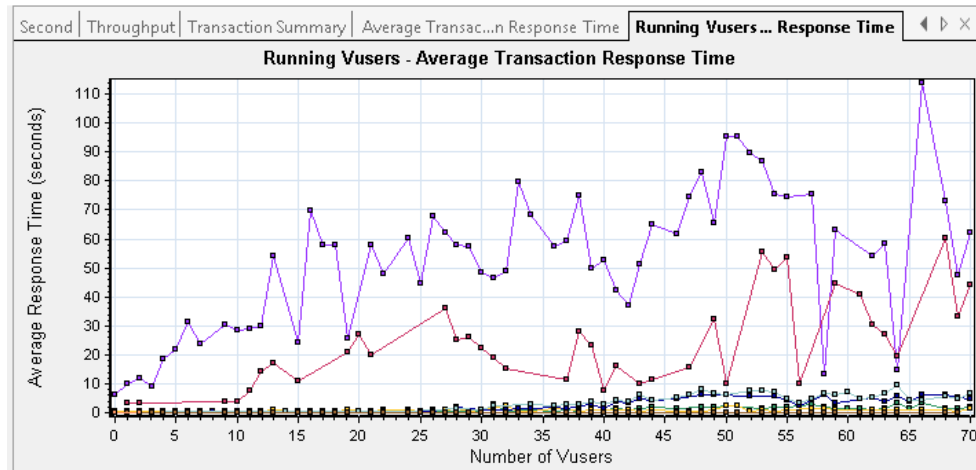
You can join two graphs together to see the effect of one graph's data upon the other graph's data. This is called correlating two graphs.

For example, you can correlate the Running Vusers graph with the Average Transaction Response Time graph to see the effect of a large number of Vusers on the average time of the transactions.

- a. Right-click the Running Vusers graph and select **Clear Filter/Group By**.
- b. Right-click the graph and select **Merge Graphs**.
- c. From the **Select graph to merge with** list, select **Average Transaction Response Time**.
- d. Under **Select type of merge**, select **Correlate**, and click **OK**.

The Running Vusers and Average Transaction Response Time graphs are now displayed

in one graph, the Running Vusers - Average Transaction Response Time graph.



4. Analyze the correlated graph.

In this graph you can see that as the number of Vusers increases, the average time of the **check_itinerary** transaction gradually increases. In other words, the average time increases as the load increases.

At 66 Vusers, there is a sudden, sharp increase in the average time. We say that the test **broke the server**. The time clearly began to degrade when there were more than 66 Vusers running simultaneously.

Saving a template

So far you have filtered a graph and correlated two graphs. The next time you analyze a scenario, you might want to view the same graphs, with the same filter and merge conditions applied. You can save your merge and filter settings into a template, and apply them in another analysis session.

To save your template:

1. Select **Tools > Templates**. The Apply/ Edit Template dialog box opens.
2. In the Templates pane, click the **New** button. The Add New Template dialog box opens.
3. Enter an appropriate name for the template and click **OK**.
4. Click **Save and close** to close the Apply/Edit Template dialog box.

The next time you open a new Analysis session and want to use a saved template:

1. Select **Tools > Templates**. The Apply/ Edit Template dialog box opens.
2. Select your template from the list, and click **Save and close**.

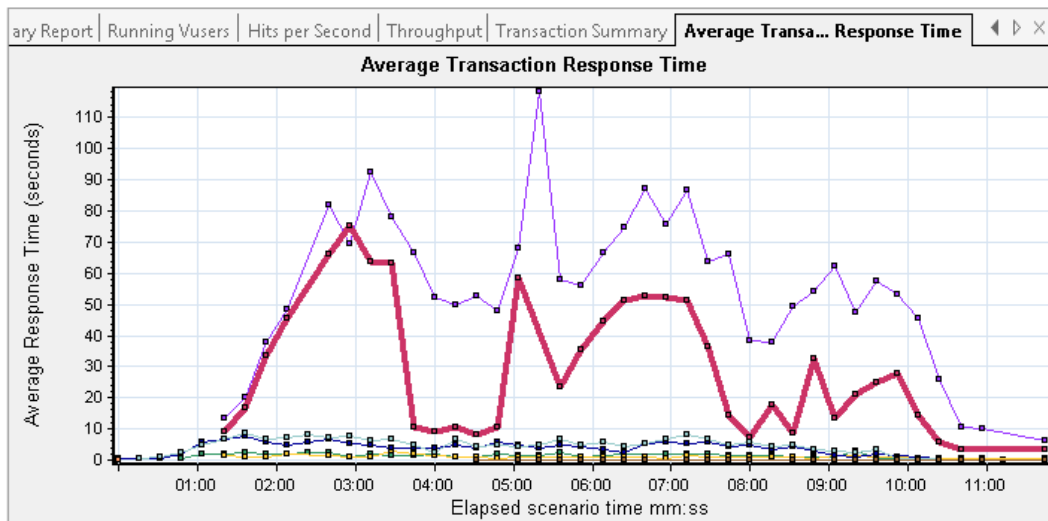
How can I pinpoint the source of the problem?

Until now, you have seen that an increase in load on the server had a negative impact on the average response time of the **check_itinerary** transaction.

You can drill down further into the **check_itinerary** transaction to see which system resources may have negatively influenced its performance.

The Auto-correlate tool can merge all the graphs that contain data that could have had an effect on the response time of the **check_itinerary** transaction, and pinpoint what was happening at the moment the problem occurred.

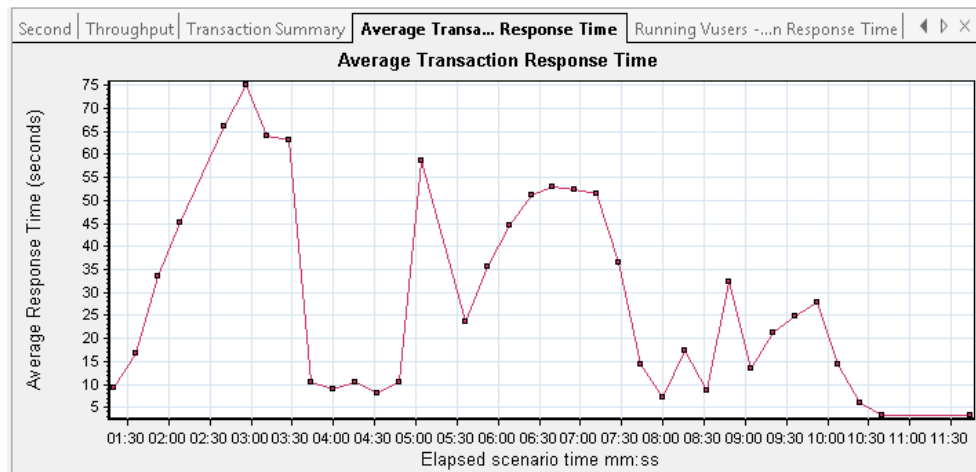
1. From the graph tree, select the **Average Transaction Response Time** graph.



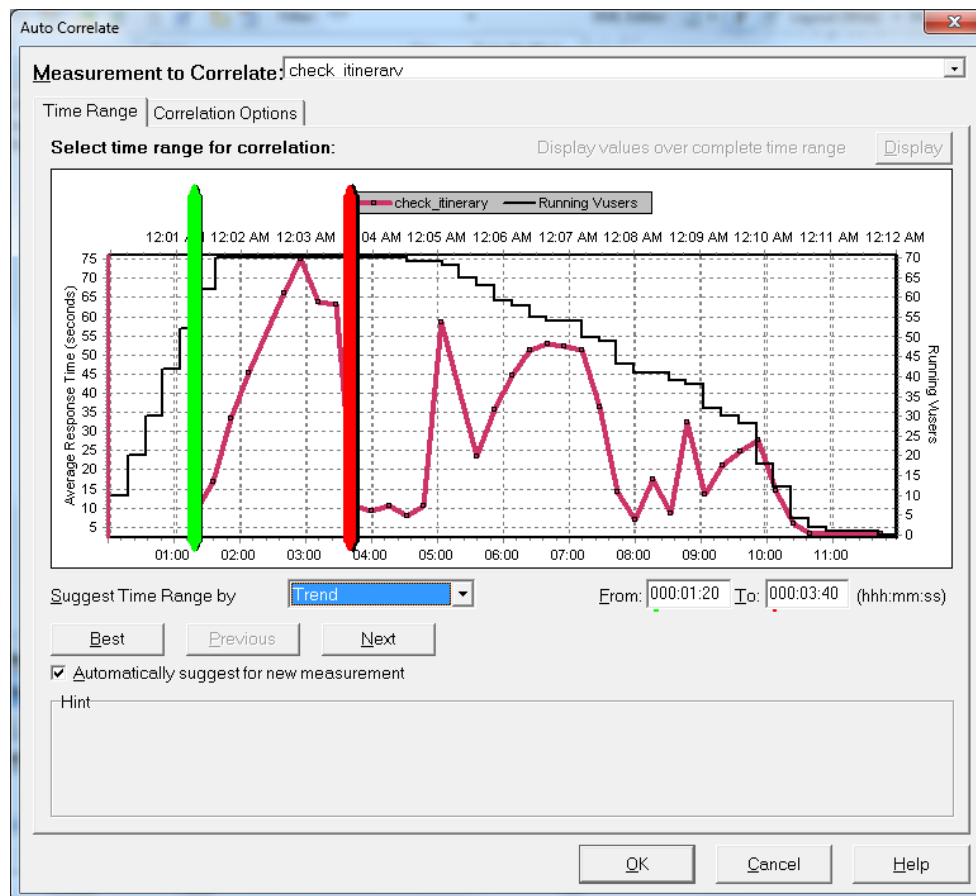
Look at the **check_itinerary** transaction, particularly at the slice of elapsed time between 1 and 4 minutes. The average response time started to increase almost immediately, until it peaked at nearly 3 minutes.

2. Filter the Average Transaction Response Time graph to display only the **check_itinerary** transaction.
 - a. Right-click the graph, and select **Set Filter/Group by**.
 - b. In the **Transaction Name / Value** cell, select **check_itinerary**.
 - c. Click **OK**.

The filtered graph displays only the **check_itinerary** transaction.

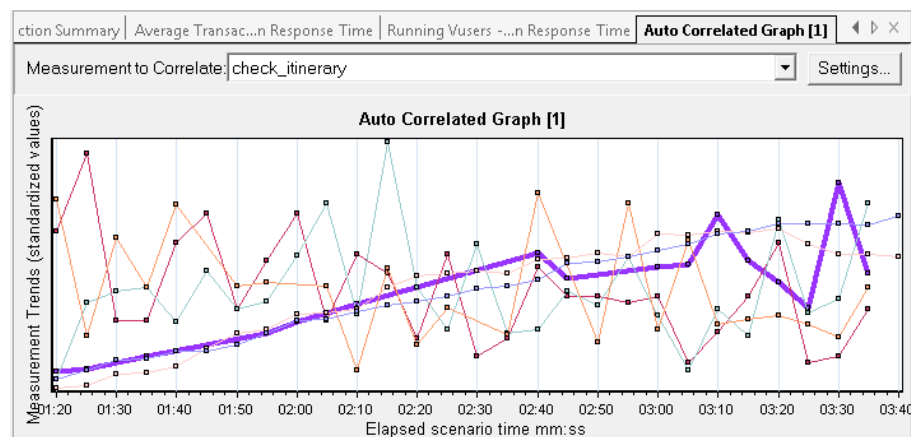


3. Auto-correlate the graph.
 - a. Right-click the graph, and select **Auto Correlate**.
 - b. In the Auto Correlate dialog box, make sure that the measurement to correlate is **check_itinerary**, and set the time range from 1:20 to 3:40 minutes - either by entering the times in the boxes, or by dragging the green and red poles into place along the **Elapsed Scenario Time** axis.



c. Click **OK**.

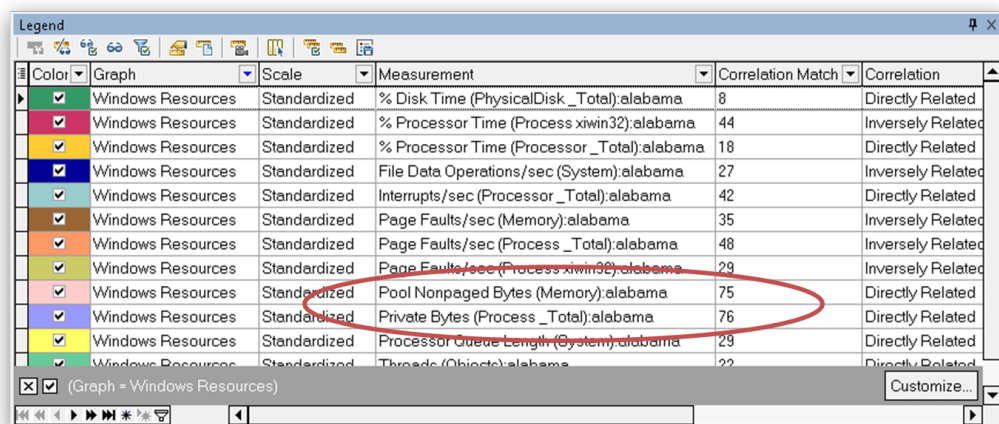
The auto-correlated graph opens in the graph viewing area. The **check_itinerary** transaction is highlighted.



The auto-correlated graph is given a default name, **Auto Correlated Graph [1]**.

4. Rename the graph.
 - a. In the Session Explorer, under **Graphs**, right-click **Auto Correlated Graph [1]**, and select **Rename Item**. The graph name becomes editable.
 - b. Type **Auto Correlated - check_itinerary** and press **Enter**, or click anywhere in the Analysis window.
5. Analyze the auto-correlated graph.

In the **Legend** pane below the graph, from the **Graph** column, scroll down to the **Windows Resources: Pool Nonpaged Bytes and Private Bytes** measurements.



Color	Graph	Scale	Measurement	Correlation Match	Correlation
✓	Windows Resources	Standardized	% Disk Time (PhysicalDisk _Total):alabama	8	Directly Related
✓	Windows Resources	Standardized	% Processor Time (Process xiwin32):alabama	44	Inversely Related
✓	Windows Resources	Standardized	% Processor Time (Processor _Total):alabama	18	Directly Related
✓	Windows Resources	Standardized	File Data Operations/sec (System):alabama	27	Inversely Related
✓	Windows Resources	Standardized	Interrupts/sec (Processor _Total):alabama	42	Directly Related
✓	Windows Resources	Standardized	Page Faults/sec (Memory):alabama	35	Inversely Related
✓	Windows Resources	Standardized	Page Faults/sec (Process _Total):alabama	48	Inversely Related
✓	Windows Resources	Standardized	Page Faults/sec (Process xiwin32):alabama	29	Inversely Related
✓	Windows Resources	Standardized	Pool Nonpaged Bytes (Memory):alabama	75	Directly Related
✓	Windows Resources	Standardized	Private Bytes (Process _Total):alabama	76	Directly Related
✓	Windows Resources	Standardized	Processor Queue Length (System):alabama	29	Directly Related
✓	Windows Resources	Standardized	Threads (Object):alabama	22	Directly Related

In the **Measurement** and **Correlation Match** columns, you can see that these memory-related measurements, have a Correlation Match of over 70% with the **check_itinerary** transaction. This means that the behavior of these elements was closely related to the behavior of the **check_itinerary** transaction during the specified time interval.

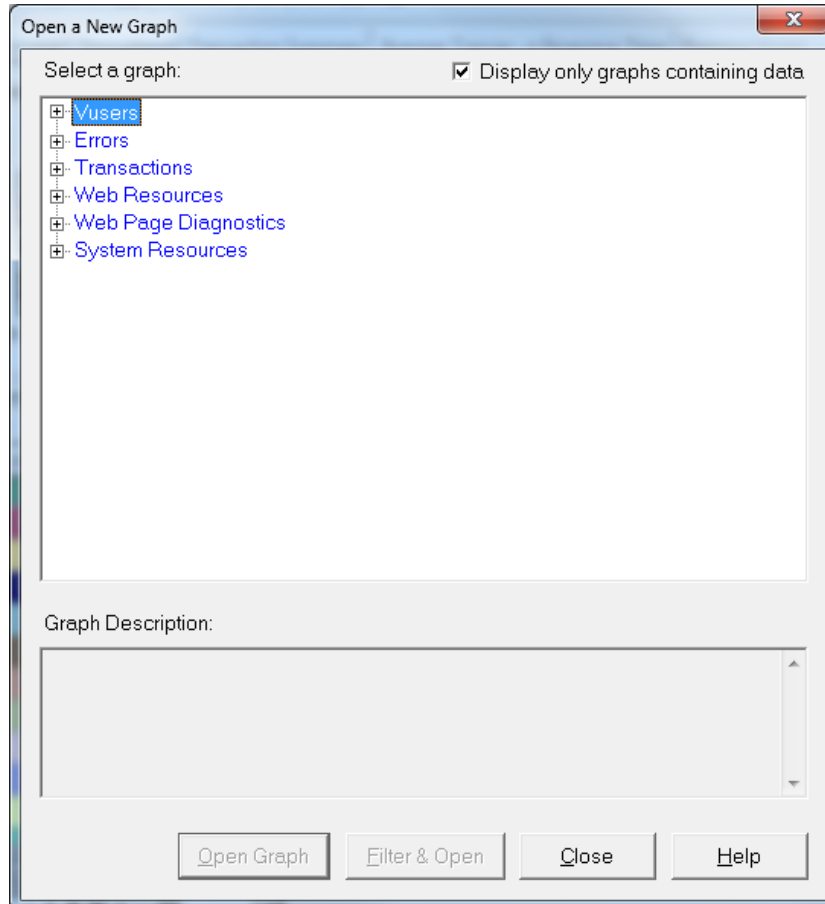
We can conclude that exactly when the **check_itinerary** transaction's response time peaked, there was a shortage of system memory resources.

What other information can I gather about my scenario run?

In addition to the graphs that appear in the graph tree at the start of an analysis session, you can display different graphs to get other information about your scenario run.

1. Click **Graph > Add New Graph**.

The Open a New Graph dialog box opens and lists the categories of graphs that contain data and can be displayed.



- **Vusers.** Displays information about the Vusers and their status.
 - **Errors.** Displays error statistics.
 - **Transactions.** Displays data about transactions and their response times.
 - **Web Resources.** Displays hits, throughput, and connection data.
 - **Web Page Diagnostics.** Displays data about each monitored Web page in your script.
 - **System Resources.** Displays system resource usage data.
2. Display a new graph.
 - a. In the Open a New Graph dialog box, click the “+” next to a category to expand it.
 - b. Select a graph and click **Open Graph**.

- c. Click **Close** to close the Open a New Graph dialog box.

Now open several additional graphs to understand more about your scenario run.

How can I publish my findings?

You can publish the findings from your analysis session in an HTML or Microsoft Word report. The report is created using a designer template, and includes explanations and legends of the presented graphs and data.

HTML Reports

The HTML report can be opened and viewed in any browser.

To create an HTML report:

1. Click **Reports > HTML Report**.
2. Specify a file name for your report, and the path where you want to save it.
3. Click **Save**.

Analysis creates the report and displays it in your Web browser. Note how the layout of the HTML report is very similar to the layout of your analysis session. You can click on the links in the left pane to see the various graphs. A description of each graph is given at the bottom of the page.

Microsoft Word Reports

You can present your analysis session in a Microsoft Word report. The Word report is more comprehensive than the HTML report, because you have the option to include general information about the scenario, measurement descriptions, and so on. You can also format the report to include your company's name and logo, and the author's details.

Like any Microsoft Word file, the report is editable, so you can add further comments and findings after you build the report.

To create a Microsoft Word report:

1. Click **Reports > New Report**.

The New Report dialog box opens.

New Report

Based on template: Customer facing (for cross session)

General | Format | Content

Title
Customer facing (for cross session)

Author
First Name:
Surname:
Job Title:
Organization:

Description
By using this template you can easily generate a cross session customer facing report that includes a wide range of performance statistics. This template will automatically include any open graph you have in your analysis session as well as the following information:
 - General details
 - Executive summary
 - Business processes
 - Workload characteristics
 - Performance overview for high level aggregated metrics and transactions
 - Worst performing URLs based on Web Page Diagnostics data

Global Settings
 Report Time Range: Whole Scenario (hhh:mm:ss)
 Granularity: 1 (sec)
 Precision: 1 (Number of digits to appear after the decimal point in none graph content items)
☐ Include Think Time
☐ Use Raw Result Time Zone

Save As Template | Generate | Cancel | Help

2. In the **General** tab:
 - a. From **Based on template**, select **Detailed report (for single run)**.
 - b. Enter a title for your report.
 - c. Enter the author's name, job title, and the company's name.

3. In the **Format** tab:

By default, the report will be built with a title page, table of contents, graph details and descriptions, and measurement descriptions. You can select options which add script details into the report, allowing you to view thumbnail images of the business process steps.

You can include a company logo by selecting **Include company logo** and browsing to the file location. The logo must be a .bmp file.

4. In the **Content** tab:
 - a. Select which sections of your scenario run and analysis session you want to include in your report.

For the purpose of this tutorial, you will add an executive summary to the Content Items list.

- b. Click the **Add** button  to open the Add Content Items window. Check **Executive**

Summary in the grid and click **OK**. The Executive Summary item is added to the list in the **Content Items** pane.

Enter the following text into the edit box:

- **Objectives:** The objectives of the test scenario were to....

- **Conclusions:** The conclusions I reached are as follows:

c. In the **Content Items** pane, select the **Largest URLs by Average Kbytes** and click the **Delete** button . This will exclude this graph from the report.

d. Change the order in which to display the items in the report.

- In the **Content Items** pane, select **Workload Characteristics**. Click on **Average Hits per Second** in the **Selected Columns** list.

- Click the Down arrow until the item appears under **Total Transactions Number**. In the report, the **Average Hits per Second** item will follow the **Total Transactions Number** item.

5. Click **Generate**.

The data is gathered and the report is created in a Microsoft Word file, which opens in Microsoft Word.

In addition to the graphs that you generated during your analysis session, the report includes an objective and a conclusion, and other sections and graphs that you chose to include while building the report.

Conclusion

In this lesson you learned the basics of defining a Service Level Agreement, analyzing a scenario run, and publishing your results in a report.

You have learned that performance problems can be pinpointed by studying various graphs that show bottlenecks on the server, possibly due to too heavy a load. You have seen that you can pinpoint the sources of these bottlenecks by configuring graphs to display correlated data.